



ISO 9001: 2015

ARTIFICIAL LIMBS MANUFACTURING CORPORATION OF INDIA
(A GOVERNMENT OF INDIA UNDERTAKING MINIRATNA COMPANY)
UNDER MINISTRY OF SOCIAL JUSTICE & EMPOWERMENT
AN ISO 9001:2008 COMPANY, G T ROAD, KANPUR- 209217
E-mail: alimco@alimco.in , Website: www.alimco.in
Toll Free No.: 1800 180 5129



NOTICE

As per instruction of competent authority, Tender Documents for Establishing a warehouse on Engineering, Procurement and Construction (EPC) basis at plot no. y-9 in industrial area trans ganga city Unnao, U.P is to be uploaded on corporation website for Suggestions/feedback/Comments from the prospective bidders/ vendors/ suppliers/OEM/experts if any, on or before 10/05/2025.

The observation/objections, if any, may be sent to email:

sm_pt@alimco.in latest by 10/05/2025 up to 17.30 Hrs.

SM(Projects), ALIMCO, KANPUR



ALIMCO

**Artificial Limbs Manufacturing Corporation of India
(A GOVERNMENT OF INDIA UNDERTAKING)**

G.T ROAD, KANPUR (U.P)

209217

TENDER NO.PT/OT/...../BD

DATED -/..... /2025

TENDER DOCUMENTS

FOR

**ESTABLISHING A WAREHOUSE ON ENGINEERING, PROCUREMENT AND
CONSTRUCTION (EPC) BASIS**

AT

PLOT NO. Y-9 IN INDUSTRIAL AREA TRANS GANGA CITY UNNAO, U.P

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NOTICE INVITING e-TENDER (NIT)

1.0 NOTICE INVITING e-TENDER (NIT)

Artificial Limbs Manufacturing Corporation of India (ALIMCO) invites Online open tenders in two parts i.e., Techno-Commercial Bid (TC-Bid) & Price Bid (P-Bid) for Establishing a Warehouse on Engineering, Procurement and construction (EPC) basis at plot no. y-9 in industrial area trans ganga city Unnao, U.P. as per schedule given below;

Bids shall be submitted online only at CPPP: <https://eprocure.gov.in/eprocure/app>.

Manual bids shall not be accepted.

1.1	Tendering Document No.	PT/.../.../BD, DATED2025
1.2	Name of the Work	Establishing a Warehouse on Engineering, Procurement and construction (EPC) basis at plot no. y-9 in industrial area trans ganga city Unnao, U.P
1.3	Estimated Cost (Incl. GST)	Rs. 3038 lacs
1.4	Period of Completion	15 Months
1.5	Earnest Money Deposit ('EMD')	Rs.5,00,000.00 (Five lakh Only) To be submitted through online payment gateway https://www.onlinesbi.sbi/sbicollect/icollecthome.htm?corpID=321478 available at www.alimco.in
1.6	Last date & time of submission of Tender Documents	Up to.....by 3.00 PM (IST)
1.7	Date & Time of Opening of Technical Bid	AT..... 11.30 AM
1.8	Last date of submission of query by bidder	At..... 3.00 PM
1.9	Pre-Tender Meeting & Venue	At.....11.30 AM The Pre-Bid meeting shall be held through VC/ physically.
1.10	Date of submission of reply by ALIMCO	At..... 3.00 PM
1.11	Date & Time of Opening of Financial Bid	To be intimated later.
1.12	Bid Validity of offer	120 days from the date of opening of tender.

Tender documents may be downloaded from ALIMCO website www.alimco.in (for reference only) and CPPP website <https://eprocure.gov.in/eprocure/app> as per the schedule.

2.0 Minimum Eligibility Criteria:

The interested **bidders / tenderers** should meet the following minimum Pre Qualifying Criteria, otherwise their Bids shall not be considered:

A. Work Experience:

Experience of having successfully completed “similar works” during the last 7 years ending last date of submission of tenders as per NIT

- a. Three similar works each costing not less than 40% of the estimated cost of the tender
OR
- b. Two similar works each costing not less than 50% of the estimated cost of the tender.
OR
- c. One similar work costing not less than 80% of the estimated cost of the tender.

“Similar works” shall mean “Warehouses/Industrial Sheds/ Building works”

The required past experience in “similar work” mentioned above at A should be supported by order/contract copies and Commission /Completion Certificates in the last 7 years, reckoned from date of opening of this tender. Satisfactory Performance Certificate from at least 1 user issued within last 1 year from date of opening of this tender mentioning the order no.& date, date of completion, final project value and name of contractor with full address.

Certificates in the name of other Companies like Subsidiary/Parent/Group Company/Own works: Any company/firm while submitting tender can use the work experience of its subsidiary company to the extent of its ownership in the subsidiary company.

On the other-hand, the companies/firms which intend to get qualified on the basis of experience of only parental company/group company/Own works shall not be considered

Work executed as labour contract shall not be considered

Joint-venture / consortia of companies are not eligible to quote for the Tender.

B. Financial Strength:

- i) The Average Annual Audited financial turnover for last 3 years shall be at least 30% of the estimated cost of the tender. A certificate authenticated by a qualified Chartered Accountant mentioning UDIN regarding average turnover is required to be furnished along with the tender.

In case of Companies/Firms less than 3 years old, the Average annual financial turnover shall be worked out for the available period only.

- ii) Net Worth of the company/firm (certified by a qualified Chartered Accountant mentioning UDIN) as on last day of preceding Financial Year, should be positive.
- iii) Copy of Bank Solvency Certificate issued from Nationalized or any Scheduled Bank for at least 30% of Estimated Cost of the tender. The certificate should have been issued within 12 months from last date of the submission of the tender.
- iv) The Bidder should have earned profit for at least one year in the last three consecutive financial years.

The bidders are required to submit Summarized pages of audited balance sheet and Profit & loss Accounts of previous three financial years.

In case audited balance sheet and Profit & loss Account of previous three financial years not available, the unaudited balance sheet and Profit & loss Account of previous three financial years duly certified by the Qualified chartered accountant mentioning UDIN is to be submitted.

3.0 The intending tenderer must read the terms and conditions of ALIMCO carefully. They should only submit tender if they consider themselves eligible and in possession of all the documents required. Information and Instructions for Tenderers posted on Website(s) shall form part of Tender Document.

4.0 The intending tenderer must have class-II/III digital signature, to submit the tender.

5.0 The Tender Document as uploaded can be viewed and downloaded free of cost by anyone including intending tenderer. But the tender can only be submitted on the e-tender website <https://eprocure.gov.in/eprocure/app> through digital signature.

6.0 Set of Contract/Tender Documents:

The following documents will constitute set of tender documents:

- a) Notice Inviting e-Tender
- b) Quoting Sheet for Tenderer
- c) Instructions to Tenderers & General Conditions of Contract
- d) Scope of Works
- e) Technical Specifications
- f) List of approved makes of materials
- g) Drawings (Page_____to_____)
- h) NOTICE INVITING e-TENDER (NIT)
- i) Acceptance of Tender Conditions (Annexure-III)
- j) Addendum/Corrigendum, if any

- k) Special instruction to Tenderer,
- l) Time lines /Mile stone schedule
- m) Pre-bid clarifications, if any

- 7.0** The tenderers are required to quote strictly as per terms and conditions, specifications, standards and List of approved makes & scope of works given in the tender documents and not to stipulate any deviations.
- 8.0** The bidders are advised in their own interest to submit their bid documents, duly signed & stamped by authorized person well in advance. However, after submission of the tender, the tenderer can re-submit revised tender any number of times before last date and time of submission of tender as notified.
- 9.0** On opening date, the tenderer can login and see the tender opening process.
- 10.0** Contractor can upload documents in the form of JPG format and PDF format.
- 11.0** Contractor is required to upload scanned copies of all the documents including valid GST registration/EPF registration, PAN No. as stipulated in the tender document.
- 12.0** If the contractor is found ineligible after opening of tenders, or his tender is found invalid, cost of tender document fee shall not be refunded.
- 13.0** Notwithstanding anything stated above, ALIMCO reserves the right to assess the capabilities and capacity of the tenderer to perform the contract, in public interest and the overall interest of ALIMCO. In case, tenderer's capabilities and capacities are not found satisfactory, ALIMCO reserves the right to reject the tender.
- 14.0** Tenderer must ensure to quote "Percentage Above or Below the Estimated Cost" as per the Financial Bid.
- 15.0** The Estimated cost of project is Rs.3038 lacs (Details as per **Annexure II**).

PROFORMA FOR FINANCIAL BID

Validate	Print	Help	Percentage BoQ				
Tender Inviting Authority: Artificial Limbs Manufacturing Corporation of India (ALIMCO) Kanpur							
Name of Work-Establishing a Warehouse on Engineering, Procurement and construction (EPC) basis at plot no. y-9 in industrial area trans ganga city Unnao, U.P							
Contract No:							
Name of the Bidder/ Bidding Firm / Company :							
PRICE SCHEDULE							
(This BOQ template must not be modified/replaced by the bidder and the same should be uploaded after filling the relevant columns, else the bidder is liable to be rejected for this tender. Bidders are allowed to enter the Bidder Name and Values only)							
NUMBER #	TEXT #	NUMBER #	TEXT #	NUMBER	NUMBER #	NUMBER #	TEXT #
Sl. No.	Item Description	Quantity	Units	Estimated Rate in Rs. P	TOTAL AMOUNT Without Taxes in Rs. P	TOTAL AMOUNT Without Taxes	TOTAL AMOUNT In Words
1	2	3	4	5	6	7	8
1	Work						
1.01	Establishing a Warehouse on Engineering, Procurement and construction (EPC) basis at plot no. y-9 in industrial area trans ganga city Unnao, U.P	1.00	Job	303800000.00	303800000.00	303800000.00	INR Thirty Crore Thirty Eight Lakh Only
Total in Figures					303800000.00	303800000.00	INR Thirty Crore Thirty Eight Lakh Only
Quoted Rate in Figures			Select		0.00	0.00	INR Zero Only
Quoted Rate in Words		INR Zero Only					

- The bidder has to fill in the space in blue colour only while quoting in CPPP.
- Rate /Amount to be filled in %age only
- Rate/ Amount shall not be disclosed at any other place except in the financial bid in CPPP or else the offer shall be treated as disqualified
- Conditional Offers shall NOT be acceptable.

16.0 Post award of Contract:

- 16.1 The items available in DSR shall be accordingly taken as above/below the DSR rate (as quoted) during finalization of detailed BoQ by the successful bidder.
- 16.2 The contractor shall prepare the detailed bill of quantity (BOQ) and specification for the complete scope of work and obtain approval from ALIMCO before commencing the execution.
- 16.3 The final estimate submitted for ALIMCO approval by the contractor should not exceed +/- 10% of contractor's quoted cost of project.

- 17.0 The tenderer(s) if required, may submit queries, if any, through E-mail and in writing to the tender inviting authority to seek clarifications within 10 days from the date of uploading of Tender on website. ALIMCO will reply only those queries which are essentially required for submission of bids.

18.0 The Pre-Bid meeting shall be held through VC/physical on date specified in NIT.

19.0 List of Documents to be scanned and uploaded on the e-tender website within the period of tender submission:

- a) Online payment receipt of payment gateway available at www.alimco.in. against the EMD as per NIT.
- b) Unconditional Letter of Acceptance of Tender Conditions (Annexure-III) (On Letter Head of the Applicant/ Bidder).
- c) Details of Similar Work Experience Certificates - FORM-A.
- d) Financial Details - FORM-B.
- e) BG for Mobilization format FORM-C.
- f) Copy of Bank Solvency Certificate- FORM-D.
- g) Documents regarding Net Worth of the Company / Firm.
- h) NIT
- i) Power of Attorney/Board Resolution of the person authorized for signing/submitting the tender
- j) EPF registration,
- k) GST registration
- l) PAN NO.
- m) All pages of the entire Corrigendum (if any).
- n) Pre-bid clarifications, if any.
- o) General Information – *Form-E*
- p) Work Experience Certificates consisting of details as mentioned in *Form-F*
- q) Form-G -Affidavit duly notarized by Notary Public on Non-Judicial Stamp Paper of Rs. 100 for correctness of Documents and Information.

NOTE:

- i. Unconditional letter of acceptance duly signed on letter head, Notarized Affidavit for correctness of document/information are mandatory documents and are needed to be checked carefully for its correctness before submission of tender. The bid shall be rejected outrightly in case of its non-submission, without seeking any further clarification/document. No claim of the bidder whatsoever shall be entertained by ALIMCO in this regard.
- ii. Tenderers are advised to upload complete details with their bids. Technical Bid Evaluation will be done on the basis of documents uploaded on the website with the bid. Please note that no fresh document other than in the form of clarification/revision in respect of an already submitted document shall be accepted .
- iii. The information should be submitted in the prescribed Performa. Bids with Incomplete/Ambiguous information are liable to be rejected.
- iv. All the uploaded documents should be in readable, printable and legible, failing which the Bids are liable for rejection.

20.0 All the uploaded documents shall be considered as duly signed by contractor/ authorized representative.

21.0 **ALIMCO** reserves the right to reject any or all tenders or cancel/withdraw the invitation for bid without assigning any reasons whatsoever thereof. ALIMCO does not bind itself to accept lowest tender.

22.0 Canvassing in connection with the tender is strictly prohibited and such canvassed tenders submitted are liable to be rejected and their earnest money shall be forfeited.

23.0 In case of any query, please contact **Shri** **Ph. No.**/ during Office hours.

(.....)

.....

INSTRUCTIONS TO TENDERER (ITT)

1.0	Online open tender is invited regarding Establishing a Warehouse on (EPC) basis at Plot no. Y-9, Industrial area Trans Ganga city Unnao, U.P
2.0	The estimated cost of Rs. 3038 Lacs shall cover all items of supplies & works whether directly stated or implied as per scope, standards & specifications, detailed drawings & designs, etc. included in this tender document, including site inspections & carrying out related preparatory works, etc.
3.0	The tender document as uploaded can be seen on website www.alimco.in or https://eprocure.gov.in/eprocure/app and can be downloaded free of cost.
4.0	Earnest Money Deposit Earnest Money Deposit amount as mentioned in “NIT” is required to be submitted along with the tender. The EMD RECEIPT shall be scanned and uploaded to the e-Tendering website within the period of tender submission. The EMD shall be payable to ALIMCO without any condition(s), recourse or reservations. (i) In case of non-submission of EMD, the Bid shall be rejected. (ii) The EMD of unsuccessful bidders in technical evaluation shall be returned within 30 days from placement of order. (iii) The EMD of the successful bidder will be discharged after the contractor has furnished the required acceptable performance guarantee. (iv) No interest shall be payable by ALIMCO on the EMD (v) EMD may be forfeited: - if the bidder withdraws the bid after bid opening during the period of validity. - Any unilateral revision in the offer made by the tenderer during the validity of the offer. - If the contractor hides/withholds (or does not disclose) any material information.

	d) Upon non acceptance of LOI/LOA, if and when placed. e) In the case of a successful bidder, if the bidder fails to Sign the Agreement with in the 30 days from the date of issue of LOA or furnish the required performance security or fail to commence the work within the stipulated time period prescribed in the contract. f) If any bidder furnishes any incorrect or false statement/ information/document.
5.0	The date of opening of financial bid shall be informed to the technically qualified bidder/tenderer.
6.0	The tender submitted shall become invalid if: (i) The tenderer is found ineligible. (ii) The tenderer does not upload all the documents as stipulated in the tender document. (iii) If any discrepancy is noticed between the documents as uploaded at the time of submission of tender
7.0	VALIDITY OF TENDER The tender for the works shall remain open for acceptance for a period of 150 (One Hundred Fifty) days from the date of opening of tender. If any tenderer withdraws his tender before the said period or issue of letter of acceptance, whichever is earlier, or makes any modifications in the terms and conditions of the tender which are not acceptable to the ALIMCO, then the ALIMCO shall, without prejudice to any other right or remedy, be at liberty to forfeit the “Earnest Money Deposit” as aforesaid. Further the tenderers shall not be allowed to participate in the retendering process of work.
8.0	ACCEPTANCE OF TENDER ALIMCO reserves the right to reject any or all the tenders in part or full without assigning any reason whatsoever. ALIMCO does not bind itself to accept the lowest tender.
9.0	The tenders shall be strictly as per the conditions of contract. Tenders with any additional condition(s)/modifications shall be rejected. Tenders in which, any of the prescribed conditions are not fulfilled or found incomplete in any respect are liable to be rejected.
10.0	The witnesses to the Tender/Contract Agreement shall be other than the tenderer/ tenderers competing for this work and must indicate full name, address, and status/occupation with dated signatures.
11.0	On acceptance of tender, the name of the accredited representative(s) of the contractor who would be responsible for taking instructions from Engineer-in-Charge or its authorized representative shall be intimated by the contractor within 07 days of issue date of work order by ALIMCO.

12.0	The time of completion of the entire work, as contained in contract shall be as mentioned in “ NIT ”, which shall be reckoned from the date of start.
13.0	Canvassing whether directly or indirectly, in connection with tenderers is strictly prohibited and the tenders submitted by the contractors who resort to canvassing will be liable for rejection.
14.0	The tender award, execution and completion of work shall be governed by tender documents consisting of (but not limited to) Letter of Award/Letter of work order, Bill of Quantities, Special Conditions of Contract, General Conditions of Contract, Specifications, Drawings. The tenderers shall be deemed to have gone through the various conditions including sub-soil water conditions, topography of the land, drainage and accessibility etc. or any other condition which in the opinion of contractor will affect his price/rates before quoting their rates. No claim whatsoever against the foregoing shall be entertained.
15.0	The drawings with the tender documents are ‘Tender Drawings’ and are indicative only. The successful bidder shall be required to carry out detailed site inspections/investigations, etc. for execution of this work and to prepare detailed ‘Good For Construction’ (‘GFC’) drawings to the satisfaction of Engineer- in- charge of ALIMCO. The structural design/ drawings & MEP (load Calculation) should be got vetted/ proof checked from the institutions like IIT’s /CBRI only, as per the approval of Engineer- in -charge of ALIMCO. The Cost of proof checking shall be borne by the Contractor.
16.0	ADDENDA/ CORRIGENDA Addenda/Corrigenda to the tender documents may be issued prior to the date of submission of the tender to clarify or effect modification in specification and/or contract terms included in various tender documents. The tenderer shall suitably take into consideration such Addenda/Corrigenda while submitting his tender. The tenderer shall return such Addenda/ Corrigenda duly signed and stamped as confirmation of its receipt & acceptance and submit along with the tender document. All addenda/ Corrigenda shall be signed and stamped on each page by the tenderer and shall become part of the tender and contract documents.
17.0	SITE VISIT AND COLLECTING LOCAL INFORMATION Before submission of tenders, the interested & eligible bidders are advised to visit the site of proposed work, its surroundings to assess and satisfy themselves about the local conditions such as the working and other constraints at site, approach roads to the site, availability of water & power supply, application of taxes, duties and levies as applicable & any other relevant information required by them to execute complete scope of work. The tenderer may obtain all necessary information as to risks, weather conditions, contingencies & other circumstances (insurgencies etc.) which may influence or affect their tender prices. Tenderer shall be deemed to have considered site conditions whether he has inspected it or not and to have satisfied himself in all respect before quoting his rates and no claim or extra charges whatsoever in this regard shall be entertained / payable by the ALIMCO at a later date.
18.0	ACCESS BY ROAD Contractor, where necessary, shall build temporary access roads to the actual site of construction for the works at his own cost to make the site accessible. The Contractor shall maintain the same in motorable condition at all the times as directed by Engineer-in-Charge at his own cost. The contractor shall be required to permit the use of any roads so constructed by

	him for vehicles of ALIMCO or any other agencies/ contractors who may be engaged on the project site, free of cost. Non-availability of access roads or approach to site, for the use of the contractor shall in no case condone any delay in the execution of work nor be the cause for any claim for compensation.
19.0	HANDING OVER & CLEARING OF SITE (i) ALIMCO shall handover the site to the Contractor free of encumbrances. However, in case of any delay in handing over of the site to the Contractor, the ALIMCO shall only consider suitable extension of time for the execution of the work. (ii) The information about the public utilities (whether over ground or underground) like electrical/ telephone/ water supply lines, OFC Cables, open drain etc. is the responsibility of contractor to ascertain the utilities that are to be affected by the works through the site investigation. (iii) The contractor shall be responsible to obtain necessary approval from the respective authorities for shifting/ re-alignment of existing public utilities. ALIMCO shall provide necessary authorization to represent ALIMCO before such authorities and only assist the contractor for liaisoning in obtaining the approval from the concerned authorities. (iv) Any services affected by the works must be temporarily supported by the contractor who must also take all measures reasonably required by the various bodies to protect their services and property during the progress of works. It shall be deemed to be the part of the contract and no extra payment shall be made to the contractor for the same.
20.0	SCOPE OF WORK (i) The “scope of work” covered in this tender shall be as per the attached scope, approved makes, specifications & standards, finishing schedules, drawings, etc. included in the tender document instructions, orders issued to the contractor from time to time during the pendency of work. The drawings for this work, which may be referred for tendering, provide only a general idea about the work to be performed under the scope of this contract. These may not be the final drawings and may not indicate the full range of the work under the scope of this contract. The work will be executed according to the detailed drawings to be prepared by the contractor as “GOOD FOR CONSTRUCTION” drawings after approval from time to time by the Engineer-in- charge of ALIMCO and according to any additions/modifications/ alterations/ deletions made from time to time, as required by any other drawings that would be issued to the contractor progressively during execution of work. It shall be the responsibility of the contractor to incorporate the changes that may be directly stated or indirectly implied in this scope of work, Standards & Specifications forming part of the tender document and/or what is envisaged at the time of tendering as being actually required to be executed.

	(ii) The contractor shall be bound to carry out and complete the stipulated work irrespective of the variation in any items.
21.0	APPROVAL OF TEMPORARY / ENABLING WORKS The setting and nature of all offices, huts, access road to the work areas and all other temporary works as may be required for the proper execution of the works shall be subject to the approval of the Engineer- in-charge. All the equipment, labour, material including cement, reinforcement and the structural steel required for the enabling/ temporary works associated with the entire Contract-shall have to be arranged by the Contractor only. Nothing extra shall be paid to the Contractor on this account.
22.0	CLARIFICATION AFTER TENDER SUBMISSION Tenderer's attention is drawn to the fact that during the period, the tenders are under consideration, the tenderers are advised to refrain from contacting by any means, the ALIMCO and/or his employees/ representatives on matters related to the tender under consideration and that if necessary, ALIMCO shall obtain clarifications in writing or as may be necessary. The tender evaluation and process of award of works is done by duly authorized Tender Scrutiny Committee and this committee is authorized to discuss and get clarification from the tenderers.
23.0	The following documents shall form the part of contract. (i) Letter of Award, along with statement of agreed variations and its enclosures, if any. (ii) Addendum / Corrigendum / Clarifications (if any) (iii) Special Instruction to tenderer (iv) Technical specifications and Standards (General, Additional and Technical Specification) as given in Tender documents. (v) Schedule of Finishes included in Tender Document (vi) Description of Work in "Scope of Work". (vii) Drawings included in tender document (viii) NIT / ITT / General Conditions of Contract. (ix) Relevant Quality Codes including National Building Code 2016, B.I.S. Codes, RDSO standards, etc. (x) Between any value written in numerals and that in words, the latter shall prevail.
24.0	Bids from bidder' against whom action to suspend business dealings has been taken by ALIMCO, shall not be accepted for any future enquiry/ bid/ tender till the expiry of period of debarment.

TIME-LINES / MILESTONE (S) :

Milestone	Physical Progress of work / Milestones	Check-Point	No. of days from the date of start	Amt. (or % of Project Value) to be withheld in case of nonachievement of milestone within scheduled time
Milestone-1	Conducting Site Survey, Soil investigation, Preparation, Designing & obtaining approval of Good for Construction (GFC), fabrication drawings, and getting them vetted by Technical / Research institutions of repute such as IIT and got approved by ALIMCO representative (s). Preparation of Schedule of quantities and to prepare detailed cost estimates and obtaining total project cost approval from ALIMCO. Obtaining Pre approvals/ Pre NOCs/Pre clearances from concerned State authorities/UPSIDA/ departments etc.	Approval of ALIMCO	D+03 Months	0.20% of Cost of Project

Milestone-2	On Completion of Boundary wall and Security Building with Entry Gate with Quality compliance as specified.	Approval of e-MB book	D+05 Months	0.20% of Cost of Project
Milestone-3	On Completion of foundation works for all structures with Quality compliance as specified.	Approval of e-MB book by ALIMCO	D+07Months	0.20% of Cost of Project
Milestone-4	On Completion of super structure works for complete Scope with Quality compliance as specified.	Approval of e-MB book by ALIMCO	D+10 Months	0.20% of Cost of Project
Milestone-6	On Completion of roads drains footpaths, Trenches/drains for rainwater, electrical cables, Drainage including Rain Water harvesting System. with Quality compliance as specified.	Approval of e-MB book by ALIMCO	D+12 Months	0.20% of Cost of Project
Milestone-7	On completion of finishing works, services like Internal & External Electrical& LV Work, Air conditioning, Development works, DG Set, Fire-fighting System, Water Supply, and testing and commissioning of all services with Quality compliance as specified.	Approval of e-MB book by ALIMCO	D+14 Months	0.20% of Cost of Project
Milestone-8	On obtaining all approvals/ NOCs/clearances from concerned State authorities/UPSIDA/ departments etc. And handing over the site	Approval by ALIMCO	D+15 Months	0.20% of Cost of Project

Note : D = The date of start of contract.

Date of start of contract shall be reckoned from 10 days after the date of issue of work order.

In case, the contractor does not achieve a particular milestone mentioned in the tender document or the scheduled / re-scheduled milestone(s), the amount shown against that milestone shall be withheld, to be adjusted against the LD levied at the final grant of Extension of Time as clause no. 7.0

Annexure II

Preliminary estimate

Name of work: Establishing a Warehouse on Engineering, Procurement and construction (EPC) basis at plot no. y-9 in industrial area trans ganga city Unnao, U.P		
ABSTRACT OF COST INFRASTRUCTURE		
S.No	Description	Amount in Rs. (Including GST).
1	Cost of Building work including Electrical and services	22,98,03,952.00
2	External Development including Electrical and services	5,21,90,912.00
3	E&M bulk services	45,68,440.00
4	Total (A) –(1+2+3)	28,65,63,304.00
5	Contingency @ 3% of Total (A)	85,96,889.00
6	Engineering Cost @ 3% of Total (A)	85,96,889.00
7	Grand Total Cost of Project Including GST	30,37,57,102.00
8	Total Cost of Project Including GST (round off)	30,38,00,000.00

The BOQ based costing of the works prepared by ALIMCO is given below for reference only:

Name of work: Construction of ALIMCO Ware house facility at Trans-Ganga City Project, Unnao						
Sl. No.	Description	Qty.	Unit	Rates per sqm in Rs.	Amount in Rs.	Ref- DPAR 2023
1	PEB framed structure (Shed Building)					
1.1	PEB framed structure (Shed Building)	6,346.00	Sqm.	20,250.00	128,506,500.00	MR
1.2	Pile Foundation up to 15m	6,346.00	Sqm.	7,290.00	46,262,340.00	1.3.7-1.3.6
				Total(1)	174,768,840.00	
2	RCC framed structure (Normal Building)					1
	RCC Framed Structure					1.1
2.1	Floor Ht. 3.60 mtr	848.00	Sqm.	30,820.00	26,135,360.00	1.1.1 (A)
2.2	Pile Foundation up to 15m only Ground Floor Area	476.00	Sqm.	7,290.00	3,470,040.00	1.3.7-1.3.6
				Total(2)	29,605,400.00	
	Total building cost marked as A = Total (1+2)				204,374,240.00	A
3	FIRE FIGHTING	7,148.92	Sqm.	1,200	8,578,704.00	1.5.3
4	FIRE ALARM	7,194.00	Sqm.	600	4,316,400.00	1.6.2
5	SERVICES (Offices)	29,605,400.00		4%	1,184,216.00	2.1
6	Electrical External Service Connections on Total (2)	29,605,400.00		3.75%	1,110,203.00	2.2.1
7	Civil external service connections on Total (2)	29,605,400.00		1.25%	370,068.00	2.2.2
8	Internal Electric Installation on Total A (1+2)	204,374,240.00		4%	8,174,969.60	MR
9	Extra for : Power Wiring and plugs on 'A' on Total (2)	29,605,400.00		4%	1,184,216.00	2.4.1
10	Lightening Conductors on Total A (1+2)	204,374,240.00		0.25%	510,935.60	2.4.2
				Total (B)	25,429,712.00	B
				Total Cost (A+B)	229,803,952.00	

S. No	Description	Qty.	Unit	Rates	Amount (in Rs.)	REF. PAR 2023
1.1	Under Ground Water Tank	75,000.00	liter	23.00	1,725,000.00	4.5
2	Development of Site					
2.1	Excavating, supplying and filling of local earth (including royalty)	13532.82	Cum.	700.50	9,479,740.41	D.S.R 2.25(a)
2.2	Internal Roads & Paths with WMM and bituminous top with Road Marking and Road Safety devices	3600.00	Sqm.	1930.00	6,948,000.00	5.2.2
2.3	150 mm thick M-30 grade Trimix Floor/cemewnt concrete for Hard Stand of Truck with 150 mm thick PCC base and kerb stone edging	3000.00	Sqm.	2690.00	8,070,000.00	4.1.8+16.43.2
2.4	Supply and Laying, spreading and compacting stone aggregate of 63 mm to 22.5 mm size stone aggregate blinding with stone Dust to WBM specifications in uniform thickness, hand picking, rolling with 3 wheeled road/vibratory roller 8-10 tonne capacity in stages to proper grade and camber, applying and brooming requisite type of screening / binding material to fill up interstices of coarse aggregate, watering and compacting to the required density. =3600.00+3000.00 =6600*0.200	1320.00	Cum.	2956.00	3,901,920.00	D.S.R (16.3.2+16.3.3+16.3.7)/3 +16.4
2.5	External Sewerage	500.00	Mtr.	3810.00	1,905,000.00	5.3
2.6	Filter Water Supply Distribution Lines upto 100mm dia	500.00	Mtr.	1930.00	965,000.00	5.4.1
2.7	Unfiltered water supply distribution lines	500.00	Sqm.	1510.00	755,000.00	5.4.3
2.8	Storm Water Drain	500.00	Mtr.	10150.00	5,075,000.00	5.5
2.9	Rain water harvesting (RWH)	200.00	Mtr.	3810.00	762,000.00	5.6
3	Boundary wall					
3.1	Boundry wall with 1500mm high wall and 600mm high MS Concertina coil including 2100mm high steel gates at every 100 meters With brickwork structure and RCC band at ground level and coping.	500.00	Mtr.	10870.00	5,435,000.00	5.8.1
3.2	From 1.5m to 3.0M hight	500.00	Mtr.	8696.00	4,348,000.00	MR
4	Horticulture Work					
	Horticulture operations including , grassing,tree plantations / shrubs and potted plants etc.	500.00	Sqm.	312.00	156,000.00	5.9.1
5	Street lighting with Led .					
5.1	Supplying, installation, testing and commissioning of LED Street/ Compound/High mast/ Pathway/Landscape Lighting for the entire Campus. Note: This is applicable for plot sizes more then 1 acre. For smaller plot sizes actual requirements may be worked out (40% area of external development) Note : - Cost for General Façade lighting, if required, with IP 66/67 LED fixtures (RGB/Tunable/Mono) along with controls (hardware and software) and cabling may be assessed on case to case basis.	13533	Sqm.	160.00	2,165,251.20	6.19.1
6	150 mm dia with 5 H.P three phase Submersibile Boring including Panel,boring pipe etc Complete with all respect.	1	Nos.	500000.00	500,000.00	MR
Total Cost					₹ 52,190,912.00	

E & M Bulk Services							
SI No.	Description	Qty	Unit	Rate	Amount	Ref. PAR 2023	Remarks
1	11kV receiving substation and 11kV HT cabling Supplying installation, testing and commissioning of 11kV substation comprising 11 kV HT pnel, inter connections, 11kV HT underground cabling to the distribution substations on ring main system substation earthing, substation safety equipment.	50	KVA	3,800.00	190000.00	6.1.1	Considering the usage the load of the plant would be approx 50KVA, there for as per the UP supply code upto 50 KVA the supply can be maode available at 415V/ so there no need to install the HT equipments.But on other hand as the solar generation panel prposed to install therefore for export purpose transformer and other HT equipments required to install.S.no 1,2,5 may be deleted if solar power plant no prposed in project.
2	SUB- STATION EQUIPMENTS Supplying installation testing and commissioning of SubStation equipments comprising H.T. Panel, Transformer, HT Cable, Bus trunking from Transformer to LT Panel, LT Panel, Automatic power factor corrector panel, Essential Panel i/c Earthing, inter connecting power cables in sub-station, safety equipments.including LT cabling from substation to the building fed by the sub station.	50	KVA	9,000.00	450000.00	6.2.1	6.2
3	DIESEL GENERATOR SET Supplying installation testing and commissioning of silent type D.G. Set, AMF Panel, Bus Trunking /cables from DG Set to essential panel, control cable, Earthing of DG Set and AMF Panel, DG set exhaust piping as per CPCB norms and minor allied works.	63	KVA	11,560.00	728280.00	6.3.1	6.3
4	CCTV System Supplying installation testing and commissioning of IP based CCTV system for building security comprising of PTZ, Fixed camera, cabling, recording, HD display system with min. display of 5" x 8" per camera hardware and software suppoert and hardware per sq. and software support.	13533	Sqm	210.00	2841892.20	6.10.1	6.10
5	WATER COOLER Water Cooler for R.O. of Storage Capacity of 25 Ltrs. and Purification Capacity 15Ltr /Hr	4	Set	50,000.00	200000.00	MR	MR
6	SIGNAGES Illuminated signages (Rate applicable on total plinth area)	7194.00	Sqm	22.00	158268.00	6.28.1	6.28.2
Total Cost					₹ 4,568,440.00		

ANNEXURE -III

ACCEPTANCE OF TENDER CONDITIONS

From: (To be submitted in ORIGINAL on the letter head of the company
by the authorized officer having power of attorney/as per Board Resolution)

ALIMCO (India) Limited,

Sub: _____ <Name of the work & NIT No.>:

Sir,

- i) This has reference to above referred tender. I/We have read/viewed all the terms & conditions and are pleased to submit our tender for the above work and I/We hereby unconditionally accept the tender conditions and tender documents in its entirety for the above work.
- ii) I/we are eligible to submit the tender for the subject tender and I/We are in possession of all the documents required.
- iii) Should this tender be accepted, I/We agree to abide by and fulfill all terms and conditions referred to above and as contained in tender documents elsewhere and in default thereof, to forfeit and pay ALIMCO, or its successors or its authorized nominees such sums of money as are stipulated in the notice inviting tenders and tender documents.

Yours faithfully, (Signature of the
tenderer with rubber stamp)

Dated _____

Tender for :**Details of similar work experience certificates**

S. No	Name of work and its location	Type of Work i.e. Residential / Non Residential Building/warehouse	Name of Client	Date and No. of Completion Certificate	Date of Start	Date of Completion	Cost of Work on completion	Reference and Page No. of Documentary Proof
1.								
2.								
3.								

Note

- a) Certified that the Completion Certificates of above works are enclosed with the Tender Documents.
- b) Details mentioned in the above Form are as per Completion Certificates and have not been presumed.
- c) If any detail is not mentioned in the Work Completion Certificate, documentary proof of the details like drawings, LOA, completion certificate/occupation certificate, copy of final bill, etc. is to be submitted and uploaded on Tender Website along with the Completion Certificate.

Signature of Bidder with Seal

Financial details**Tender for :**

S.No	Description	1st FY Rs. (In Lacs)	2nd FY Rs. (In Lacs)	3rd FY Rs. (In Lacs)
(i)	Profit/Loss	a	b	c
(ii)	Gross annual turnover of previous 3 years as per balance sheet ending as on last day of preceding financial year			
(iii)	Average annual turnover for previous 3 Financial years as per balance sheet (Rs. In Lacs) = $(a+b+c)/3$			
(iv)	Net worth (paid up capital + reserves) as on last day of the preceding financial year.			
(v)	Bank Solvency amount as mentioned in bank Solvency Certificate.			

Summarized pages of audited balance sheet and Profit & loss Account of previous three financial years, is to be submitted.

In case audited balance sheet and Profit & loss Account of previous three financial years not available, the unaudited balance sheet and Profit & loss Account of previous three financial years duly certified by the chartered accountant is to be submitted.

Signature of Chartered account

Account with Seal

Membership Number:

UDIN:

Signature of Bidder with Seal

FORMAT FOR BANK GUARANTEE TOWARDS SECURITY DEPOSIT/PERFORMANCE GUARANTEE/ADVANCE

(To be executed by any Nationalized Bank in India, on a Rs. 200 non-judicial stamp paper under bank's covering letter mentioning address of the bank)

In consideration of M/s Artificial Limbs Manufacturing Corporation of India, hereinafter referred to as 'ALIMCO', which expression unless repugnant to the context and meaning thereof shall include its successors and assigns having agreed to exempt M/s _____ (Name of the vender/supplier/contractor) hereinafter referred to as supplier/contractor which expression unless repugnant to the context and meaning thereof shall include its successors and assigns] from depositing _____ with ALIMCO a sum of Rs. _____ towards security / performance guarantee in lieu of the said Supplier/contractor having agreed to furnish a bank guarantee for the said sum of Rs. _____

As required under the terms and conditions of contract / work order / purchase order no. _____

Dated _____ [hereinafter referred as the order'] placed by ALIMCO on the said supplier / contractor.

We, _____ the bank [hereinafter referred to as 'the bank' which expression shall include Its successors and assigns] do hereby undertake to pay ALIMCO an amount not exceeding Rs. _____ on the demand made by ALIMCO on us due to a breach committed by the said supplier / Contractor of the terms and conditions of the order.

We _____ the bank hereby undertakes to pay the amount under the guarantee without any demur merely on a demand from ALIMCO stating that there is a breach by the supplier / contractor of any of the terms and conditions contained in the order or by the reasons of the supplier's / contractor's failure to comply with the terms and conditions as stipulated in the order or amendment(s) thereto.

The demand made on the bank shall be conclusive as to the breach of the terms and conditions of the order and as regard to the amount due and payable by the bank under this guarantee, notwithstanding any dispute or disputes raised by the said supplier / contractor regarding the validity of such breach and we agree to pay the amount so demanded by ALIMCO without any demur. However, our liability under this guarantee shall be restricted to an amount not exceeding Rs. _____.

We, _____ the bank further agree that the guarantee herein contained shall remain in full force and effect during the period that would be taken for the performance of the said order and that it shall continue to be enforceable till the dues of ALIMCO under or by virtue of the said order have been fully paid and its claim satisfied or discharged or till ALIMCO certifies that the terms and conditions of the order have been fully and properly carried out by the supplier / contractor and accordingly discharge the guarantee.

We_____the bank, undertake to pay to ALIMCO any money so demanded notwithstanding any dispute or disputes raised by the said supplier / contractor in any suit or proceedings pending before any court or tribunal relating thereto as our liability under this present being absolute and unequivocal. The payment so made by us under this bond shall be valid discharge of our liability for payment there under and the said supplier / contractor shall have no claim against us for making such payment.

We_____the bank further agree that ALIMCO shall have full liberty, without our consent and without affecting in any manner our obligation hereunder to vary any of the terms and conditions of the order or to extend time of performance by the said supplier / contractor from time to time or to postpone for any time or from time to time any of the powers exercisable by the ALIMCO against the said supplier / contractor and to forbear or enforce any of the terms and conditions relating to the order and shall not be relieved from our liability by reason of any such variation or extension being granted to the said supplier / contractor or for any forbearance, act or omission on the part of ALIMCO or any indulgence by ALIMCO to the supplier / contractor or by any such matter or thing whatsoever which under the law relating to sureties would but for this provisions have effect of so relieving us. Our liability under this guarantee is restricted to Rs._____and shall remain in force up to _____ unless demand or claim under this guarantee is made on us in writing within 12 months from the date of expiry viz. _____. We shall be discharged from all liabilities under this guarantee thereafter.

This guarantee will not be discharged due to change in the constitution in the bank or the said supplier / contractor.

The bank hereby agrees to address all the future correspondence in regard to this bank guarantee to Chairman & Managing Director, Artificial Limbs Manufacturing Corporation of India.

We,_____the bank lastly undertake not to revoke this guarantee during its currency except with the previous consent of ALIMCO in writing.

Signed on the _____ day of _____

Witness

Signature for the Bank

Dispatch number of the bank /Date

SOLVENCY CERTIFICATE ON THE LETTER HEAD OF THE BANK

This is to state that to the best of our knowledge and information that M/s..... having /registered office address is customer of bank and has been maintain his accounts with our branch since as per records available with the bank, M/s..... can be treated as solvent up to a limit of RS.....(Rupees in words)

It is clarified that the above information is furnished and this certificate is being issued at specific request of the customer.

**Name ,Designation ,
Signature with Seal**

Note: The certificate should have been issued within the 12 months from last date of the submission of the tender.

General Information

1.	Name of the Applicant/Company	
2.	Address for correspondence	
3.	Official E-mail for communication	
4.	Contact person: Telephone Number: Mobile Number: Fax Nos:	
5.	Type of Organization: a. An Individual b. A proprietary firm c. A Firm in partnership: (Attach Copy of partnership) d. A limited company (Attach copy of Article Association) e. Any other (Mention type)	
6.	Place and Year of incorporation	
7.	Name of directors/ Partners /Proprietor/ Owner in the Organization	
8.	Bank details: Name of the bank, Address of the bank Branch, Account Number RTGS, IFC Code	

Signature of Bidder with Seal

Name of the Client with Address, E-mail & phone No.

No:

Date:

WORK EXPERIENCE CERTIFICATE

Name of the Contractor :.....

1	Name of the project / Project Location	
2	Name and address of the client	
3	Agreement amount	
4	Cost of work on completion	
5	Date of start	
6	Stipulated date of completion	
7	Actual date of completion	
10	Brief details of work:	

**Name & Designation
Signature with seal
Of issuing Authority**

Date :

AFFIDAVIT

(To be submitted by bidder on non-judicial stamp paper of Rs.100/ (Rupees Hundred only) duly attested by Notary Public)

Affidavit of Mr.S/o
R/o

I, the deponent above named do hereby solemnly affirm and declare as under:

1. That I am the Proprietor/Authorized signatory of M/s
Having its Head Office/Regd. Office at
.....
2. That the information/documents/Experience certificates/ Bank Guarantee(s) submitted by M/s..... along with the tender for (Name of work) To ALIMCO are genuine and true and nothing has been concealed.
3. I shall have no objection in case ALIMCO verifies those from issuing authority(ies). I shall also have no objection in providing the original copy of any of the document(s), in case ALIMCO demands so for verification.
4. I hereby confirm that in case, any document, information & / or certificates submitted by me found to be incorrect / false / fabricated, ALIMCO at its discretion may disqualify / reject / terminate the bid/contract and also place under Holiday list as per ALIMCO Policy.

I,, the Proprietor / Authorised signatory of M/s..... do hereby confirm that the contents of the above Affidavit are true to my knowledge and nothing has been concealed there from. and that no part of it is false.

DEPONENT

Verified atthis.....day of

DEPONENT

ATTESTED BY (NOTARY PUBLIC)

SPECIAL INSTRUCTIONS TO TENDERER

1. Working Drawings

The successful bidder should provide two sets of hard copies and soft copy (PDF & Auto cad) of all working drawing. Successful bidder shall submit monthly programme indicating their planning for execution of works.

2. Bench Mark

Contractor shall construct all other survey pillars required by them for survey work with reference to the bench mark pillar. The contractor shall take all precautions to protect the survey pillars from being damaged / dislocated during progress of work by providing suitable barbed wire fencing around the survey pillars.

3. Specification

Specifications for construction work and materials shall be as per "Specification of Work" (latest edition) published by CPWD shall be followed.

4. Sequences of work

Successful bidder shall plan and sequence all works so as to achieve the desired progress keeping in mind overall safety and stability at all point of time. If due to particular design or specification or availability of machines or any other reason, a particular sequence of operation is demanded by the engineer due to which some interruption to any one or more types of work or items of execution are inherent, then no claim for such interruption shall be entertained and contractor shall have to follow the sequence as instructed by the Engineer.

Execution of work shall be through meticulous planning & arranging the man, material and resource etc. to carry out the work within constraints & strict requirement of ALIMCO and adherence to time schedule.

5. Damages

Any existing services of other properties if damaged or demolished during the course of execution either wilfully or negligently, shall be repaired and re-erected by the contractor at his own cost. The contractor will take all precautions against damages due to rain and storm and no liability shall rest with the purchaser for any loss for redoing of any work due to this account. Existing underground services coming in the way of trench or foundation excavation etc. have to be adequately supported by the contractor at his own cost so as to avoid any damage to those services. In case of any accidental damage, the contractor shall arrange to rectify and reinstall those services immediately at his own cost.

6. Quality assurance

To ensure quality of construction, the contractor shall ensure inspection and testing of all materials and works at their own cost conforming to specifications.

All certificates and guarantees obtained by the contractor from different suppliers of materials, sub-contractors doing part job, manufacturers of doors, windows etc. shall be passed on the purchaser.

All defective materials / works which don't conform to the specification shall be dismantled and removed from site by the contractor at his own cost.

Vetting / inspection of any site work or material from IIT/ NIT/ Govt Engineering college or any govt recognize agency as decided by ALIMCO shall be under scope of Contractor.

The contractor shall provide a site laboratory with equipment's, maintain and engage qualified technical personnel at the site to run the site laboratory at their own cost. All routine tests of cement, aggregate, sand, bricks, concrete cubes, moisture content for

timber etc. Shall be conducted at site and equipment required for the same shall be kept in site laboratory.

For tests which are not possible to be carried out at site laboratory, contractor shall arrange to send requisite sample to approved laboratory outside the site and get those tested as required entirely at his cost. All required test including design mix of concrete shall be carried out by contractor at his own cost as directed by engineer in charge.

Before starting any concreting or erection work at site, the stage passing certificates have to be obtained from the engineer in the formats to be mutually discussed and agreed.

Before covering up any work within excavated pits, clearance has to be taken from engineer in respect of completion of said works as per drawing and specifications.

Third Party Inspection (TPI) for any bought out capital / services items shall be under scope of Contractor at their own cost.

7. Progress meeting :

The Contractor will be required to attend progress meetings on site progress on weekly basis unless otherwise specified by ALIMCO Engineer-in-charge, to report on progress and to discuss any issues. Meetings will be held at a location to be specified by Engineer-in-charge. The Contractor shall arrange for the attendance of any of his staff members as required by Engineer-in-charge. The meeting will be chaired by ALIMCO team. The Contractor may be required by ALIMCO to attend other meetings, as necessary.

CLAUSES OF CONTRACT

1.0 DEFINITIONS

The Contract means the documents forming the tender and acceptance thereof and the formal agreement executed between the competent authority on behalf of **ARTIFICIAL LIMBS MANUFACTURING CORPORATION OF INDIA (ALIMCO)** and the contractor, together with the documents referred to therein including these conditions, the specifications, designs, drawings and instructions issued from time to time by the Engineer-in-Charge and all these documents taken together, shall be deemed to form one contract and shall be complementary to one another.

1.1 ARTIFICIAL LIMBS MANUFACTURING CORPORATION OF INDIA, hereinafter called 'ALIMCO' propose to get the works executed as mentioned in the Contract.

1.2 In the contract, the following expressions shall, unless the context otherwise requires, have the meanings, hereby respectively assigned to them: -

- a) **APPROVAL** means approved in writing including subsequent written confirmation of previous verbal approval.
- b) **CONTRACTOR** shall mean the individual, firm, LLP or company, whether incorporated or not, undertaking the works and shall include the legal personal representative of such individual or the persons composing such firm or LLP or company, or the successors of such firm or company and the permitted assignees of such individual, firm or company.
- c) **CONTRACT VALUE** means the sum for which the tender is accepted as per the work order.
- d) **DRAWINGS** mean the drawings referred to in the contract document including modifications if any and such other drawings as may from time to time be furnished and/ or approved by ALIMCO.
- e) **DATE OF COMMENCEMENT OF WORK: The date of start of contract shall be reckoned from 10 days after the date of issue of work order.**
- f) **ENGINEER-IN-CHARGE** means the Engineer of ALIMCO, the name of whom is intimated through work order, or his representative duly authorized by him in writing who shall supervise the work and issue necessary instructions to the contractor in matters so delegated to him and/or take contractually permitted actions under such delegated responsibilities as they fall in purview of the Engineer in Charge under relevant clauses of GCC
- g) **LANGUAGE:** All documents and correspondence in respect of this contract shall be in English Language.
- h) **“LETTER OF AWARD”** shall mean ALIMCO’s letter or notification conveying its acceptance of the tender subject to such conditions as may have been stated therein.
- i) **MONTH** means English Calendar month. ‘Day’ means a Calendar day of 24 Hrs each.
- j) Artificial Limbs Manufacturing Corporation of India (ALIMCO) is the only non-Profit motive manufacturing organization, registered under section 8 (Not for Profit) of Indian Companies act, 2013 is working under the aegis of Ministry of Social Justice& Empowerment, Government of India and its head quarter is located at Kanpur or its Administrative officers or its engineer or other employees authorized

to deal with any matter with which these persons are concerned and authorized on its behalf.

- k) **OWNER/ CLIENT** means ALIMCO .
- l) **SCHEDULE(s)** referred to in these conditions shall mean the standard schedule of rates of the government mentioned in the Memorandum (Annexure-I) with the amendments thereto issued up to the date of receipt of the tender.
- m) **SITE** means the lands and other places on, under, in or through which the works are to be executed or carried out and any other lands or places provided by ALIMCO/client/owner or used for the purpose of the contract.
- n) **TENDER** means the Contractor's priced offer to ALIMCO for the execution and completion of the work and the remedying of any defects therein in accordance with the provisions of the Contract, as accepted by the Award letter. The word TENDER is synonymous with Tender and the word TENDER DOCUMENTS with "Tendering Documents" or "offer documents".
- o) **WRITING** means any manuscript typed written or printed statement under or over signature and/or seal as the case may be.
- p) **Works or Work** shall unless there be something either in the subject or context repugnant to such construction, be construed and taken to mean the works by or by virtue of the contract contracted to be executed whether temporary or permanent, and whether original, altered, substituted or additional.
- q) Headings in the clauses/ conditions of tender documents are for convenience only and shall not be used for interpretation of the clause/condition.
- r) Words imparting the singular meaning only also include the plurals and vice versa where the context requires. Words importing persons or parties shall include firms and corporations and organizations having legal capacities.
- s) **Market Rate** shall be the rate as decided by the Engineer-in-Charge on the basis of the prevailing cost of materials and labour at the site where the work is to be executed plus the percentage mentioned elsewhere in the tender document to cover, all overheads and profits.

2.0 PERFORMANCE BANK GUARANTEE:

- 2.1 "Within 15 (Fifteen) days from the date of issue of work order, the contractor shall submit to ALIMCO irrevocable performance bank guarantee of 5% of the contract value, in the form Bank guarantee pledged in favor of ALIMCO from any Nationalized Bank or all Commercial schedule bank, for the due and proper execution of the Contract. The Performance Guarantees shall be initially valid up to the stipulated date of completion of defect liability period including execution period of the contract plus 60 days beyond that. In case the time for completion of works gets extended, the contractor shall get the validity of Performance Guarantee extended to cover such extended time for completion defect liability period.
- 2.2 ALIMCO reserve the right of forfeiture of the performance guarantee in the event of termination of contract or non-fulfilment under any of the clauses/ conditions of contract.
- 2.3 PBG shall be returned after successful completion/ testing/ commissioning and handing over and defect liability period the project to the entire satisfaction of ALIMCO and finalization of Extension of Time by Engineer-in-charge.
- 2.4 In case the contractor fails to submit the performance guarantee(s) of the requisite

amount within the stipulated period or extended period, work order automatically will stand withdrawn and EMD of the contractor shall be forfeited.

- 2.5** In case part scope of work has been satisfactorily completed and handed over to ALIMCO by the contractor and the remaining work/buildings are under hold/suspended for more than 3 months due to some hindrance beyond control of contractor or due to non-availability of site/clearance from ALIMCO, the performance bank guarantee submitted by the contractor and available with ALIMCO as on date, on specific request of the contractor, may be released and returned back to the contractor in lieu of submission and due verification of a fresh PBG of the equivalent amount towards balance work.

3.0 SECURITY DEPOSIT

- 3.1** The Security deposit shall be deducted from each running bill of the contractor @ 10% (Ten per cent only) till the realization of the total amount of SD @ 5% of contract value.
- 3.2** The release/refund of security deposit of the contractor shall be subject to the observance/compliance of the conditions as under and whichever is later:
- a. Project completion and handing over of the site.
 - b. The contractor has to produce a clearance certificate from the labour office. As soon as the work is completed and handed over, the contractor shall apply for the labour clearance certificate to the Labour Officer under intimation to the Engineer-in-Charge.
- 3.3** ALIMCO reserves the right of part or full forfeiture of security deposit in addition to other claims in the event of contractor's failure to fulfill any of the contractual obligations or in the event of termination of contract as per terms and conditions of contract.

4.0 MOBILIZATION ADVANCE

In case the contractor wishes to avail Advance of 10% of cost of Project Value, they have to submit an undertaking to the effect that an Advance Bank Guarantee (BG) of 110% of the ADVANCE value, valid for successful completion of the project shall be submitted within 1 months from the date of releasing the PO/Work Order of the project in the format enclosed at Form-D.

The interest on the advance shall bear simple interest at the rate of 12 (Twelve) percent per annum and shall be calculated from the date of payment to the date of recovery, both days inclusive, on the outstanding amount of advance. It shall be ensured that at any point of time, Bank Guarantee is available with ALIMCO for the amount of outstanding advance.

The recovery of such sums advanced shall be made by the deduction from the contractors bills commencing after first ten per cent (10%) of the gross value of the work is executed and paid, on prorata percentage basis to the gross value of the work billed beyond 10% in such a way that the entire advance is recovered by the time eighty per cent (80%) of the gross value of the contract is executed and paid, together with interest due on the entire outstanding amount up to the date of recovery of the instalment.

Further, contractor has to open a separate Bank account for the work and the initial deposit will be deposited in that account. The Party will meet the expenditure from this deposit for execution of the Project and shall submit monthly bills for recouping of expenditure.

The interest accrued to this account will be refunded to the ALIMCO at quarterly basis.

5.0 DEVIATIONS / VARIATIONS EXTENT AND PRICING

5.1 The Engineer-in-Charge shall have power

- a) To make any alterations in, omissions from, additions to or substitutions for, the original specifications, drawings, designs and instructions that may appear to him to be necessary during the progress of the work.
- b) To omit part of the works in case of non-availability of a portion of the site or for any other reasons and the contractor shall be bound to carry out the works in accordance with any instructions given to him in writing signed by the Engineer-in-Charge and such alterations, omissions, additions, or substitutions shall form part of the contract as if originally provided therein and any altered, additions or substituted works which the contractor may be directed to do in the manner specified above as part of the work, shall be carried out by the contractor on the same conditions in all respects including price on which he agreed to do the main work except as hereunder provided:

5.2 In the case of extra item(s), the contractor may within fifteen days of receipt of order claim rates, supported by proper analysis, for the work and the engineer-in-charge shall within one month of the receipt of the claims supported by analysis, after giving consideration to the analysis of the rates submitted by the contractor, determine the rates on the basis of the market rates and agree with the contractor that extra items shall be paid in accordance with the rates so determined & mutually agreed.

In the case of substituted items (items that are taken up with partial substitution or in lieu of items of work in the contract), the rate for the agreement item (to be substituted) and substituted item shall also be determined in the manner as mentioned in the following Para:

- (a) If the market rate for the substituted item so determined is more than the market rate of agreement item (to be substituted), the rate payable to the contractor for the substituted item shall be the rate for the agreement item (to be substituted) so increased to the extent of the difference between the market rates of substituted item and the agreement item (to be substituted).
- (b) If the market rate for the substituted item so determined is less than the market rate of the agreement (to be substituted), the rate payable to the contractor for the substituted item shall be the rate for the agreement item (to be substituted) so decreased to the extent of the difference between the market rates of substituted item and the agreement item (to be substituted)

5.3 The contractor shall send to the Engineer-in-Charge once every three months, an up-to-date account giving complete details of all claims for additional payments to which the contractor may consider himself entitled and of all additional work ordered by the Engineer-in-Charge which he has executed during the preceding quarter failing which the contractor shall be deemed to have waived his right. However, the Engineer-in-charge may authorize consideration of such claims on merits.

5.4 Market Rates to be determined as per various sub-clauses under the clause 6.0 shall be on the basis of prevailing rates of Material excluding GST (unless mentioned otherwise), Relevant Labour authority rate for Labour, market rates of

T&P etc. plus 15% towards Contractors' Profits and Overheads.

The following factors may be considered in the justification of rates on which Contractor's overhead & profit shall not be applicable:

- a) Buildings and Other Construction Worker Cess as applicable in the state of work place
- b) GST on works contract

6.0 ESCALATION

No claim on account of any escalation on whatsoever ground shall be entertained at any stage of works. Price quoted by contractor shall be firm and fixed for entire contract period as well as extended period for completion of the works. No escalation shall be payable on this contract

7.0 Liquidated Damages(LD)

If the contractor fails to maintain the required progress in terms of clauses of tender to complete the work and clear the site on or before the contract or extended date of completion, they shall, without prejudice to any other right or remedy available under the law to the ALIMCO on account of such breach, pay as agreed compensation the amount calculated at the rates stipulated below (plus GST extra) as the Engineer in charge (whose decision in writing shall be final and binding) may decide on the amount of awarded value of the work for every completed day / week (as applicable) that the work remains incomplete.

i) LD for delay of work @ 0.5% of the awarded value of contract, per week of delay

Provided always that the total amount of LD for delay to be imposed under this Condition shall not exceed 10% of the Awarded Value of work for which a separate period of completion is originally given. The amount of LD may be adjusted or set-off against any sum payable to the Contractor under this or any other contract with ALIMCO.

In case, the contractor does not achieve a particular milestone mentioned elsewhere in the tender document or the re-scheduled milestone(s), the amount shown against that milestone shall be withheld, to be adjusted against the LD levied at the final grant of Extension of Time. With-holding of this amount or failure to achieve a milestone, shall be automatic without any notice to the Contractor. However, the withheld amount against the milestone shall be released on achieving that milestone subsequently. In case the contractor fails to achieve a milestone and subsequent milestone(s), the amount mentioned against each milestone(s) not completed shall be withheld combinedly. Further, no interest, whatsoever, shall be payable on such withheld amount.

8.0 ACTION IN CASE WORK NOT DONE AS PER SPECIFICATIONS

All works under or in course of execution or executed in pursuance of the contract, shall at all times be open and accessible to the inspection and supervision of the Engineer-in-charge, authorized subordinates in charge of the work and all the superior officers, officer of the Quality Assurance Unit of the ALIMCO or any organization engaged by the ALIMCO for Quality Assurance, and the contractor shall, at all times, during the usual working hours and at all other times at which reasonable notice of the visit of such officers has been given to the contractor, either himself be present to receive orders and instructions or have a responsible

agent duly accredited in writing, present for that purpose. Orders given to the Contractor's agent shall be considered to have the same force as if they had been given to the contractor himself.

If it shall appear to the Engineer-in-charge or his authorized subordinates in-charge of the work or to the officer of Quality Assurance or his subordinate officers or the officers of the organization engaged by the ALIMCO for Quality Assurance, that any work has been executed with unsound, imperfect, or unskillful workmanship, or with materials or articles provided by him for the execution of the work which are unsound or of a quality inferior to that contracted or otherwise not in accordance with the contract, the contractor shall, on demand in writing which shall be made within twelve months of the completion of the work from the Engineer-in-Charge specifying the work, materials or articles complained of notwithstanding that the same may have been passed, certified and paid for forthwith rectify, or remove and reconstruct the work so specified in whole or in part, as the case may require or as the case may be, remove the materials or articles so specified and provide other proper and suitable materials or articles at his own charge and cost. In the event of the failing to do so within a period specified by the Engineer-in-Charge in his demand aforesaid, then the contractor shall be liable to pay compensation at the same rate as under clause 7.0 of the contract (for non-completion of the work in time) for this default. In such case the Engineer-in-Charge may not accept the item of work at the rates applicable under the contract but may accept such items at reduced rates as the Engineer in charge may consider reasonable during the preparation of on account bills or final bill if the item is so acceptable without detriment to the safety and utility of the item and the structure or he may reject the work outright without any payment and/or get it and other connected and incidental items rectified, or removed and re-executed at the risk and cost of the contractor. Decision of the Engineer-in-Charge to be conveyed in writing in respect of the same will be final and binding on the contractor.

9.0 ACTION IN CASE OF BAD WORK

If it shall appear to the Engineer-in-Charge or his authorized representative in charge of the work where the work is being executed, that any work has been executed with unsound, imperfect, or unskillful workmanship or with materials of any inferior description, or that any materials or articles provided by him for the execution of the work are unsound or of a quality inferior to that contracted for or otherwise not in accordance with the contract, the contractor shall on demand in writing which shall be made within twelve months of the completion of the work from the Engineer-in-Charge specifying the work, materials or articles complained of notwithstanding that the same may have been passed, Certified and paid for forthwith rectify, or remove and reconstruct the work so specified in whole or in part as the case may require or as the case may be, remove the materials or articles so specified and provide

other proper and suitable materials or articles at his own proper charge and cost, and in the event of his failing to do so within a period to be specified by the Engineer-in-Charge in his demand aforesaid while the contractor failure to do so shall continue, the Engineer-in-Charge may rectify or remove and re-execute the work or remove and replace with others, the material or articles complained of as the case may be at the risk and expense in all respects of the contractor.

10.0 CANCELLATION/DETERMINATION OF CONTRACT IN FULL OR PART

Subject to other provisions contained in this clause the Engineer-in-Charge may, without prejudice to his any other rights or remedy against the contractor in respect of any delay, inferior workmanship, any claims for damages and / or any other provisions of this contract or otherwise, and whether the date of completion

has or has not elapsed, by notice in writing absolutely determine the contract in any of the following cases:

- i) If the contractor having been given by the Engineer-in-Charge a notice in writing to rectify, reconstruct or replace any defective work or that the work is being performed in an inefficient or otherwise improper or un-workmanlike manner shall omit to comply with the requirement of such notice for a period of seven days thereafter; or
- ii) If the contractor has, without reasonable cause, suspended the progress of the work or has failed to proceed with the work with due diligence so that in the opinion of the Engineer-in-Charge (which shall be final and binding) he will be unable to secure completion of the work by the date for completion and continues to do so after a notice in writing of seven days from the Engineer-in-Charge; or
- iii) If the contractor fails to complete the work within the stipulated date or items of work with individual date of completion, if any stipulated, on or before such date(s) of completion and does not complete them within the period specified in a notice given in writing in that behalf by the Engineer-in-Charge; or
- iv) If the contractor persistently neglects to carry out his obligations under the contract and / or commits default in complying with any of the terms and conditions of the contract and does not remedy it or take effective steps to remedy it within 7 days after a notice in writing is given to him in that behalf by the Engineer-in-Charge; or
- v) If the contractor shall offer or give or agree to give to any person in ALIMCO service or to any other person on his behalf any gift or consideration of any kind as an inducement or reward for doing or forbearing to do or for having done or forborne to do any action relation to the obtaining or execution of this or any other contract for ALIMCO; or
- vi) If the contractor shall enter into a contract with ALIMCO in connection with which commission has been paid or agreed to be paid by him or to his knowledge, unless the particulars of any such commission and the terms of payment thereof have been previously disclosed in writing to the Engineer-in-Charge; or
- vii) If the contractor shall obtain a contract with ALIMCO as a result of wrong tendering or other non-bona-fide methods of competitive tendering or commits breach of Integrity Pact; or
- viii) If the contractor being an individual, or if a firm, any partner thereof shall at any time be adjudged insolvent or have a receiving order or order for administration of his estate made against him or shall take any proceedings for liquidation or composition (other than a voluntary liquidation for the purpose of amalgamation or reconstruction) under any Insolvency Act for the time being in force or make any conveyance or assignment of his effects or composition or arrangement for the benefit of his creditors or purport so to do, or if any application be made under any Insolvency Act for the time being in force for the sequestration of his estate or if a trust deed be executed by him for benefit of his creditors; or
- ix) If the contractor being a company, shall pass a resolution or the Court shall make an order for the winding up of the company, or a receiver or manager on behalf of the debenture holders or otherwise shall be appointed or circumstances shall arise which entitle the Court or debenture holders to appoint a receiver or manager; or

- x) If the contractor shall suffer an execution being levied on his goods and allow it to be continued for a period of 21 days, or.
- xi) If the contractor assigns, transfers, sublets (engagement of labour on a piece-work basis or of the labour with materials not to be incorporated in the work, shall not be deemed to be subletting) or otherwise parts with or attempts to assign, transfer sublet or otherwise parts with the entire works or any portion thereof without and prior written approval of the Engineer-in-Charge.

When the contractor has made himself liable for action under any of the cases aforesaid, the Engineer-in-Charge may without prejudice to any other right or remedy which shall have accrued or shall accrue hereafter to ALIMCO, by a notice in writing to cancel the contract as whole or only such items of work in default from the Contract, the Engineer-in-charge shall have powers:

- a) To determine or rescind the contract as aforesaid in full or part (of which termination or rescission notice in writing to the contractor under the hand of the Engineer-in-Charge shall be conclusive evidence) and get the same executed at the risk & cost of the contractor. Upon such determination or rescission the already retained security deposit recovered under the contract and performance guarantee shall be liable to be forfeited and un-used materials, construction plants, implements, temporary buildings, etc. shall be taken over and shall be absolutely at the disposal of the ALIMCO. or
- b) To employ labour and to supply materials to carry out the work or any part of the work debiting the contractor with the cost of the labour and the price of the materials of the amount of which cost and price certified by the Engineer-in-Charge shall be final and conclusive) and/ or
- c) After giving notice to the contractor to measure up the work of the contractor and to take such whole, or the balance or part thereof as shall be un- executed or delayed with reference to the Conditions of Contract clause no. 21.0 and/ or relevant clause of Special Conditions of Contract, out of his hands and to give it to another contractor to complete.
- xii) Any sums in excess of the amounts due to ALIMCO and unsold materials, constructional plant etc. shall be returned to the contractor, provided always that if cost or anticipated cost of completion by ALIMCO of the works or part of the works is less than the amount which the contractor would have been paid had he completed the works or part of the works, such benefit shall not accrue to the contractor.
- xiii) In the event of anyone or more of the above courses being adopted by the Engineer-in-Charge the contractor shall have no claim to compensation for any loss sustained by him by reasons of his having purchased or procured any materials or entered into any engagements or made any advances on account or with a view to the execution of the work or the performance of the contract. And in case action is taken under any of the provision aforesaid the contractor shall not be entitled to recover or be paid any sum for any work thereof or actually performed under this contract unless and until the Engineer-in-Charge has certified in writing the performance of such work and the value payable in respect thereof and he shall only be entitled to be paid the value so certified.

TAKEN UNDER CLAUSE 10.0

In any case in which any of the powers conferred upon the Engineer-in-Charge by relevant clause thereof, shall have become exercisable and the same are not exercised, the non-exercise thereof shall not constitute a waiver of any of the conditions hereof and such powers shall notwithstanding be exercisable in the event of any future case of default by the contractor and the liability of the contractor for compensation shall remain unaffected. In the event of the Engineer-in-Charge putting in force all or any of the powers vested in him under any clause he may, if he so desires after giving a notice in writing to the contractor, take possession of (or at the sole discretion of the Engineer-in-Charge which shall be final and binding on the contractor) use as on hire (the amount of the hire money being also in the final determination of the Engineer-in-Charge) all or any tools, plant, materials and stores, in or upon the works, or the site thereof belonging to the contractor, or procured by the contractor and intended to be used for the execution of the work or any part thereof, paying or allowing for the same in account at the contract rates, or in the case of these not being applicable, at current market rates to be certified by the Engineer-in-Charge, whose certificate thereof shall be final and binding on the contractor and/or direct the contractor, clerk of the works, foreman or other authorized agent to remove such tools, plant, materials, or stores from the premises (within a time to be specified in such notice) in the event of the contractor failing to comply with any such requisition, the Engineer-in-Charge may remove them at the contractor's expense or sell them by auction or private sale on account of the contractor and his risk in all respects and the certificate of the Engineer-in-Charge as to the expenses of any such removal and the amount of the proceeds and expenses of any such sale shall be final and conclusive against the contractor.

12.0 CARRYING OUT PART WORK AT RISK & COST OF CONTRACTOR

If contractor:

- (i) At any time makes default during currency of work or does not execute any part of the work with due diligence and continues to do so even after giving a notice in writing of 7 days in this respect from the Engineer-in-Charge; or
- (ii) Commits default in complying with any of the terms and conditions of the contract and does not remedy it or takes effective steps to remedy it within 7 days even after a notice in writing is given in that behalf by the Engineer-in-Charge; or
- (iii) Fails to complete the work(s) or items of work with individual dates of completion, on or before the date(s) so determined, and does not complete them within the period specified in the notice given in writing in that behalf by the Engineer-in-Charge.

The Engineer-in-Charge without invoking action under clause 10.0 of contract may, without prejudice to any other right or remedy against the contractor which have either accrued or accrue thereafter to ALIMCO, by a notice in writing to take the part work/part incomplete work of any item(s) out of his hands and shall have powers to:

- (a) Take possession of the site and any materials, constructional plant, implements, stores, etc., thereon; and/or
- (b) Carry out the part work / part incomplete work of any item(s) by any means at the risk and cost of the contractor.

The Engineer-in-Charge shall determine the amount, if any, is recoverable from the contractor for completion of the part work/ part incomplete work of any

item(s) taken out of his hands and execute at the risk and cost of the contractor, the liability of contractor on account of loss or damage suffered by ALIMCO because of action under this clause shall not exceed 10% of the tendered value of the work.

In determining the amount, credit shall be given to the contractor with the value of work done in all respect in the same manner and at the same rate as if it had been carried out by the original contractor under the terms of his contract, the value of contractor's materials taken over and incorporated in the work and use of plant and machinery belonging to the contractor. The certificate of the Engineer-in-Charge as to the value of work done shall be final and conclusive against the contractor provided always that action under this clause shall only be taken after giving notice in writing to the contractor. Provided also that if the expenses incurred by the department are less than the amount payable to the contractor at his agreement rates, the difference shall not be payable to the contractor.

Any excess expenditure incurred or to be incurred by ALIMCO in completing the part work/ part incomplete work of any item(s) or the excess loss of damages suffered or may be suffered by ALIMCO as aforesaid after allowing such credit shall without prejudice to any other right or remedy available to ALIMCO in law or per as agreement be recovered from any money due to the contractor on any account, and if such money is insufficient, the contractor shall be called upon in writing and shall be liable to pay the same within 30 days.

If the contractor fails to pay the required sum within the aforesaid period of 30 days, the Engineer-in-Charge shall have the right to sell any or all of the contractors' unused materials, constructional plant, implements, temporary building at site etc. and adjust the proceeds of sale thereof towards the dues recoverable from the contractor under the contract and if thereafter there remains any balance outstanding, it shall be recovered in accordance with the provisions of the contract.

In the event of above course being adopted by the Engineer-in-Charge, the contractor shall have no claim to compensation for any loss sustained by him by reason of his having purchased or procured any materials or entered into any engagements or made any advance on any account or with a view to the execution of the work or the performance of the contract.

13.0 TERMINATION OF CONTRACT ON DEATH OF CONTRACTOR

Without prejudice to any of the right or remedies under this contract if the contractor dies, the Engineer in-charge shall have the option of terminating the contract without compensation to the contractor.

14.0 TIME ESSENCE OF CONTRACT & EXTENSION FOR DELAY

- 14.1 The time allowed for execution of the Works as specified in the Memorandum (Annexure-I) or the extended time in accordance with these conditions shall be the essence of the contract. The execution of the works shall commence from such time period as mentioned in Memorandum (Annexure-I) or the date on which the Engineer-in-Charge issues written orders to commence the work. If the Contractor commits default in commencing the execution of the work as aforesaid, the ALIMCO shall without prejudice to any other right or remedy available in law, be at liberty to forfeit the earnest money & performance guarantee absolutely.

The Critical milestones to be achieved during execution of the contract within overall scheduled completion period and the amount to be withheld in case of non-achievement of the same are stated at Annexure-I.

- 14.2 The Contractor shall submit, as per the time-schedule as intimated/directed by

Engineer-in-charge of ALIMCO in LOA or afterwards in the kick-off meeting, a Time and Progress Chart (CPM/ PERT/Quantified Bar chart) and get it approved by the Engineer-in-Charge. The Chart shall be prepared in direct relation to the time stated in the contract documents for completion of items of the works/milestone(s) included in 'SCC' or 'Annexure-I to ITT'. It shall indicate the forecast (mile-stones) of the dates of commencement and completion of various items, trades, sections of the work and may be amended as necessary by agreement between the Engineer-in-Charge and the Contractor within the limitations of time stipulated in the Contract documents and further to ensure good progress during the execution of the work. The physical progress report including photographs shall be submitted by the contractor on the prescribed format & the intervals (not exceeding one month) as decided by the Engineer-in-Charge.

- 14.3 If the work(s) be delayed by any of the following events/eventualities which directly or indirectly make it legally or physically impossible to carry-on of supplies or execution of works on the site, and which are unforeseeable or unusual as per norms/trends/practices in the trade:
- i. Force-Majeure or
 - ii. Abnormally bad weather, or
 - iii. Serious loss or damage by fire, or
 - iv. Civil commotion, rioting, war or local commotion of workmen, strike or lockout, affecting any or the trades employed on the work, or other similar issues of law & order,
 - v. Delay on the part of other contractors or tradesmen engaged by Engineer-in-Charge in executing work not forming part of the Contract, or
 - vi. Non-availability of stores, which are responsibility of the ALIMCO or, lack of statutory approvals, that are beyond the scope of works of the contractor
 - vii. Non-availability or break down of tools and plant to be supplied or supplied by ALIMCO or,
 - viii. Any other cause which, in the absolute discretion of the ALIMCO, is beyond the Contractor's control, then upon the happening of any such event causing delay, the Contractor shall immediately give notice thereof in writing to the Engineer-in-Charge within 07 days but shall nevertheless use constantly his best endeavor to prevent or make good the delay and shall do all that may be reasonably required to the satisfaction of the Engineer-in-Charge to proceed with the works.
- 14.4 Request for rescheduling of Milestones and extension of time, to be eligible for consideration, shall be made by the Contractor in writing within fourteen days of the happening of the event causing delay along with the reasons in the prescribed form. The Contractor may also, if practicable, indicate in such a request the period for which rescheduling of Milestones/ extension is desired. In any such case ALIMCO may give a fair and reasonable extension of time for completion of work/rescheduling of Milestone(s). Such extension shall be communicated to the Contractor by the Engineer-in-Charge in writing within a reasonable time from the receipt of such request. Non application by the contractor for extension of time/rescheduling of Milestone(s) shall not be a bar for giving a fair and reasonable extension/rescheduling of Milestone(s) by the Engineer-in-Charge and the extension of time/rescheduling of Milestone(s) so given by the Engineer-in-Charge

shall be binding on the contractor.

15.0 TIME SCHEDULE & PROGRESS

- 15.1 Time allowed for carrying out all the works as entered in the tender shall be as mentioned in the "NIT" which shall be reckoned from the 10th day from the date on which the work order is issued to the Contractor. Time shall be the essence of the contract and contractor shall ensure the completion of the entire work within the stipulated time of completion.
- 15.2 The contractor shall also furnish, as per the time-schedule as intimated/directed by Engineer-in-charge of ALIMCO in LOA or afterwards in the kick-off meeting, a CPM network/ PERT chart/ Bar Chart for completion of work within stipulated time. This will be duly got approved from, Engineer-in-charge of ALIMCO and shall be in conformity with Execution Milestones included in 'Annexure-I of ITT'. This approved Network/ PERT Chart shall form a part of the agreement. Achievement of milestones as well as total completion has to be within the time period allowed.
- 15.3 Contractor shall mobilize and employ sufficient resources for completion of all the works as indicated in the agreed BAR CHART/PERT Network. No additional payment will be made to the contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule even though the time schedule is approved by the Engineer-in-Charge.
- 15.4 During the currency of the work the contractor is expected to adhere to the time schedule on milestone and total completion and this adherence will be a part of Contractor's performance under the contract. During the execution of the work contractor is expected to participate in the review and updating of the Network/BAR CHART undertaken by the ALIMCO. These reviews may be undertaken at the discretion of Engineer-in-charge either as a periodical appraisal measure or when the quantum of work order on the contractor is substantially changed through deviation orders or amendments. The review shall be held at site or any of the offices of ALIMCO at the sole discretion of ALIMCO. The contractor will adhere to the revised schedule thereafter. The approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time to the contractor.
- 15.5 Contractor shall submit (as directed by Engineer-in-Charge) progress reports on a computer-based program (program and software to be approved by Engineer-in-Charge) highlighting status of various activities and physical completion of work. The contractor shall send completion report with as built drawings to the office of Engineer-in-Charge, of ALIMCO in writing within a period of 30 days of completion of work.

The photographs of the project taken on last day of every month indicating progress of work (in soft copies) shall be attached along with the physical progress reports to be submitted to Engineer-in-charge.

16.0 TAXES AND DUTIES

- 16.1 The contract price is inclusive of all taxes, duties, cess and statutory levies payable under any law (as applicable on the date of submission of bid) by the contractor in connection with execution of the contract.

The contract price will be adjusted prospectively for any increase / decrease in the GST rate on works contract notified by Government of India.

- 16.2 Notwithstanding anything contained in clause 17.1, the contractor shall ensure payment of appropriate taxes, cess, levies, etc. on the items of supplies and works made good under the contract. The contractor shall take/obtain registration under the applicable enactment levying tax/levy/cess on supply of goods or services under the contract and issue invoice having all the particulars prescribed under the applicable provisions of the law, including description of goods/services, rate and amount of tax paid or payable on the supplies made under the contract, so that ALIMCO can avail credit of such tax, wherever applicable. The contractor shall comply with all applicable provision of Goods and Service Tax (GST) levied by Union Government and State Governments (CGST, UTGST, SGST and IGST). The contractor shall get himself registered and discharge his obligations for payment of taxes, filing of returns etc. under the appropriate provisions of law in respect of all the taxes, duties, levies, cess, etc.

ALIMCO would have right to seek necessary evidence that the contractor is registered under the law and duly discharging its obligations under the tax law, enabling ALIMCO to avail input tax credit.

Whenever interest is payable on reversal of Input Tax Credit (ITC) for non-payment of value and tax of supply under GST Act, due to any default on contractor's part (i.e due to poor performance/quality or non-fulfilment of terms of contract), the amount of interest so levied on ALIMCO shall be recovered from the contractor.

- 16.3 In case any law requires ALIMCO to pay tax/levy/cess on the contract price under any law for the time being in force, (such as GST, royalty on minerals, works contract taxes on reverse charge basis, etc.) the amount of tax/levy/cess so deposited by ALIMCO on this behalf would be considered as paid to the contractor and, accordingly, the price payable to the contractor would stand reduced to that extent.
- 16.4 In case the contractor does not deposit or file the tax/levy/cess with appropriate authority/department, and which is payable-by or levied on contractor on account of execution of this contract, or if any such tax/levy/cess (such as GST, WCT, royalty on minerals, etc.) is deposited-by or recovered-from ALIMCO but is actually payable by or levied-on the contractor, or, if the contractor has not provided the invoice or other relevant document pertaining to deposit/credit of such tax/levy/cess to ALIMCO, showing/proving deposit of appropriate the amount of tax/levy/cess, or has not uploaded the document in computerised tax network as per prevailing law, leading to non-availability of inputs credit of the tax to ALIMCO, the amount equivalent to such tax shall be deducted from the contract price.
- 16.5 Stamp duty and registration charges, if any, payable on the executed contract document, shall be borne by the contractor.
- 16.6 Tax deduction at source, if any, shall be made by ALIMCO as per law applicable from time to time from the amount payable to the contractor.

17.0 INCOME TAX DEDUCTION (TDS)

Income tax deductions shall be made from all payments made to the contractor including advances, in respect of the work/ project undertaken by the contractor, in accordance with the provisions of the Income Tax Act and Rules made there under prevailing and in force from time to time.

18.0 ROYALTY ON MATERIALS:

- 18.1 The contractor shall deposit royalty and obtain necessary permit for supply of *bajri*, stone, *kankar*, sand and other materials etc. from the local authorities and quoted rates shall be inclusive of royalty.
- 18.2 The contractor shall be deemed to have inspected the site, its surrounding and acquainted itself with the nature of the ground, accessibility of the site and full extent and nature of all operations necessary for the full and proper execution of the contract, space for storage of materials, constructional plant, temporary works, restrictions on the plying of heavy vehicles in area, supply and use of labour, materials, plant, equipment and laws, rules and regulations, if any, imposed by the local authorities.
- 18.3 The rates and prices to be tendered in the bill of quantities are for completed and finished items of works and complete in all respects. It will be deemed to include all constructional plant, labor, supervision, materials, transport, all temporary works, erection, maintenance, contractor's profit and establishment/overheads, together with preparation of designs & drawings pertaining to casting yard, shop drawing, fabrication drawing (if required), staging form work, stacking yard, etc. all general risk, all taxes, royalty, duties, cess, octroi and other levies, insurance liabilities and obligations set out or implied in the tender documents and contract.
- 18.4 If any temporary/ permanent structure is encountered or safety of such structure in the vicinity is endangered due to execution of the project, the contractor has to protect the structures by any means as per direction of Engineer-in-Charge. If any damage is caused to any temporary or permanent structure(s) in the vicinity due to execution of the project, the contractor has to make good the same by any means as per direction of Engineer-in-Charge. The contractor should inspect the site of work from this point of view. The cost to be incurred in this regard shall be deemed to be included in his quoted rates of BOQ items and the contractor shall not be entitled for any extra payment in this regard.

19.0 INSURANCE OF WORKS ETC

- 19.1 Contractor is required to take Contractor's All Risk policy or erection all risk policy (as the case may be) from an approved insurance company in the joint name with first name of ALIMCO and bear all costs towards the same for the full period of execution of works including the defect liability period for the full amount of contract against all loss of damage from whatever cause arising other than excepted risks for which he is responsible under the terms of the contract and in such manner that the ALIMCO and the contractor are covered during the period of construction of works and/or also covered during the period of defect liability for loss or damage

- a) The work and the temporary works to the full value of such works.
- b) The materials, constructional plant, centering, shuttering and scaffolding materials and other things brought to the site for their full value.

The contractor is required to submit the original policy document and the receipt for payment of the current premium to ALIMCO.

In case, the contract value gets enhanced by more than 25 percent due to any deviation /variation/ extra items, the contractor shall submit additional insurance policy for the enhanced contract value.

19.2 INSURANCE UNDER WORKMEN COMPENSATION ACT

Contractor is required to take adequate insurance coverage as prescribed under the Workman Compensation Act, 1923 amended from time to time from an approved insurance company and pay premium charges thereof.

The contractor is required to submit the original policy document and the receipt for

payment of the current premium to ALIMCO.

If the Contractor fails to effect or keep in force or provide adequate cover in the Insurance policies mentioned in this sub-clause, or any other insurance which he might be required to effect under the Contract, then in such cases, the Employer may effect and keep in force any such insurance or further insurance and the cost and expenses incurred by him in this regard shall be deductible from payments due to the Contractor or from the Contractor's Performance Guarantee.

19.3 THIRD PARTY INSURANCE

Contractor is required to take third party insurance cover for an amount of 5%(five percent) of contract value from an approved insurance company for insurance against any damage, injury or loss which may occur to any person or property including that of ALIMCO/owner/client, arising out of the execution of the works or temporary works.

The contractor is required to submit the original policy document and the receipt for payment of the current premium to ALIMCO.

Engineer-In charge has to ensure that Insurance policies are submitted by the contractor within 15 days from the date of issue of work order. In case of failure of the contractor to obtain contractors all risk policy, insurance under workman compensation act and third party insurance as described above, ALIMCO reserves the right of forfeiture of the Performance Bank Guarantee.

If the Contractor could not effect a comprehensive insurance cover against risks which he may be required to effect under the terms of the contract, then he shall give his attention to get the best insurance cover available and even in case of effecting a wider insurance cover than the one which the subsidiary of the General Insurance Company could offer, such an insurance is ought to be done after the ALIMCO's approval, by or through the subsidiary of the General Insurance Company.

The contractor shall at all times indemnify ALIMCO and Owner against all claims, damages or compensation under the provision of Payment of wages act-1936, Minimum Wages Act-1948 Code on Wages, 2019, Employer's liability Act-1938, the workmen's compensation Act-1947, Industrial Disputes Act-1947 and Maternity Benefit Act-1961 or any modifications thereof or any other law in force or as consequence of any accident or injury to any workman or other persons in or about the works, whether in the employment of the contractor or not, against all costs, charges and expenses of any suit, action or proceedings arising out of such incident or injury and against all sum or sums which may with the consent of the contractor be paid to compromise or compound any such claim. Without limiting his obligations and liabilities as above provided, the contractor shall insure against all claims, damages or compensation payable under the Workmen's Compensation Act 1923 or any modification thereof or any other law relating thereto.

20.0 Measurements, Certificates and Payments:

20.1 Measurements:

- a. The mode of measurements to be followed for the building and other civil works/ services shall be in accordance with CPWD specifications and relevant Indian standards (latest edition) for all the works.
- b. In the absence of mode of measurements for the particular work in the said the provisions contained in the relevant Indian Standard Code of Practice (latest edition) shall be followed. the same shall be determined by the Engineer in accordance with the good engineering practice and manufacturer's recommendations. The decision of the Engineer shall be final and binding on the Contractor.

20.2 Record and Measurements:

- a. Measurements shall be taken jointly by the Engineer or his authorized representative and by the Contractor or his authorized representative from time to time and at such intervals as in the opinion of the Engineer shall be proper, having regard to the progress of the Works. It shall be clearly noted that the responsibility of recording measurements and timely submission of bills rests with the contractor and any delay on this account will not be entertained.
- b. The final bill must invariably be preceded by a thorough check of measurements of the whole of the Work performed.
- c. Before taking final measurements of any Work, the contractor shall give a reasonable notice to the Engineer. If the Contractor fails to submit the final measurements and prolongs it unreasonably, then in any such event the measurements taken by the Engineer or by the person deputed by him shall be taken to be the correct measurement of the Work and shall be final and binding on the Contractor and no claim whatsoever shall thereafter be entertained regarding the accuracy and classification of measurement.
- d. The Contractor shall, without extra charges provide with every appliance, labour and things necessary for measurement.

20.3 Payment:

- a. Payment against bills for the work done shall be made to the Contractor based on detailed measurements and certification of bills by the Engineer.
- b. Bills shall be prepared and submitted by the contractor. Joint measurements shall be taken continuously and need not be connected with billing stage. Based on these joint measurements recorded in the register, the contractor shall submit the bills in both soft and hard copies (three copies) in the prescribed format along with the detailed measurements item wise. All the pages of the measurement sheets shall be serially numbered and signed by the contractor. The bills along with measurement sheet shall be computer generated and also suitably programmed for effecting the necessary corrections easily. The bills shall be accompanied with necessary documents such as abstract of quantities, variation statements, reconciliation of materials if any and fulfilling other statutory obligations such as ESI/ PF/ Insurance/ labour license etc. Payment against bills for the work done will be made after the verification and certification by the Engineer of the bill submitted by the contractor.
- c. 50% of the amount admissible will be paid by the Company within 7 days of presentation of the bill and the balance 50% shall be paid within 30 days of receipt of all documents and clarifications, if any, on the bills submitted and after complying with all the statutory provisions by the contractor with respect to the bills.
- d. Normally, bills will be entertained once in a month. However, Minimum value /invoice should not be less than 25 Lakh. Payments made to the Contractor shall be without prejudice to the final making up of the accounts. payment cannot by itself imply his having handed over any part of the Works and the liability of the Contractor shall continue to exist till the work is completed, inspected, approved and taken over by the Engineer. It is the sole responsibility of the Contractor to see that the Works are completed and handed over in a satisfactory manner.

20.4 Completion Certificate:

The Works shall be deemed to have been completed in all respects on the day the Engineer -in -charge certifies that the works have been so completed in accordance with this Contract, takes over the completed Works and issues a certificate to that effect.

20.5 Final Payments:

- a. Based on the measurement of Work performed, the Contractor shall submit his final bill for the Works within 3 months of completion of work. The bill shall be based only on Works as measured and at accepted tender rates including rates for any additional or extra work which might have been approved by the Company. All deductions due under the Contract shall be incorporated. The final bill shall be accompanied by:
 - i) A copy of the completion certificate issued by the Engineer -in -charge.
 - ii) A "No Claim Certificate" in the prescribed form or a list of claims if any, not included in the final bill with full details.
 - b. The Engineer shall examine and certify the final bill for payment after satisfying that the Works have been satisfactorily completed and that all properties, works and things removed or disturbed or damaged in consequence of the Work have been properly replaced and made good and all expenses and demands incurred or made by or on the Company or in respect of any damage or loss by, from or consequence of the Works have been satisfied, all materials have been returned and the site cleared. In respect of list of claims given by the Contractor, the Company shall examine and either accept or repudiate in whole or in part and convey its decision in writing to the Contractor.
- 20.6 Deduction of Income Tax:**
- a. Income Tax on the gross amount billed will be deducted from Contractor's bills as per Norms of the Income Tax Act.
 - b. The successful contractor should inform the assessing Income Tax Officer concerned (within one month from the date of issue of work order) about the award of work to him. The successful contractor shall furnish their Income Tax Permanent Account Number (PAN) to ALIMCO while furnishing their bills for payment.

21.0 MEASUREMENTS OF WORKS

Engineer-in-charge shall, except as otherwise provided, ascertain and determine by measurement, the value of work done in accordance with the contract.

Except where any general or detailed description of the work expressly shows to the contrary, measurement shall be taken in accordance with the procedure set forth in the CPWD Specification. In the case of items which are not covered by specifications, mode of measurement as specified in the Technical Specifications of the contract and if for any item no such technical specification is available, then a relevant standard method of measurement issued by the Bureau of Indian Standard shall be followed.

Provided further that, in case of Cancellation/termination of Contract in Full or in Part in accordance with clauses of tender, following methodology shall be adopted in respect of measurements in addition to what has been mentioned in foregoing: -

- a) All measurements and levels shall be taken jointly by the Engineer-in-charge or his authorized representative and by the contractor or his authorized representative from time to time during the progress of the work and such measurements shall be signed and dated by the Engineer-in-Charge and the contractor or their representatives in token of their acceptance. If the contractor objects to any of the measurements recorded, a note shall be made to that effect with reason and signed by both the parties.
- b) If for any reason the contractor or his authorized representative is not available and the work of recording measurements is suspended by the Engineer-in-Charge or his representative, the Engineer-in-Charge and ALIMCO shall not entertain any claim from contractor for any loss or damages on this account. If the contractor or his authorized representative does not remain present at the time of such measurements after the contractor or his authorized representative has been given

a notice in writing three (3) days in advance or fails to countersign or to record objection within a week from the date of the measurement, then such measurements recorded in his absence by the Engineer-in-Charge or his representative shall be deemed to be accepted by the Contractor. The contractor shall, without extra charge, provide all assistance with every appliance, labour and other things necessary for measurements and recording levels.

22.0 Measurement Book in electronic form (E-MB) shall be filled through Excel.

23.0 WITHHOLDING AND LIEN IN RESPECT OF SUMS DUE FROM CONTRACTOR

23.1 Whenever any claim or claims for payment of a sum of money arises out of or under the contract or against the contractor, ALIMCO shall be entitled to withhold and also have a lien to retain such sum or sums in whole or in part from the security, if any, deposited by the contractor and for the purpose aforesaid, ALIMCO shall be entitled to withhold the security deposit, if any, furnished as the case may be and also have a lien over the same pending finalization or adjudication of any such claim. In the event of the security being insufficient to cover the claimed amount or amounts or if no security has been taken from the contractor, ALIMCO shall be entitled to withhold and have a lien to retain to the extent of such claimed amount or amounts referred to above, from any sum or sums found payable or which may at any time thereafter become payable to the contractor under the same contract or any other contract pending finalization of adjudication of any such claim.

23.2 It is an agreed term of the contract that the sum of money or moneys so withheld or retained under the lien referred to above by the Engineer-in-Charge or ALIMCO will be kept withheld or retained as such by the Engineer-in-Charge or ALIMCO till the claim arising out of or under the contract is determined by the competent court and that the contractor will have no claim for interest or damages whatsoever on any account in respect of such withholding or retention under the lien referred to above and duly notified as such to the contractor. For the purpose of this clause, where the contractor is a partnership firm or a limited company, the Engineer-in-Charge or the ALIMCO shall be entitled to withhold and also have a lien to retain towards such claimed amount or amounts in whole or in part from any sum found payable to any partner/limited company, as the case may be whether in his individual capacity or otherwise. ALIMCO shall have the right to cause an audit and technical examination of the works and the final bills of the contractor including all supporting vouchers, abstract, etc., to be made after payment of the final bill and if as a result of such audit and technical examination any sum is found to have been overpaid in respect of any work done by the contractor under the contract or any work claimed to have been done by him under the contract and found not to have been executed, the contractor shall be liable to refund the amount of over-payment and it shall be lawful for ALIMCO to recover the same from him in the manner prescribed in clause of tender or in any other manner legally permissible; and if it is found that the contractor was paid less than what was due to him under the contract in respect of any work executed by him under it, the amount of such under payment shall be duly paid by ALIMCO to the contractor, without any interest thereon whatsoever.

24.0 WORK TO BE EXECUTED IN ACCORDANCE WITH SPECIFICATIONS, DRAWINGS, AND ORDERS ETC.

All items of work in the bill of quantities/ schedule of quantities shall be carried out as per the CPWD/ MORTH (as the case may be) specifications,

drawings and instructions of the Engineer-in-Charge of ALIMCO and the rates shall include for supply of required materials including proper storage, consumables, skilled & unskilled labour, supervision and tools, tackles, plant & machinery complete as called for in the detailed specifications and conditions of the contract. Latest updated CPWD specification shall be followed for execution of work.

25.0 MATERIALS TO BE PROVIDED BY THE CONTRACTOR

The contractor shall, at his own expense, provide all materials, required including Cement & Steel for the works. The contractor shall at his own expense and without delay; supply to the Engineer-in-Charge samples of materials to be used on the work and shall get the same approved in advance. All such materials to be provided by the Contractor shall be in conformity with the specifications laid down or referred to in the contract. The contractor shall, if requested by the Engineer-in-Charge furnish proof, to the satisfaction of the Engineer-in-Charge regarding compliance of the material so procured. The contractor shall submit the samples of materials to be tested or analysed and bear all charges and cost of testing unless specifically provided for otherwise elsewhere in the contract or specifications. The Engineer-in-Charge or his authorized representative shall at all times have access to the works and to all workshops and places where work is being prepared or from where materials, manufactured articles or machinery are being obtained for the works and the contractor shall afford every facility and every assistance and cost in obtaining the right and visit to such access. The Engineer-in-Charge shall have full powers to require the removal from the premises of all materials which in his opinion are not in accordance with the specifications and in case of default, the Engineer-in-Charge shall be at liberty to employ at the expense of the contractor, other persons to remove the same without being answerable or accountable for any loss or damage that may happen or arise to such materials. The Engineer-in-Charge shall also have full power to require other proper materials to be substituted thereof and in case of default, the Engineer-in-Charge may cause the same to be removed from the supplies and all costs which may require such removal and substitution shall be borne by the contractor.

The contractor shall ensure that the materials are brought to site in original sealed containers or packing bearing manufacturer's markings and brands (except where the quantity required is a fraction of the smallest packing).

The contractor shall produce receipted vouchers showing quantities of the materials to satisfy Engineer-In-Charge that the materials comply with the specifications. These vouchers shall be endorsed, dated and signed by the contractor. A certified copy of each such vouchers signed both by ALIMCO and the Contractor shall be kept on record.

26 MATERIALS AND SAMPLES

26.1 The materials/products used on the works shall be one of the approved make/ brands out of list of manufacturers / brands /makes given in the tender documents. The contractor shall submit samples/ specimens out of approved makes of materials/ products to the Engineer-in-Charge for prior approval. In exceptional circumstances Engineer-in-Charge may allow alternate equivalent makes/ products/ materials at his sole discretion. The final choice of brand / make shall remain with the Engineer-in-Charge, whose decision in this matter shall be final and binding and nothing extra on this account shall be payable to the Contractor. In case single brand/ make are mentioned, other equivalent makes/ brands may be considered by the Engineer-in-Charge. In case of variance in CPWD/IS/BIS Specifications from approved products/makes specification, the specification of approved product/make shall prevail for which nothing shall be paid extra to the Contractor. In case no make or brand of any

materials, articles, fittings and accessories etc. is specified, the same shall comply with the relevant Indian Standard Specifications and shall bear the ISI/BIS mark. ALIMCO shall have the discretion to check quality of materials and equipment to be incorporated in the work, at source of supply or site of work and even after incorporation in the work. ALIMCO shall also have the discretion to check the workmanship of various items of work to be executed in this work. The contractor shall provide the necessary facilities and assistance for this purpose.

- 26.1 The above provisions shall not absolve the contractor from the quality of final product and in getting the material and workmanship quality checked and approved from the Engineer-in-Charge of ALIMCO.
- 26.2 The contractor shall well in advance, produce samples of all materials, articles, fittings, accessories etc. that he proposes to use and get them approved in writing by ALIMCO. The materials, articles etc. as approved shall be labelled as such and shall be signed by ALIMCO and the Contractor's representative.
- 26.3 The approved samples shall be kept in the custody of the Engineer-in-Charge of ALIMCO till completion of the work. No payment will be made to the contractor for the samples or samples destroyed in testing.
- 26.4 The brands of all materials, articles fittings etc. approved together with the names of the manufacturers and firms from which supplies have been arranged shall be recorded in the site order book.
- 26.5 The contractor shall set up and maintain at his cost, a field testing laboratory for all day to day tests at his own cost to the satisfaction of the Engineer-in-Charge. This field testing laboratory shall be provided with equipment and facilities to carry out all mandatory field tests as per MORTH/CPWD (as the case may be) specifications. The laboratory building shall be constructed and installed with the appropriate facilities, Temperature and humidity controls shall be available wherever necessary during testing of samples. All equipment shall be provided by the Contractor so as to be compatible with the testing requirements specified. The Contractor shall maintain all the equipment in good working condition for the duration of the contract. The Contractor shall provide approved qualified personnel to run the laboratory for the duration of the Contract. The number of staff and equipment available must at all times be sufficient to keep pace with the sampling and testing programme as required by the Engineer-in-charge. The Contractor shall fully service the site laboratory and shall supply everything necessary for its proper functioning, including all transport needed to move equipment and samples to and from sampling points on the site, etc. The Contractor shall re-calibrate all measuring devices whenever so required by the Engineer-in-charge and shall submit the results of such calibration without delay. All field test shall be carried out in the presence of ALIMCO's representative. All costs towards samples, materials, collection, transport, manpower, testing etc. shall be borne by the Contractor and are deemed to be included in the rates quoted by him in the bill of quantities.

The contractor(s) shall display the calibration certificate of each equipment at the location of equipment & shall get recalibrated at least one week before its expiry date.

27.0 CONTRACTOR TO SUPPLY TOOLS & PLANTS

The contractor shall provide at his own cost all materials, machinery, tools & plants as require for completion of work. In addition to this, appliances, implements, other plants, ladders, inspection lifts, cordage, tackle, scaffolding and temporary works

required for the proper execution of the work. The contractor shall also supply without charge the requisite number of persons with the means and materials, necessary for the purpose of setting out works, and counting, weighing and assisting the measurement or examination at any time and from time to time of the work or materials. Failing his so doing, the same may be provided by the Engineer-in-Charge at the expense of the contractor and the expenses may be deducted, from any money due to the contractor, under this contract or otherwise and/or from his security deposit or the proceeds of sale thereof, or of a sufficient portions thereof.

28.0 MOBILIZATION OF MEN, MATERIALS AND MACHINERY:

- 28.1 All expenses towards mobilization at site and de-mobilization including bringing in equipment, work force, materials, dismantling the equipments, clearing the site etc. shall be deemed to be included in prices quoted and no separate payment on account of such expenses shall be entertained.
- 28.2 The contractor shall mobilize all tools, machinery, equipment, for timely and safe completion of work as per applicable quality standards and specifications mentioned in the contract. The contractor shall ensure un-interrupted supplies of water & electricity at site for construction and related purposes, and shall obtain temporary water & electricity connections at its own. For small activities other than concreting, the contractor shall install water- purification equipment, to bring it in conformity with required quality standards.
- 28.3 It shall be the responsibility of the contractor to obtain the approval for any revision and/or modification desired by him from ALIMCO before implementation.
- 28.4 It is mandatory for the contractor to provide safety equipment and gadgets to his all workers, supervisory and Technical staff engaged in the execution of the work while working.
- 28.5 All designs, drawings, bill of quantities, etc., except Bar Bending Schedule, Shop & Fabrication drawings, for all works shall be supplied to the contractor for their scope of work all buildings services and development works by ALIMCO in phased manner as the works progress. However, it shall be the duty and responsibility of the contractor to bring to the notice of the ALIMCO in writing as to any variation, discrepancy or any other changes required and to obtain revised drawings and designs and / or approval of the ALIMCO in writing for the same.
- 28.6 One copy of contract documents including drawings furnished to the contractor shall be kept at the site and the same shall at all reasonable times be available for inspection.
- 28.7 All materials, construction plants and equipment etc. once brought by the contractor within the project area, will not be allowed to be removed from the premises without the written permission of the Engineer-in-charge. Similarly, all enabling works built by the contractor for the main construction undertaken by him, shall not be dismantled and removed without the written authority of the ALIMCO.

29.0 QUALITY ASSURANCE PROGRAMME

To ensure that the services under the scope of this contract are in accordance with the specifications, the Contractor shall adopt Quality Assurance Programme to control such activities at the necessary points. The contractor shall prepare and finalize such Quality Assurance Programme within 15 days from date of issue of work order. ALIMCO shall also carryout quality audit and quality

surveillance of systems and procedures of Contractor's quality control activities. A Quality Assurance Programme of Contractor shall generally cover the following:

- a) His organization structure for the management and implementation of the proposed Quality Assurance Program.
- b) Documentation control system.
- c) The procedure for purpose of materials and source inspection.
- d) System for site controls including process controls.
- e) Control of non-conforming items and systems for corrective actions.
- f) Inspection and test procedure for site activities.
- g) System for indication and appraisal of inspection status.
- h) System for maintenance of records.
- i) System for handling, storage and delivery.
- j) A quality plan detailing out quality practices and procedures, relevant standards and acceptance levels for all types of work under the scope of this contract.

All the quality reports (i.e. checklists & registers) shall be maintained /submitted by the Contractor as per ALIMCO 's approved QA-QC plan . Checklists & Registers enclosed in the approved QA-QC plan shall be followed while carrying out Construction activities(items). If any item is not covered by the Checklist/Register, the Format for the same may be developed and submitted to Engineer-in-Charge for approval and the same shall be adopted. These filled in reports shall be duly signed by representatives of contractor and ALIMCO. All the costs associated with Printing of Formats and testing of materials required as per technical specifications or by Engineer-in-charge shall be included in the Contractor's quoted rates in the Schedule/ Bill of quantities.

30.0 CONTRACT COORDINATION PROCEDURES, COORDINATION MEETINGS AND PROGRESS REPORTING

The Contractor shall prepare and finalize in consultation with ALIMCO, a detailed contract coordination procedure within 15 days from the date of issue of work order for the purpose of execution of the Contract. The Contractor shall have to attend all the meetings at any place in India at his own cost with ALIMCO, Consultants of ALIMCO during the currency of the Contract, as and when required and fully cooperate with such personal and agencies involved during these discussions. The Contractor shall not deal in any way directly with the Consultants of ALIMCO and correspondence if required at any time with Consultants shall be through ALIMCO only. During the execution of the work, Contractor shall submit at his own cost a detailed Monthly progress & programme report to the Engineer-in-charge of ALIMCO by 5th of every month. The format of monthly progress & programme report shall be as approved by Engineer-in-Charge of ALIMCO.

31.0 COMPLETION CERTIFICATE AND COMPLETION PLANS

- 31.1 Within ten days of the completion of the work, the contractor shall give notice of such completion to the Engineer-in-Charge and within thirty days of the receipt of such notice, the Engineer-in-Charge shall inspect the work and if there is no defect in the work, shall furnish the contractor with a final certificate of completion,

- 31.2 But no final certificate of completion shall be issued, nor shall the work be considered to be complete until the contractor shall have
- a) Removed from the premises on which the work shall be executed all scaffolding, surplus materials, rubbish and all huts and sanitary arrangements required for his/their work people on the site in connection with the execution of the works as shall have been erected or constructed by the contractor(s)
 - b) Cleaned off the dirt from all wood work, doors, windows, walls, floor or other parts of the building, in, upon, or about which the work is to be executed or of which he may have had possession for the purpose of the execution; thereof,
 - c) Not until the work shall have been measured by the Engineer-in-Charge.

31.3 If the contractor shall fail to comply with the requirements of this Clause 33.2 above then,

(A) The Engineer-in-Charge may at the expense of the contractor

(i) remove such scaffolding, surplus materials and rubbish etc.,

and/or

(ii) dispose of the same as he thinks fit and clean off such dirt as aforesaid,

AND,

(B) The contractor shall have no claim in respect of scaffolding or surplus materials as aforesaid except for any sum actually realized by the sale thereof less actual cost incurred on removal of materials / debris / malba etc.

31.4 The Contractor shall be responsible for handing over of project including signing of inventories by the ALIMCO and shall obtain final work completion certificate from ALIMCO in the given format. Necessary support for the same, shall however, be provided by ALIMCO. No payment of final bill shall be released to the contractor until final work completion certificate is obtained from ALIMCO.

The contractor shall submit completion plan as required vide General Specifications for Electrical and other works as applicable within thirty days of the completion of the work.

32.0 PROHIBITION OF UNAUTHORISED CONSTRUCTION & OCCUPATION

No unauthorized buildings, construction of structures should be put up by the contractor anywhere on the project site, neither any building built by him shall be occupied in un-authorized manner by him or his staff.

It shall be the responsibility of the contractor to see that the building under construction is not occupied by anybody in un-authorized manner during construction, and is handed over to the Engineer-in-Charge with vacant possession of complete building. If such building though completed is occupied illegally, then the Engineer-in-Charge shall have the option to refuse to accept the said building/buildings in that position. Any delay in acceptance on this account will be treated as the delay in completion and for such delay, a levy of compensation upto 5% of tendered value of work may be imposed by the Engineer-in-Charge whose decision shall be final both with regard to the justification and quantum and shall be binding on the contractor.

However, the Engineer-in-Charge, through a notice, may require the contractor to remove the illegal occupation any time on or before construction and delivery.

33.0 FORECLOSURE OF CONTRACT BY ALIMCO

If at any time after the commencement of the work the ALIMCO shall for any reason whatsoever is required to abandon the work or is not require the whole work thereof as specified in the tender to be carried out, the Engineer-in-Charge shall give notice in writing of the fact to the contractor, who shall have no claim to any payment of compensation whatsoever on account of any profit or advantage which he might have derived from the execution of the work in full, but which he did not derive in consequence of the foreclosure of the whole or part of the works.

34.0 DEFECTS LIABILITY PERIOD

The contractor shall be responsible for the rectification of defects in the works for a period of **Thirty Six (36)** months from the date of taking over of the works by the ALIMCO or clients whichever is later. Any defects discovered and brought to the notice of the contractor forthwith shall be attended immediately to and rectified by him at his own cost and expense. In case the contractor fails to carry out these rectifications, the same may without prejudice to any other right or remedy available, be got rectified by ALIMCO at the cost and expense of the contractor

35.0 FORCE MAJEURE

Any delay in or failure to perform of either party, shall not constitute default so as to give rise to any claim for damages, to the extent such delay or failure to perform is caused by an act of God, or by fire, explosion, flood or other natural catastrophe, governmental legislation, orders or regulation etc. Failure of the client / owner to hand over the entire site and / or release funds for the project, to ALIMCO, shall also constitute force majeure. The time for performance of the obligation by the parties shall be deemed to be extended for a period equal to the duration of the force majeure event. Both parties shall make their best efforts to minimize the delay caused by the force majeure event. If the failure / delay of the client / owner in handing over the entire site and / or in releasing the funds continues even on the expiry of the stipulated date of completion, ALIMCO, may, at the request of the contractor, foreclose the contract without any liability to either party. In the event of such foreclosure, the contractor shall not be entitled to any compensation whatsoever. If prior to such foreclosure the contractor has brought any materials to the site, the Engineer-in-Charge shall always have the option of taking over of all such materials at their purchase price or at the local current rates, whichever is lower.

36.0 NO COMPENSATION CLAUSE

The contractor shall have no claim whatsoever for compensation or idle charges against ALIMCO on any ground or for any reason, whatsoever.

37.0 DIRECTION FOR WORKS

- 37.1 All works under the contract shall be executed under the direction and subject to approval in all respect of the Engineer-in-Charge of ALIMCO who shall be entitled to direct at whatever point or points and in whatever manner works are to be commenced and executed.
- 37.2 The Engineer-in-Charge and his representative shall communicate or confirm their instructions to the contractor in respect of the execution of work during their site inspection in a 'Works Site Order Book' maintained at the site office of Engineer-in-Charge. The contractor or his authorized representative shall confirm receipt of such instructions by signing against the relevant orders in the book.

38.0 WORK ON SUNDAYS, HOLIDAYS AND DURING NIGHT

For carrying out work on Sunday and Holidays or during night, the contractor will approach the Engineer-in-Charge or his representative at least two days in advance and obtain his permission. The Engineer-in-Charge at his discretion can

refuse such permission. The contractor shall have no claim on this account whatsoever. If work demand, the contractor shall make arrangements to carry out the work on Sundays, Holidays and in two, three shifts with the approval of Engineer-in-Charge at no extra cost to ALIMCO.

39.0 WATER AND ELECTRICITY

The contractor shall make his own arrangement for Water (conforming to quality /standard prescribed for the particular use that water is brought into) & Electrical Power for construction and related purposes, make standby arrangement at his own cost, and pay requisite charges.

40.0 LAND FOR LABOUR HUTS/SITE OFFICE AND STORAGE ACCOMMODATION

- 40.1 The contractor may construct temporary office, storage within the site premises where the space is available at site. Accommodation and labour huts is not allowed within the site premises where the space is available at site. In case, where surplus land is not available within the site and/or not permitted by the client, the contractor shall arrange the land for temporary office, storage, accommodation and labour huts at his own cost and is responsible for taking the clearance of local authorities, if required, for setting up/construction of labour camp and same is deemed to be included in the rates quoted by the contractor for the works. The contractor shall ensure that the area of labour huts is kept clean and sanitary conditions are maintained as laid down by the local authorities controlling the area. The land for the above purposes shall be so placed that it does not hinder the progress of work or access to the worksite. The vacant possession of the land used, for the purpose shall be given back by contractor after completion of the work.

The security deposit of the contractor shall be released only after contractor demolishes all structures including foundations and gives back clear vacant possession of this land.

- 40.2 In the event the contractor has to shift his labour campus at any time during execution of the work on the instructions of local authorities or as per the requirement of the work progress or as may be required by ALIMCO, he shall comply with such instructions at his cost and risk and no claim whatsoever shall be entertained on this account.

41.0 WATCH, WARD AND LIGHTING OF WORK PLACE

The contractor shall at his own cost take all precautions to ensure safety of life and property by providing necessary barriers, obstruction, lights, watchmen etc. during the progress of work as directed by Engineer-in-charge.

42.0 CEMENT AND CEMENT GODOWN

Cement shall be procured by Contractor conforming to BIS: 8112 and / or BIS: 1489 Specification latest edition or higher Grade as per approved list attached. The cement shall be procured directly from the reputed manufacturers as per approved list of ALIMCO. Relevant vouchers and test certificates will be produced as and when required. The cement shall be stored by the contractor in such suitable covered and lockable stores, well protected from climate and atmospheric effects. The cement go-down shall be constructed by the contractor as per the drawing in CPWD specifications at his own cost. The cement will remain under double lock, one from ALIMCO and other from Contractor. The cement in bags shall be stored in go-downs in easy countable position. Cement bags shall be used on first in first out basis. Cement stored for beyond 90 days will be required to be tested at contractor's cost, before use in works.

43.0 STEEL & STEEL STOCKYARD

Steel conforming to BIS specifications (latest edition) shall be procured by the contractor directly from reputed manufacturers/producers as per approved list of ALIMCO. Relevant vouchers & test certificates will be produced by the contractor. Reinforcement steel, structural steel shall be stored and stacked in such manner so as to facilitate easy identification, removal etc. The contractor shall take proper care to prevent direct contact between the steel and the ground/ water for which he shall provide necessary arrangement at his own cost including ensuring proper drainage of area to prevent water logging as per directions of the Engineer-in-Charge. Steel shall also be protected, by applying a coat of neat cement slurry over the bars for which no extra payment shall be made. Test certificates for each consignment of steel shall be furnished and tests to be got carried out from the authorized laboratory as per the directions of Engineer-in-Charge, before incorporating the materials in the work.

44.0 BITUMEN WORK

The contractor shall collect the total quantity of tar or bitumen required for the work as per standard formula, before the process of painting started and shall hypothecate it to the Engineer-in-Charge. Although the materials are hypothecated to ALIMCO the Contractor undertakes the responsibility for their proper watch, safe custody and protection against all risks. The materials shall not be removed from site of work without the written consent of the Engineer-in-charge.

If any bitumen or tar remains unused on completion of the work on account of lesser use of materials in actual execution for reasons other than authorized changes of specifications and abandonment of portion of work, a corresponding deduction equivalent to the cost of unused materials as determined by the Engineer-in-Charge shall be made and the material return to the contractors.

45.0 WATER PROOF TREATMENT AND ANTI-TERMITE TREATMENT

- 45.1 The water proof treatment shall be of type and specifications as given in the schedule of quantities and/or specifications/ design-basis-report mentioned in tender documents.
- 45.2 The water-proofing treatment of basement, roofs, water retaining areas and termite infestation shall be and remain fully effective for a period of not less than 10(Ten) years to be reckoned from the date of expiring of the Defect Liability period, prescribed in the contract.
- 45.3 Re-treatment if required shall be attended to and carried out by the Contractor within fifteen days of the notice from Engineer-in-Charge of ALIMCO.
- 45.4 The ALIMCO reserves the right to get the quality of treatment checked in accordance with recognized test methods and in case it is found that the chemicals with the required concentration and rate of application have not been applied, or the water proofing treatment is not done as per specifications, the contractor will be required to do the re-treatment in accordance with the required concentration & specifications at no extra cost failing which no payment for such work will be made. The extent of work thus rejected shall be determined by ALIMCO.
- 45.5 Water proofing and Anti-termite treatment shall be got done through approved / specialized agencies only with prior approval of Engineer-in-Charge.
- 45.6 The contractor shall make such arrangement as may be necessary to safe guard the workers and residents of the building against any poisonous effect of the chemicals

used during the execution of the work.

- 45.7 During the execution of work, if any damage shall occur to the treatment already done, either due to rain or any other circumstances, the same shall be rectified and made good to the entire satisfaction of Engineer-In-Charge by the contractor at his cost and risk.
- 45.8 The contractor shall make his own arrangement for all equipments required for the execution of the job.

46.0 INDIAN STANDARDS

Wherever any reference is made to any IS in any particular specifications, drawings or bill of quantities, it means the Indian Standards editions with up to date amendments issued till last date of receipt of tender documents.

47.0 CENTERING & SHUTTERING

Marine plywood or steel plates or any material mentioned elsewhere in the tender document or as approved by Engineer-in-Charge shall be used for formwork. The shuttering plates shall be cleaned and oiled before every repetition and shall be used only after obtaining approval of ALIMCO's Engineers at site. The number of repetitions allowed for plywood and steel shuttering shall be at the discretion of Engineer-in-Charge of ALIMCO depending upon the condition of shuttering surface after each use and the decision of Engineer-in-Charge in this regard shall be final and binding on the contractor. No claim whatsoever on this account shall be admissible.

48.0 RECORDS OF CONSUMPTION OF CEMENT & STEEL

- 48.1 For the purpose of keeping a record of cement and steel received at site and consumed in works, the contractor shall maintain a properly bound register in the form approved by the ALIMCO, showing columns like quantity received and used in work and balance in hand etc. This register shall be signed daily by the contractor's representative and ALIMCO's representative.
- 48.2 The register of cement & steel shall be kept at site in the safe custody of ALIMCO's Engineer during progress of the work. This provision will not, however, absolve the contractor from the quality of the final product.

49 TESTS AND INSPECTION

The contractor shall carry out the various mandatory tests as per specifications and the technical documents that will be furnished to him during the performance of the work. All the tests on materials, as recommended by CPWD, MORTH and relevant Indian Standard Codes or other standard specifications (including all amendments current at the last date of submission of tender documents) shall be got carried out by the contractor at the field-testing laboratory or any other recognized institution/ laboratory, at the direction of the ALIMCO. All testing charges, expenses etc. shall be borne by the contractor. All the tests, either on the field or outside laboratories concerning the execution of the work and supply of materials shall be got carried out by the contractor or ALIMCO at the cost of the Contractor.

50.0 WORKS TO BE OPEN TO INSPECTION

All works executed or under the course of execution in pursuance of this contract shall at all times be open to inspection and supervision of the ALIMCO.

The work during its progress or after its completion may also be inspected, by Chief Technical Examiner of Government of India (CTE) and/ or an inspecting authority of State Government of State in which work is executed and/or by third party checks by owner/ clients. The compliance of observations/improvements as suggested by the inspecting officers of ALIMCO/CTE/ State authorities/ Owners shall be obligatory on the part of the Contractor at the cost of contractor.

51.0 SETTING OUT OF THE WORKS

The contractor shall be responsible for the true and proper setting out of the works and for the correctness of the position, levels, dimensions and alignment of all parts of the works. If at any time during the progress of works, shall any error appear or arise in the position, levels, dimensions or alignment of any part of the works, the contractor shall at his own expenses rectify such error to the satisfaction of Engineer-in-charge. The checking of any setting out or of any line or level by the engineers of ALIMCO shall not in any way relieve the contractor of his responsibility for the correctness.

52.0 NOTICE BEFORE COVERING UP THE WORK

The contractor shall give not less than seven day notice before covering up or otherwise placing beyond the reach of measurement any work, to the Engineer-in-charge in order that the same may be inspected and measured. If any work is covered up or placed beyond the reach of inspection/measurement without such notice or his consent being obtained the same shall be uncovered at the contractor expenses and he shall have to make it good at his own expenses.

53.0 SITE CLEARANCE

The contractor shall ensure that the working site is kept clean and free of obstructions for easy access to job site and also from safety point of view. Before handing over the work to the ALIMCO the contractor shall remove all temporary structures like the site offices, cement go-down, stores, labour hutments etc., scaffolding rubbish, debris etc. left over materials tools and plants, equipment etc., clean the site to the entire satisfaction of the Engineer-in-charge. If this is not done the same will be got done by ALIMCO at contractors risk and cost.

The contractor shall clean all floors, remove cement/ lime/ paint drops and deposits, clean joinery, glass panes etc., touching all painter's works and carry out all other necessary items of works to make the premises clean and tidy before handing over the building, and the rates quoted by the contractor shall be deemed to have included the same.

54.0 SET-OFF OF CONTRACTOR'S LIABILITIES

ALIMCO shall have the right to deduct or set off the expenses incurred or likely to be incurred by it in rectifying the defects and/or any claim under this agreement against the Contractor from any or against any amount payable to the contractor under this agreement including security deposit and proceeds of performance guarantee.

55.0 POSSESSION PRIOR TO COMPLETION

ALIMCO shall have the right to take possession of or use any completed or partially completed work or part of the work. Such possession or use shall not be deemed to be any acceptance of any work not completed in accordance with the contract agreement. If such prior possession or use by ALIMCO delays the progress of work an equitable adjustment in the time of completion will be made and the contract agreement shall be deemed to be modified accordingly. The decision of ALIMCO in such case shall be final binding and

conclusive.

When the whole of the works or the items or the groups of items of work have been completed the contractor will give a notice to that effect to the Engineer in writing. The Engineer shall within 7 days of the date of receipt of such notice inspect the works and give instructions in writing to the contractor specifying the balance items of work which are required to be done by the contractor and shall also notify the contractor of any defect in the works affecting completion.

The contractor shall during the course of execution prepare and keep updated a complete set of 'as built' drawings to show each and every change from the contract drawings, changes recorded shall be countersigned by the Engineer-in-Charge and the contractor. Four copies of 'as built' drawings shall be supplied to ALIMCO by the contractor within 30 days of the completion. All costs incurred in this respect shall be borne by the contractor.

56.0 EMPLOYMENT OF PERSONNEL

56.1 The contractor shall employ only Indian Nationals as his representatives, servants and workmen after verifying their antecedents and loyalty. He shall ensure that no personnel of doubtful antecedents and any other nationality in any way is associated with the works.

56.2 In case ALIMCO observed misconduct negligence or incompetence etc. on the part of any representative, agent, servant and workmen or employees etc. Of the contractor, the ALIMCO shall have full power and without giving any reason to the contractor, instruct the contractor to remove such engineer /staff / worker from site and provide suitable replacements. The decision of the Engineer-in-charge shall be final and binding on the contractor. The contractor shall not be allowed any compensation on this account.

57.0 TECHNICAL STAFF FOR WORK

57.1 The contractor shall employ at his cost the adequate number of technical staff during the execution of this work depending upon the requirement of work. The technical staff should be available at site, whenever required by ALIMCO to take instructions.

57.2 Within 15 days of date of issue of work order, the contractor shall submit a site organizational chart and resume including details of experience of the Project-in-Charge and other staff proposed to be deputed by him and the technical team shall be deputed by them on the Project after getting approval from Engineer-in-Charge. If desired by the contractor at later date, the Project-in-Charge and other staff whose resume is approved by ALIMCO can be replaced with prior written approval of ALIMCO and replacement shall be with equivalent or superior candidate only. Decision of Engineer-in-Charge shall be final and binding on the contractor.

57.3 Even after approving the site organizational chart, the Engineer-in-Charge due to technical reasons and exigency of work can direct the contractor to depute such additional staff as in view of Engineer-in-Charge is necessary and having qualification and experience as approved by the Engineer-in-Charge. The contractor shall not be paid anything extra whatsoever on account of deployment of additional staff and decision of the Engineer-in-Charge shall be final and binding on the contractor.

57.4 The decision of the Engineer-in-charge as to number of Technical Staff to be adequate for the project and the period for which the desired strength of technical staff was not employed by the contractor and as to the reasonableness of the amount to be deducted on this account shall be final and binding on the contractor as to the amount and the contractor's liability to pay the said amount.

58.0 VALUABLE ARTICLES FOUND AT SITE

All gold, silver and other minerals of any description and all precious stones, coins, treasure, relics, antiques and all other similar things which shall be found in, under or upon the site, shall be the property of the owner/ ALIMCO.

59.0 MATERIALS OBTAINED FROM DISMANTLEMENT TO BE OWNER'S PROPERTY

All materials like stone, boulders and other materials obtained during the work of dismantling, excavation etc. will be considered ALIMCO/owner property and such materials shall be disposed-off to the best advantage of

ALIMCO/owner according to the instructions in writing issued by the Engineer-in-charge.

60.0 FURNISHED OFFICE ACCOMMODATION & MOBILITY COMMUNICATION TO BE ARRANGED BY CONTRACTOR

On acceptance of tender, the contractor at his own cost will preferably construct Pre-engineered/ Porta cabin's office at site suitably equipped with basic facilities like electricity and drinking water supply and vehicle for the supervisory staff with driver as per the requirement of the project. The contractor shall maintain the aforesaid facilities intact/operational during the tenancy of the contract or maximum up to 6 months beyond the stipulated contractual completion date if the work is delayed due to any reasons. Operation and maintenance cost on the above facilities shall be borne by the contractor.

The contractor shall also make sufficient arrangement for photography/ videography so that photographs video can be taken of any specific activity at any point of time.

The contractor shall make all arrangements for ground breaking ceremony/inaugural function etc. for the project as required and the cost towards it deemed to be included in his rates/offer. Any expenditure already incurred/to be incurred by ALIMCO, shall be recovered from the contractor.

61.0 LABOUR LAWS

61.1 LABOUR LAWS TO BE COMPLIED BY THE CONTRACTOR

The contractor shall obtain a valid license under the contract labour (Regulation & Abolition) Act 1970 and the contract labour Act (Regulation & Abolition) Central Rules 1971 and amended from time to time, and continue to have a valid license until the completion of the work including defect liability period.

The contractor shall also comply with the provisions of the building and other Construction Workers (Regulation of Employment & Conditions of Service) Act, 1996 and the building and other Construction Workers Welfare Cess Act, 1996.

The contractor shall not engage any labour below the age of 18 under any circumstances. The provisions under Child Labour (Prohibition and Regulation) Amendment Act, 2016 should be strictly adhered to. In case of any non-compliance to the requirements of Labour laws, the contractor shall be liable for all consequences or any penalty imposed in this regard.

Any failure to fulfil above requirement shall attract the penal provisions of this contract arising out the resultant for non-execution of the work before the commencement of work.

61.2 Payment of wages:

(i) The contractor shall pay to labour employed by him either directly or through

subcontractors, wages not less than fair wages as defined in the provisions of the Contract Labour (Regulation and Abolition) Act, 1970 and the contract Labour (Regulation and Abolition) Central Rules, 1971, wherever applicable.

- (ii) The contractor shall transfer/ credit the wages/salary of all labourers/workers preferably in their bank accounts. He shall be responsible for opening of bank accounts of all labourers/workers employed by the contractor at work site in this regard.
- (iii) In respect of all labour directly or indirectly employed in the works for performance of the contractor's part of this contract, the contractor shall comply with or cause to be complied with the Labour Regulations in regard to payment of wages, wage period, deductions from wages recovery of wages not paid and deductions unauthorisedly made, maintenance of wage books or wage slips, publication of scale of wages and other terms of employment, inspection and submission of periodical returns and all other matters of the like nature or as per the provisions of the Contract Labour (Regulation and Abolition) Act, 1970, and the Contract Labour (Regulation and Abolition) Central Rules, 1971, wherever applicable.
- (iv) (a) The Engineer-in-Charge concerned shall have the right to deduct from the moneys due to the contractor any sum required or estimated to be required for making good the loss suffered by a worker or workers by reason of non-fulfilment of the conditions of the contract for the benefit of the workers, non-payment of wages or of deductions made from his or their wages which are not justified by their terms of the contract or non-observance of the Regulations.
- (b) Under the provision of Minimum Wages (Central) Rules, 1950, the contractor is bound to allow to the labours directly or indirectly employed in the works one-day rest for 6 days continuous work and pay wages at the same rate as for duty. In the event of default, the Engineer-in-Charge shall have the right to deduct the sum or sums not paid on account of wages for weekly holidays to any labours and pay the same to the persons entitled thereto from any money due to the contractor by the Engineer-in-Charge concerned
- (v) The contractor shall comply with the provisions of the Payment of Wages Act, 1936, Minimum Wages Act, 1948/ Code on Wages, 2019, Employees Liability Act, 1938, Workmen's Compensation Act, 1923, Industrial Disputes Act, 1947, Maternity Benefits Act, 1961, and the Contractor's Labour (Regulation and Abolition) Act 1970, or the modifications thereof or any other laws relating thereto and the rules made there under from time to time.
- (vi) The contractor shall indemnify and keep indemnified ALIMCO against payments to be made under and for the observance of the laws aforesaid and the ALIMCO Contractor's Labour Regulations without prejudice to his right to claim indemnity from his sub-contractors.
- (vii) The laws aforesaid shall be deemed to be a part of this contract and any breach thereof shall be deemed to be a breach of this contract.

61.3 **LABOUR SAFETY PROVISION**

The contractor shall be fully responsible to observe the labour safety provisions:

The contractor shall at his own cost take all precautions to ensure safety of life and property by providing necessary barriers, lights, watchmen etc. During the progress of work as directed by Engineer-in-charge

In case of all labour directly or indirectly employed in work for the performance on the contractor's part of this contract, the contractor shall comply with all rules framed by Govt. from time to time for the protection of health and sanitary arrangements for workers.

61.4 OBSERVANCE OF LABOUR LAWS

61.4.1 The contractor shall be fully responsible for observance of all labour laws applicable including local laws and other laws applicable in this matter and shall indemnify and keep indemnified ALIMCO against effect or non-observance of any such laws. The contractor shall be liable to make payment to all its employees, workers and sub-contractors and make compliance with labour laws. If ALIMCO is held liable as "Principal Employer" to pay contributions etc. under legislation of Government or Court decision in respect of the employees of the contractor, then the contractor would reimburse the amount of such payments, contribution etc. to ALIMCO and/ or same shall be deducted from the payments, security deposit etc. of the contractor.

61.4.2 The Contractor shall submit proof of having valid EPF registration certificate. He shall within 7 days of the close of every month, submit to ALIMCO a statement showing the recoveries of contributions in respect of each employee employed by or through him and shall furnish to ALIMCO such information as the ALIMCO is required to furnish under the provisions of para 36 B of the EPF scheme 1952 to the EPF authorities and other information required by EPFO authorities from time to time. He shall also submit a copy of challan every month in token of proof of having deposited the subscription and contribution of workers engaged on the project.

The contractor shall also ensure the compliance of EPF & MP Act, 1952 by the sub-contractors, if any, engaged by the contractor for the above said work.

The contractor shall submit affidavit to indemnify and save harmless the Corporation from and against all actions, suits, proceedings, losses, costs, damages, charges, claims and demands of every nature and description brought or recovered against the Corporation by reasons of any act or omission of the Contractor, his agents or employees in connection with complying the provisions of the Employees Provident Fund & Miscellaneous Provisions Act, 1952 as amended from time to time. All sums payable by way of compensation

/ penalty / damages / interest on the outstanding amounts payable by the Contractor shall be considered as reasonable and be payable by the Contractor to the Corporation immediately and if the Contractor does not pay the amount immediately the same will be deducted from the security deposit or earnest money or any other amount available with the Corporation or any money payable to the Contractor by the Corporation.

Contractor should submit a Compliance Certificate along with the details of employees as per the provisions of the EPF and ESI Act as amended from time to time.

61.4.3 The Contractor shall submit proof of having valid ESI registration for Construction site workers located in the ESI implemented areas for every construction site worker before his/her engagement on the ALIMCO site of works as per requirement of ESI act, 1948 amended upto date and rules made thereunder.

The contractors are required to comply with all the relevant provisions of ESI act, 1948 as amended from time to time and deposit of his contribution as may be required under the above said act to the ESI authorities at required intervals / time of deposit and submit the proof to ALIMCO.

The contractor shall at all times indemnify ALIMCO and Owner against all claims,

damages or compensation under the provision of ESI Act, 1948 or any modifications thereof or as consequence of any accident or injury to any workman or other persons in or about the works, whether in the employment of the contractor or not, against all costs, charges and expenses of any suit, action or proceedings arising out of such incident or injury and against all sum or sums which may with the consent of the contractor be paid to compromise or Compound any such claim.

61.5 MINIMUM WAGES ACT

The contractor shall comply with all the provisions of the minimum wages Act, 1948 Code on Wages, 2019, contract labour Act (Regulation & Abolition) 1970, and rules framed there under and other labour laws/local laws affecting contract labour that may be brought into force from time to time.

61.6 LABOUR RECORDS

The contractor shall submit by the 4th & 19th of every month to the Engineer-in-Charge of ALIMCO a true statement, showing in respect of the second half of the proceeding month and the first half of the current month, respectively, of the following data:

- a) The number of the labour employed by him (category-wise).
- b) Their working hours.
- c) The wages paid to them.
- d) The accidents that occurred during the said fortnight showing the circumstances under which they happened and the extent of damage and injury caused.
- e) The number of female workers who have been allowed Maternity Benefits and the amount paid to them.
- f) Any other information required by Engineer-in-Charge

The decision of the ALIMCO shall be final in deducting from any bill due to the contractor default, amount levied as fine and is binding on the contractor.

61.7 If It appears to the Engineer-in-Charge that the contractor(s) is/are not properly observing and complying with the provisions of the ALIMCO Contractor's Labour Regulations and Model Rules and the provisions of the Contract Labour (Regulation and Abolition) Act 1970, and the Contract Labour (R&A) Central Rules 1971, for the protection of health and sanitary arrangements for work-people employed by the contractor(s) (hereinafter referred as "the said Rules") the Engineer-in-Charge shall have power to give notice in writing to the contractor(s) requiring that the said Rules be complied with and the amenities prescribed therein be provided to the work-people within a reasonable time to be specified in the notice. The contractor(s) shall provide each hut with proper ventilation.

- (i) Water Supply - The contractor(s) shall provide adequate supply of water for the use of labourers.
- (ii) Drainage - The contractor(s) shall provide efficient arrangements for drainage away of sullage water so as to keep the camp neat and tidy.
- (iii) The contractor(s) shall make necessary arrangements for keeping the camp area sufficiently lighted to avoid accidents to the workers.
- (iv) Sanitation - The contractor(s) shall make arrangements for conservancy and sanitation in the labour camps according to the rules of the Local Public Health and Medical Authorities.

62.0 LABOUR CESS

The rates of the contractor shall be inclusive of labour cess. Labour cess shall be deposited by the contractor and proof of the same submitted to ALIMCO.

The contractor shall also be responsible for maintaining register of beneficiaries i.e. the workers in such form as may be prescribed by the competent authority & the same shall be kept open at all reasonable times for inspection of relevant authority and officials of ALIMCO.

In the event of contractor failing to comply with the above clause(s) in part or in full, ALIMCO, without prejudice to any other rights or remedy available under law or any other clause(s) of contract, shall be at absolute liberty to forfeit any sum or sums that are payable or could become payable on account of execution of contract work and decision of Engineer-in-charge shall be final & binding in this regard on the contractor.

63.0 RECOVERY OF COMPENSATION PAID TO WORKMEN

In every case in which by virtue of the provisions sub-section (1) of Section 12, of the Workmen's Compensation Act, 1923, ALIMCO is obliged to pay compensation to a workman employed by the contractor, in execution of the works, ALIMCO will recover from the contractor, the amount of the compensation so paid; and, without prejudice to the rights of the ALIMCO under sub-section (2) of Section 12, of the said Act, ALIMCO shall be at liberty to recover such amount or any part thereof by deducting it from the security depositor from any sum due to the contractor whether under this contract or otherwise. ALIMCO shall not be bound to contest any claim made against it under sub-section

(1) of Section 12, of the said Act, except on the written request of the contractor and upon his giving to ALIMCO full security for all costs for which ALIMCO might become liable in consequence of contesting such claim.

64.0 ENSURING PAYMENT AND AMENITIES TO WORKERS IF CONTRACTOR FAILS

In every case in which by virtue of the provisions of the Contract Labour (Regulation and Abolition) Act, 1970, and of the Contract Labour (Regulation and Abolition) Central Rules, 1971, ALIMCO is obliged to pay any amounts of wages to a workman employed by the contractor in execution of the works, or to incur any expenditure in providing welfare and health amenities required to be provided under the above said Act or under the ALIMCO Contractor's Labour Regulations, or under the Rules framed by Government from time to time for the protection of health and sanitary arrangements for workers employed by ALIMCO's Contractors, ALIMCO will recover from the contractor, the amount of wages so paid or the amount of expenditure so incurred; and without prejudice to any other right or remedy available under this contract, ALIMCO shall be at liberty to recover such amount or any part thereof by deducting it from the security deposit or from any sum due by ALIMCO to the contractor whether under this contract or otherwise. ALIMCO shall not be bound to contest any claim made against it under sub-section (1) of Section 20, sub-section (4) of Section 21, of the said Act, except on the written request of the contractor and upon his giving to the ALIMCO full security for all costs for which ALIMCO might become liable in contesting such claim.

65.0 CHANGE IN FIRM'S CONSTITUTION TO BE INTIMATED

Where the contractor is a partnership firm, the prior approval in writing of the Engineer-in-Charge shall be obtained before any change is made in the constitution of the firm. Where the contractor is an individual or a Hindu undivided

family business concern such approval as aforesaid shall likewise be obtained before the contractor enters into any partnership agreement where under the partnership firm would have the right to carry out the works hereby undertaken by the contractor. If prior approval as aforesaid is not obtained, the contract shall be deemed to have been assigned in contravention of Clause 62.0. hereof and the same action may be taken, and the same consequences shall ensue as provided in the said Clause 62.0.

66.0 INDEMNITY AGAINST PATENT RIGHTS

The contractor shall fully indemnify the ALIMCO from and against all claims and proceedings for or on account of any infringement of any patent rights, design, trademark or name or other protected rights in respect of any construction plant, machine, work or material used for in connection with the works or temporary works.

67.0 LAW COVERING THE CONTRACT

This contract shall be governed by the Indian laws for the time being in force.

68.0 LAWS, BYE-LAWS RELATING TO THE WORK

- 68.1 The contractor shall fully comply with the DIPP's PPP-MII order no P-45021/2/2017/E II dated 15.06.17 or any further revision at any later date during the entire tenancy of the contract.

In case of procurement for a value in excess of Rs 10 crore, undertakings shall be provided from a statutory auditor or cost auditor of the company (in the case of companies) or from a practicing chartered accountant (in case of tenderer other than companies).

- 68.2 The contractor shall strictly adhere by the provisions, for the time being in force, of law relating to works or any regulations and bylaws made by any local authority or any water & lighting agencies or any undertakings within the limits of the jurisdiction of which the work is proposed to be executed. The contractor shall be bound to give to the authorities concerned such notices and take all approvals as may be provided in the law, regulations or bylaws as aforesaid, and to pay all fees and taxes payable to such authorities in respect thereof.

69.0 CONTRACT AGREEMENT

The Contractor shall enter into a Contract Agreement with the ALIMCO within 30 (Thirty) days from the date of issue of work order failing which no payment shall be released to the contractor. The cost of stamp papers, stamp duty, registration, if applicable on the contract, shall be borne by the Contractor. In case, the contractor does not sign the agreement as above or start the work within 10 (Ten) days of the issue of date of issue of work order, his earnest money is liable to be forfeited and date of issue of work order consequently will stand withdrawn.

70.0 MANNER OF EXECUTION OF AGREEMENT

- 70.1 The agreement as per prescribed Performa as enclosed shall be signed at the office of the ALIMCO within 30 (Thirty) days from the date of issue of date of issue of work order. The Contractor shall provide for signing of the Contract, appropriate Power of Attorney and the requisite documents/ materials. Unless and until a formal contract is prepared and executed, the date of issue of work order read in conjunction with the Tendering Documents will constitute a binding contract.

- 70.2 The agreement will be signed in five originals and the Contractor shall be provided with one signed original and the other four originals will be retained by the ALIMCO
- 70.3 The Contractor shall provide free of cost to the ALIMCO all the Engineering data, drawings and descriptive materials submitted along with the tender, in at least three (3) copies to form an integral part of the Agreement within seven 7 days after date of issue of work order.
- 70.4 Subsequent to signing of the Agreement, the Contractor at his own cost shall provide to the ALIMCO with at least three (3) true hard bound copies of Agreement within thirty (30) days of its signing.

71.0 JURISDICTION

The agreement shall be executed at Kanpur on non-judicial stamp paper purchased in Kanpur and the courts in Kanpur alone will have jurisdiction to deal with matters arising there from, to the exclusion of all other courts.

72.0 ARBITRATION

In case of any dispute, arising during the operation of the contract, efforts shall be made to resolve it by mutual discussions failing which the matter shall be referred to the Chairman cum Managing Director of ALIMCO G.T.Road, Kanpur-209217 or his nominee whose decision shall be final and binding on both the parties. Arbitration proceedings shall be held at Kanpur as per provisions of the Arbitration & Conciliation Act 1996 and the Kanpur City Court shall have the jurisdiction in the matter.

SCOPE OF WORK

1.0 SCOPE OF WORK:

The work shall be executed on EPC mode involving getting all local body clearances, Survey, Soil Investigation, structural design & drawings, approval of fabrication / shop drawings and As Built drawings, procurement & constructions & commissioning of buildings & services and obtaining completion certificate in accordance with layout plan and architectural / structural drawings.

This facility shall employ Divyangjans and shall also be used by Divyangjans. Hence necessary ramps, steel hand rails, accessible pathways, accessible toilets etc. to be provided make facility accessible for Divyangjans.

The works shall include all civil works including PEB/RCC buildings, roads, hard standings, foot paths, sewer lines, plumbing, drainage works, sanitary and other fixtures. Internal & external electrical works /DG set/AC works, CCTV camera, Firefighting with wet riser, sprinkler & fire alarm, water supply system transformers, LT electricity panel, Rainwater harvesting system, public address system Networking, Switches, Active & passive, horticulture work, furniture, Electrical, Mechanical, all Safety Aspect sand any other works Required for the Construction and Completion of the works in all respects and to the Satisfaction of the Engineer-in-charge.

The design of the structure for structural and functional requirement shall be done for combined effect of applicable dead load, imposed load, earthquake forces, wind loads and other loads as per applicable Indian Standards and National Building Code of India and actual site conditions. It would be the responsibility of bidder to get the Design and Drawings of structure vetted by Technical / Research institutions of repute such as IITs only.

1. Following units are included in the scope of this tender package:
 - a. Ware house (PEB structure)
 - b. Security block (RCC framed structure)
 - c. Admin building (RCC framed structure)
 - d. Boundary wall
 - e. Bituminous Roads with marking, foot paths, RCC drains, Trenches.
 - f. Signages (SS and LED)
 - g. Area development
2. Following supporting facilities are also included in the scope of this tender: -
 - a. Plant and Machineries.
 - b. Power Distribution System for all newly constructed unit and infrastructure by stepping down incoming 11 kV power supply at consumer premises.
 - c. Illumination for the area and units
 - d. Fire alarm & system. Conduiting for IT Infrastructure and thereafter installation and commissioning of complete system
 - e. standby power supply arrangement by 415 V, DG Sets 1x100 kVA Minimum capacity including cold starting kit radiator cooled.
 - f. Roads & storm water drain
 - g. U/G Water tank, Drinking water and Fire Hydrant System
 - h. Plumbing & sanitation work.
 - i. Ventilation.
 - j. Pump House & Electrical Sub Station
 - k. Arboriculture work including land scaping.
 - l. Bore/tube well.
 - m. Miscellaneous / other items / facilities / services as per requirement and as specified

2.0 Brief scope of Internal & External, Water supply pump set and water cooler with purifier:

Planning, designing, supply, installation, testing and commissioning of complete Internal and External Electrical Works which includes all copper wiring in MS conduits, LED energy efficient luminaires, occupancy & daylight sensors for all internal and external lighting as per directions of Engineer-in-Charge, fans, modular switches, sockets, Light Plugs, Power Plugs, Computer Plugs, industrial sockets, Modular Telephone/TV/Information Sockets, DBs, MCBs, MCCBs, RCCBs, suitable size cables for distribution of normal, Emergency & UPS Power, Electrical Panels with all accessories, exit signages etc. cable trays, floor trunking (wherever required), raceways, floor trunking box, stainless steel floor trunking box covers with required openings, earthing i/c all materials, labours, T&P, equipment and all accessories, lightning arrestor (all of the above) as per NBC- 2016, cable TV wiring in MS conduit, Shielded CAT6A cable for CCTV System, call bell system, addressable Electrical LED signages & LED Board (including photo luminescent), external services connection for all E&M services includes installation of all external of flood lights /street/compound lighting using LED light fittings, UG cabling Suitable Capacity 3 phase Pumps (Working+Standby) for Water Supply System consisting of pump sets and accessories, connecting piping work, control panel, interconnecting pipes, cables and switches, metering, all related electrical and Instrumentation work, water cooler with purifier, feeder pillar/panel with all control metering and automation switch gears and wiring and accessories, dry run protection phase reversal protection, overload & short circuit protection, suitable supports and protections, earthing, valves, connection inter connections etc. complete as required. The suitable size of trench/pipes shall be provided for entry of various services in the buildings e.g. data cable, TV cable, External CCTV, Telephone cable, power cable, earthing etc.

3.0 Brief scope of Fire Fighting with Down Comer System & Fire Extinguishers:

The work shall consist of planning, designing, supplying, installation, testing & commissioning of Fire fighting with Down Comer system i/c external fire hydrant system. The system shall be as per CPWD specifications, NBC 2016 and Local bylaws and as per clearance issued by Local Fire Service Department. The scope of work also includes planning, designing, preparation of drawings, obtaining approvals from Department, supply, installation, testing and commissioning of Fire fighting with Down Comer system in the entire building as per requirement and for all electrical panel, as per Fire bye-laws including applicable codes and standards. Any other Firefighting system in addition to mentioned above, if required as per local bye laws shall also be within the scope of work etc. complete as required as per tender document. The firefighting system shall be designed as per NBC 2016 as per direction of Engineer-in-charge.

4.0 Brief scope of Addressable Automatic Fire alarm & PA System:

The work shall consist of planning, designing, supplying, installation, testing & commissioning of Automatic Intelligent Addressable Fire alarm system. The system shall be as per CPWD specifications, NBC 2016 and Local bylaws and as per clearance issued by Local Fire Service Department. The scope of work also includes planning, designing, preparation of drawings, obtaining approvals from Department, supply, installation, testing and commissioning of PA system as per Fire bye-laws including applicable codes and standards.

5.0 Brief scope of D.G. set :

The work shall cover complete design, supply, transportation, foundation, installation, testing and commissioning of DG Sets (1x100 kVA Minimum capacity including cold starting kit). All minor civil works including Foundation, electrical work including DG Synchronizing panel (for future expansion) /AMF panel and other works such as cooling of DG Sets with water circulation pumps, pipe line cooling tower, ventilation and acoustic treatment of DG Sets room (if any), extending exhaust pipe of suitable diameter duly insulated

and cladded with Aluminium sheet up to above the building as per latest CPCB – IV+ /local body norms, testing, installation and commissioning of DG set shall be carried out by the tenderer as per specification. The tender should quote for complete job to be executed under the works contract.

NOTE : This scope of work also includes the making arrangement for placing of the equipment with suitable foundation and enclosure for protection of equipment as per CPWD manual for area requirement for E&M services & as per approval of Engineer-in-Charge.

6.0 Brief scope of IP Based CCTV System:

Design, Supply, Installation, testing & commissioning of CCTV system to monitor & supervise the internal & external areas for security purpose, as well as to record and inform officials on unwanted, untoward incidents. It is also essential to have recorded images to be stored at least for min 30 days of all critical areas to facilitate investigations of a reported incidents. The hardware required for the system including Servers VMS & Recording, Workstations, Monitors, shielded CAT-6A Patch Cable/ Fiber Optic Cable, camera, Cables, connectors, active switches, patch panels, networking racks, conduits, power supplies etc. is in the scope of work.

7.0 Brief scope of LT Panel Room in Admin Building :

Design, Supply, installation, testing and commissioning of power supply to the building which includes, APFC Panels as per site requirement/directions of Engineer-in-Charge, LT Panels, Sub-LT Panels,—Electrical Distribution Panels, power distribution throughout the building with Copper / aluminium conductor XLPE insulated, PVC inner and outer sheathed armoured UG Cable, Fire Survival Cables as per provisions of NBC-2016, subpanels, Feeder Pillars etc. as per specifications given. The work includes supplying and laying of two number HT Cables from meter board of supply company to the LT Panel inside the premises.—The work also includes earthing and safety equipment as per CPWD specifications.

NOTE: This scope of work also includes the making arrangement for placing of the equipment with suitable foundation and enclosure for protection of equipment as per CPWD manual for area requirement for E&M services & as per approval of Engineer-in-Charge.

If any item required to make the building / scheme habitable and functional, is not specifically mentioned in the scope of services, the same is deemed to be included within the scope of this tender and nothing extra shall be paid on this account.

8.0 Brief scope of Street Lighting & Compound Lighting:

Octagonal GI poles/ Ornamental poles with sliding door, single/double bracketed arm along the road as per IS standards. Ornamental/ Decorative poles and bollards with LED fittings may also be provided as per the site/client requirements.

Note:

1. The above-mentioned scope of work is indicative in nature and is not exhaustive, the actual work shall be carried out as per the CPWD specification of relevant package/subhead and NBC 2016 after obtaining approval of Engineer-in-Charge.
2. All the drawings of each Sub-Head shall have to be vetted by the IIT's only.
3. All the E&M Items shall be brought at site only after the approval of TDS & physical sample of the item/equipment and same shall be preserved at site lab/sample room. The Test Report from OEM/Third Party/Both for each/any E&M item/equipment can be sought by the Engineer-in-Charge.
4. The payment for the tests shall be borne by agency, and no reimbursement shall be made by the department. If the agency fails to submit the test reports from OEM/Third Party/Both for each/any E&M item/equipment as sought by the Engineer-in-Charge, then the department will get all the required tests done and the amount paid to the test agency by the department will

be recovered from the contractor bill.

9.0 Amenities to be provided :

1. All works have to be executed as per specifications provided in the bid document, CPWD Specifications, National building Code 2016 (in case of difference, if any, stringent/higher specification of the two shall be followed) and ECBC 2017 Code. In absence of CPWD Specifications, IS Codes, National Building Code 2016 sound engineering practices shall be adopted as per order of precedence defined in the contract.

There may be several incident works, which are not mentioned in the contract document/specifications but will be necessary to complete the item in all respect. All these incidental works/ costs which are not mentioned, but are necessary to complete the work shall be deemed to have been included in the overall amount quoted by the contractor for various components of work. No adjustment of rates shall be made for any variation in quantum of incidental works due to variation/change in actual working drawings. Also, no adjustment of rates shall be made due to any change in incidental works or any other deviation in such element of work (which is incidental to the items of work and are necessary to complete such items in all respects) on account of the directions of Engineer-in-charge. Nothing extra shall be payable on this account.

In case, some items are missing in the scope of work or specifications in the bidding documents the same shall be taken from the specification mentioned in similar type of items mentioned for similar type of building in the scope of work or shall be executed as given in the CPWD Specifications, NBC-2016, IS Codes or according to sound engineering practices so as to make the building including related services fully functional. No claim what so ever may be entertained at later stage. All cost of providing and making building with services works fully complete in all aspect unless specifically mentioned in the contract document and making buildings with services fully are included in the cost tendered for this work.

2. The **scope of work also includes Defect Liability Period: 3 (three) years** commencing from the date of Handing over of site.
3. Scope of work also include making the building completely functional including obtaining all pre & post construction statutory approvals as applicable.

The contractor shall obtain all mandatory approval and Non-Objection Certificate / Consent for Establishment / Clearance etc. from UPSIDA, local body authorities like local Fire Department, Local Town Planning Authority, Local Ground Water Authority, Local Electricity Supply Authority, Local Pollution Control Board, Forest Department, Environment Clearance, Central Electricity Authority, Department of Explosives for boiler and storage of diesel etc.

4. The responsibility of planning, procurement, construction, safety, quality, all statutory approvals and Liasoning work applicable to project and risk of engineering lies with the contractor.
5. Successful contractor for work will comply and coordinate with local statutory norms / departments during execution and shall obtain all statutory / regulatory bodies approval.
6. The above-mentioned Intent of Work shall be supplemented with scope of work indicated in preliminary Drawings as well as with the stipulations of complete Tender document.

7. All items of work shall be completed in all respect with necessary services and any items not covered in the technical specification but essential for proper execution of work shall be deemed to be included in the scope of the Bidder.
8. The contractor has to prepare all the documents as required and submit directly to the local statutory bodies. The contractor shall at his own cost collect field samples and carry out all necessary tests required for submission of necessary applications. All such applications /approvals shall be in the name of the ALIMCO.
9. The contractor has to comply and, if necessary, resubmit applications which are required by the local bodies.
10. If required the contractor has to appoint at his own cost consultants for obtaining local body approval.

10.0 DESCRIPTION OF UNITS, PLANT AND EQUIPMENT

UNITs: -

The units envisaged in this tender consists of the following units:

A. Ware house for storing Raw Material Shed /Finished Good Shed

(Suggestive Drawing Enclosed)

One warehouse with air ventilation system as per GRIHA norms is envisaged to store raw material and finished goods in Structural building (Pre- Engineered Building). as per drawing.

B. Admin Building

(Suggestive Drawing enclosed)

Two Stories civil building (G+1) with foundation strength for G+ 2 (RCC Structure) as per drawing.

C. Security & Time Office Building

(Suggestive Drawing enclosed)

Two Stories civil building (G+1) with foundation strength for G+ 2 (RCC Structure) as per drawing.

D. LT Panel Room in Admin Building

E: Underground Sump with Pump Room

F. Boundary wall as per drawing.

(Suggestive Drawing Enclosed)

11.0 The following facilities / systems shall be provided as services works: -

- A. Drinking Water and Fire Hydrant System Drinking Water System: shall be provided to cater the drinking water requirement of Buildings / Units of ALIMCO. The system comprises Borewell, pipe work, pumps, RO units and water coolers in the building.
For fire-fighting purpose, all the buildings / shed shall be covered by fire hydrant system. Fire water pipes shall be laid on ring main concept with provision of hydrants.
- B. Ventilation System: Ware house for Raw material / Finished good shed shall be provided with roof mounted natural air driven ventilation system. Substations and other buildings / toilets shall be provided with suitable exhaust fans for ventilations

- C. Arboriculture Work: Plantation & Landscaping shall be provided along the roads and areas surrounding the warehouse / non warehouse buildings and open space as per
- D. Landscape Design. Planting of trees shall be as per Environment norms and native species shall be planted considering evergreen easy growing. Implementation of all the arboriculture shall be as per project plan.
- E. Other Systems: Other systems like electrical power supply system, IT infrastructure, illumination & Area Lighting . Roads, Rain water harvesting and Drainage System etc.

12.0 General Scope of Work

- 12.1 Scope of work for all supplied equipment / item (excluding civil & structural work) shall include following as applicable in addition to specification & scope of work given in the respective chapters:
- 12.2 The successful tenderer shall submit schedule of submission of drawings, documents and load data referred in the specification within 15 (fifteen) days of signing of Contract Agreement.
- 12.3 Six copies of erection schedule & manual, assembly / erection drawing, erection sequence and special precaution to be followed.

13 Site Services:

The Bidder shall also consider and make site services as given following for successful execution of project. Rates for the same deemed to include in given quoted rates.

Obtaining all pre, during & post construction statutory approvals, Clearance from Forest department, environmental (PCB), UPSIDA, directorate of industries etc. as applicable.

Obtaining approvals/ NOCs/ clearances for sanctioning of permanent power connection/ load and providing electricity to the plant from the State Power Corporation, obtaining necessary statutory permissions including liaisoning with the concerned State authorities/ departments, Execution & complete monitoring / supervision at site including verification of the bills of the contractors etc.

Liaisoning with local State Electricity Board/Company, safety office, Regulatory Commission, PCB, Renewable energy department etc for receiving 33/11 kV supply at ALIMCO premises with net metering provision for import/export of energy generated as per latest regulatory board guidelines if required.

Temporary Site office accommodation for project Monitoring agency for the project duration: The office accommodation shall be furnished & adequate furniture like table, chair, almirah, filing cabinets etc., Internet, Laptop / Desktop Computers, printers, all consumable like office Stationery (For Temporary Site office) shall be provided. Water coolers with RO units shall be provided for drinking water requirement.

Temporary Site office accommodation for ALIMCO with furniture, Desktop/Laptop Computer, Printer, Internet. Transport arrangement for movement of Engineer- in -chief and site officials for official work.

TECHNICAL SPECIFICATIONS

1.0 GENERAL ARCHITECTURAL CONSIDERATIONS

1.1 PEB

PEB building should also meet the requirements of various aspects of the room such as electrical, illumination, ventilation, exhaust, thermal, safety, security etc.

Accordingly, this shall include all items such as windows with shades, doors with shades/ lock/ key, rolling shutters with lock/ key, ventilator provisions, exhaust provisions, rain water gutters/ downcomers/ pipes etc. Raw materials for steel sections, sheets etc shall be of reputed make such as Jindal Steel and Power, RINL, SAIL, TISCO, JSW Steel, Essar Steel, Ispat Industries, Lloyd Steel Industries.

Insulated wall cladding or roofing shall consist of double skin metal cladding with Poly Urethane Foam (PUF). PUF must be made of continuous method PU foam and must be CFC free, self-extinguishing, fire retardant type with density $40 \pm 2 \text{ kg/m}^3$ and thermal conductivity $0.019\text{--}0.022 \text{ W/(m.K)}$ at 10°C . The PUF panels shall be a factory-made item ready for installation at site.

PUF panels shall be made of t roughed permanently colour coated metal sheets of steel for roofing and side cladding (internal and external) shall conform to the requirements of Table-1 and IS: 513 for Hot -dip Zinc coated or Al/ Zn coated sheets. The insulation material thickness and details shall be as specified at relevant para in the specification.

PUF insulated panels Metal Sheet for Roofing and side cladding consist of external sheet as troughed permanently colour coated sheet & internal sheet as plain permanently colour coated sheet . Chemical composition of Troughed permanently colour metal sheet for roofing and side cladding shall conform to the provisions of same reference code to which the mechanical properties conform to. Plain permanently colour coated steel metal sheet for ridge and hips, flashing, trimming, closure for vertical and horizontal joints, capping etc. shall conform to the same requirements as those of t roughed permanently colour coated metal sheet for roof and side cladding.

1.2 Roof Insulation & Type:

Both metal sheets shall have an under insulation of minimum 80 mm thick PUF with density $40 \pm 2 \text{ kg/m}^3$ and thermal conductivity $0.019\text{--}0.022 \text{ W/(m.K)}$ at 10°C with gutters and down take pipes along with Flashing & Top cap of required size and colour complete with all necessary hardware complete. Roof shall be projected at -least 300 mm from the wall.

1.3 Wall Insulation:

All voids of external and internal metallised walls shall have an under insulation of minimum 60 mm thick PUF with density $40 \pm 2 \text{ kg/m}^3$ and thermal conductivity $0.019\text{--}0.022 \text{ W/(m.K)}$ at 10°C with proper supports etc. as approved.

Both the walls should be separated by spacers system made up of cold formed steel bars and fastened through zinc / zinc-t in coated self-drilling screws. The external wall of Inverter room / security room facing the transformer area shall be as per IS: 1646 - Code of practice for fire safety of buildings (general): electrical installations.

2.0 FINISHING

The surfaces of brick work, RCC, CC etc. shall be treated and finished with Cement Plaster. The use of PP Cement shall be preferred or OPC with minimum 30% replacement with BIS-recommended waste by weight of cement shall be used. The cement plaster shall be provided as under:

- a) Internal wall faces: 12mm thickness (minimum) cement plaster
- b) External wall faces: 15mm thickness (minimum) cement plaster
- c) Ceiling and RCC works: 6mm thickness (minimum) cement plaster.
- d) The junction of RCC work and brick walls shall be covered with 24-gauge chicken wire mesh fixed with screws/washers to avoid cracks in plaster work.
- e) Water proofing material in cement plaster work shall be mixed in proportion recommended by the manufacturers.
- f) The trenches / open drains: 15mm plaster finished with cement plaster 1:4 (1cement: 4 fine sand) with floating coat of neat cement & adding of Water proof compound.
- g) Provide drip course/ groove in plastered surface or moulding to R.C.C. projections.

3.0 PAINTING

The plastered surfaces shall be finished as per the finishing schedule/tender drawings.

All interior wall and ceiling finishes such as primers, paints, putty, etc. shall have low VOC content and shall be lead free. Painting on doors, windows, Grills, MS work, structural steel, rolling shutters, railing and other members requiring painting and polishing etc., wherever required, shall be treated with primer coat and finished with painting/polishing of approved shade and manufacture, as per CPWD Specifications, to meet the functional requirements.

- a) All paint work on concrete and plaster surfaces shall include application of white cement-based putty as base preparation, application of primer in compatibility with the respective type of paint and painting with 2 or more coats of paint as per technical specifications.
- b) All paint work on structural components (excluding Stainless steel) shall include application of primer in compatibility with the respective type of paint and painting with 2 or more coats of paint as per technical specifications.
- c) The soffits of all projections, jambs, parapet walls terrace (in side) shall be finished with exterior paint as per finishing schedule.
- d) All paints shall be Eco friendly paint with low Volatile Organic Compounds (not exceeding 50 g/L) of approved shade & make.
- e) For painting on steel work and ferrous metals, BS: 5493 and IS: 1477 shall be followed. The type of surface preparation, thickness and type of primer, intermediate and finishing paint shall be according to the painting system adopted.

3.1 EXTERNAL PAINT

- a) External Acrylic Emulsion Paint over weather coat primer on plastered surfaces. All exterior coatings shall have VOC content as per norms.
- b) Granite/Stone Cladding.
- c) Exterior Texture Paint.

4.0 FLOORING

The flooring shall be as per the finishing schedules, tender drawings or as specified.

The edges of steps in the staircases, counters, kitchen platform, window sills, facias and similar location shall be edge moulded as required. Staircase Tread should have Anti- Skid Grooves as specified in the tender drawing or as directed by Engineer-in charge.

The types of flooring shall be as per finishing schedule / tender drawings.

All tiles shall be fixed using cement based high polymer modified quick set tile

adhesive (water based) conforming to IS: 15477, using 5 kg adhesive per sq.m. of tile area with epoxy, in avg. 3mm thickness. The VOC of adhesives shall be less than 65 g/l as per

specification of manufacturer.

The flooring work shall be carried out as per CPWD specification:

Concrete Flooring 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate) finished with a floating coat of neat cement

Polished Kota Stone Flooring – 25 mm thick over 20 mm (average) thick base laid over and jointed with grey cement slurry mixed with pigment to match the shade of the slab including rubbing and polishing complete with base of cement mortar 1 : 4 (1 cement : 4 coarse sand).

Polished Granite Flooring - 18mm thick gang saw cut mirror polished premoulded and repolished) machine cut with joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing, edge moulding and polishing to edge to give high gloss finish.

Vitrified Tile Flooring-Glazed Vitrified tiles Matt/Antiskid finish Vitrified floor tiles in of sizes 600X600,800X800,1200X1200 (thickness to be specified by the manufacturer) with water absorption less than 0.08%and conforming to IS : 15622 , of approved make, in all colours and shades, laid on 20mm thick cement mortar 1:4 (1 cement : 4 coarse sand) jointing with grey cement slurry @ 3.3kg/sqm including grouting the joints with white cement and matching pigments etc., complete.

Tactile-The Tactile Tiles must confirm to IS 13801:1993 and mandatory to place in all building as per norms and design.

Skirting/Dado:

Skirting in respect of above shall be of the same material and specifications and the height as specified.

The dado work in the toilets/washroom/kitchen/pantry or as specified shall be with glazed/digital ceramic wall tiles of size 300mm X 600mm and of height as specified in arch drawing, conforming to IS: 13754.

5.0 DOOR & WINDOWS

The doors and windows shall be provided as per the requirements indicated in the finishing schedules/tender drawings/ and CPWD specifications. The various types of Doors and Windows shall be as under:

5.1 Wooden

Flush doors shutter with Pressed steel frame conforming to IS: 4351: ISI marked 30 mm thick flush door shutters including ISI marked Stainless Steel butt hinges with necessary screws conforming to IS2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters with necessary screws as per CPWD specification. Pressed steel door frames, manufactured from commercial mild steel sheet of 1.60 mm thickness, including hinges, jamb, lock jamb, bead and if required angle threshold, as per CPWD specification. Hardware: All hardware for doors and windows shall be of stainless steel as per direction of engineer in charge.

5.2 MS Works

Doors frame: Powder coated Pressed Steel Door Frame (Mild Steel Sheet 1.6 mm thick) conforming to IS: 4351, manufactured from commercial mild steel sheet of 1.60 mm thickness, including hinges, jamb, lock jamb, bead and if required angle threshold, as per CPWD specification. Windows/Ventilators: Powder coated Pressed Steel Frame (Mild Steel Sheet 1.6 mm thick) double rebate frame with 1 glass shutter and 1 mosquito net (SS wire gauge) shutter and MS grill in between.

Hardware: All hardware for doors and windows shall be Powder coated MS or as specified.

M.S. Grill fabricated from MS flats, square or round bars etc. including priming coat with approved steel primer.

5.3 Powder coated Aluminium door-window

aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass/ stainless steel screws.

5.4 Wood work for door

Sal wood work in frames of doors, windows, clerestory windows and other frames, wrought framed and fixed in position with hold fast lugs or with dash fasteners of required dia & length.

5.5 Fly proof stainless steel grade 304 wire gauge

0.2 mm wire in windows Aluminium

5.6 Water proofing

All items for water proofing treatment for Roof Slab, Underground /overhead Water Tanks shall be guaranteed for TEN YEARS, to be reckoned from the date of expiring of the Defect Liability period prescribed in the contract.

6.0 Soil & waste pipes : CPVC /UPVC Pipe.

6.1 Pipes Internal :

CPVC Composite Pressure Pipes conforming to IS having thermal stability for hot & cold water supply, capable to with stand temperature up to 800 C, including all special fittings of composite material as per CPWD Specification. Soil waste pipes: shall be of 110/160 mm dia with the wall thickness of 3.8 mm SWR pipes of UPVC of finolex / kissan/ supreme or its equivalent with all fitting etc as required.

6.2 Mirror :

Providing and fixing of required size bevelled edge mirror of superior glass (of approved quality) complete with 6 mm thick hard board ground fixed to wooden cleats with C.P. brass screws and washers complete.

6.3 Sanitary fixtures :

Long body pillar cocks as per norms of Physically challenged persons

6.4 Tac tiles:

S S 316 grade/Heavy duty industrial tac tiles.

7.0 TECHNICAL SPECIFICATION :

The technical specification of **CPWD SPECIFICATION 2019, VOLUME I & II** would be applicable and referred.

Samples and makes of each item shall be prepared for approval of ALIMCO before commencement of work.

The contractor shall prepare all the needed shop drawings /sample well in advance and get them approved before placing the order and execution of the item.

8.0 E&M Works

1. All Electrical(E) & Mechanical(M) works shall be as per the guidelines/ specifications mentioned herein, which shall represent the minimum requirement compliant to the related CPWD specifications and relevant Indian standards/ International standards as applicable including installation, testing and commissioning.
2. The requirement of the energy efficiency measures shall be complied as per the provisions of the local body and provisions as per minimum "Green rating" as per Green Rating Manual 2021 and ECBC 2017 [minimum ECBC building] (as applicable for individual service/ item).
3. Equipment's capacity/selection and design shall also be made as per good engineering practices, based on design/site / client requirements, CPWD Specifications, Relevant BIS standards, NBC -2016, Local by-laws, Local Fire authority Rules as modified up to date.
4. For the purpose of the interpretation of the specifications, the order of preference shall be:
 - a. CPWD specifications
 - b. Bureau of Indian Standards (BIS)
 - c. Manufacturer's standard practice/ specifications
 - d. International Standards
5. Provisions contained in Handbook on barrier free built environment and corresponding provision of NBC 2016 shall be made.

Item No.	Description	Brief Specifications
1	Internal Electrical Installations	As per CPWD General Specification for Electrical Works Part I (Internal) – 2023 amended upto date.
1.1	Conduits & Cable Trays	The work will be carried out in recessed/surface MS conduit wiring system with modular socket in accordance of CPWD General Specifications for Electrical Works Part-I (Internal)- 2023 amended upto date; Hot Dip Galvanized GI Cable trays (perforated or ladder type) for cable laying. Data and communication cables shall be laid in the recessed/surface MS conduit or laid in the metallic raceways/ channels with cover and matching junction boxes.
1.2	Wires & Cables	FRLS PVC insulated Copper conductor cables will be used for points, circuit & sub-main wiring; Power & light point wiring shall be with copper conductor of size as per CPWD specifications & relevant BIS standards; RG-6 Cable TV wiring; Cat-6/ 6A Cabling / telephone wiring as required for EPABX / Data networking requirements;
1.3	Switches & Sockets	All switches, sockets, stepped type electronic fan regulators, bell push and accessories along with matching mounting boxes shall be of modular type.
1.4	MCB DBs & LT Panels	Meter Boards & Main Distribution Boards as per CPWD specification / requirement; Size of distribution board shall be as per number of

		light / power circuits. All distribution boards shall be MS powder coated MCB based, Single phase/ 3 phase and suitable in-comer as per individual DB requirement, double door type with RCCB of required capacity and MCCB/MCB of suitable rating. LT panels shall be as per CPWD specifications & IS/IEC 61439 with MCB/MCCB/ACB as required.
1.5	Fittings	Lighting fixtures, LED type, in all spaces, common areas on all floors, staircases, corridors, lobbies, services rooms, basement, terrace etc shall be provided. For basement and wet areas, the degree of protection for the fittings shall be minimum IP 65. The minimum illumination requirement and other parameters shall be as per the NBC 2016 amended upto date. All LED Indoor lighting fixtures should be following below parameters
	Description	Parameters
	CRI	> 80
	THD	<10%
	System efficacy	minimum 100 lm/W at temp 0 to +45deg. C and humidity 10% to 90%
	System life time (Burning Hour with driver)	50000 hours @ L70 as per LM 70 and testing as per LM 80
	Driver voltage	220-240V
	power factor	>0.95
	efficiency	> 85%
	Surge protection	(L-N) 1KV, (E-L/N) 2 KV
	Color Temperature	As per site requirement.
1.6	Fans/ Exhaust fans	Ceiling fans [BLDC] (min 5-star rating as per Bureau of Energy Efficiency) with two module step regulator & exhaust fans/ fresh air fans in all residential units, services rooms, bathrooms/ toilets and in common areas shall be provided as per CPWD specifications and related BIS.
1.7	Earthing	Copper/GI plate earth pits and complete earthing system as required as per CPWD specifications.
1.8	Lightening Arrestor	Lightning Protection System as per IS/IEC 62305 standard & National Building Code (NBC 2016).
1.9	External Lighting & Power distribution	As per CPWD General Specification for Electrical Works Part II (External) – 1994 and relevant BIS, amended upto date.
1.10	Cables	XLPE armoured cable copper/ aluminium between outdoor feeder pillar to termination box of pole etc. and wiring of suitable size between termination box to fitting i/c their termination. The power cabling should be sized to ensure that the distribution losses do not exceed 3% of total power usage.
1.11	Feeder Pillars	Outdoor type Feeder Pillars shall be suitable for 3 phase, 50Hz, 415 volts, A.C. system. Rating and size of Feeder pillar shall be as per designed load and requirement; The feeder pillars shall be as per CPWD specifications with MCB/MCCB/ACB as required. To be provided with astronomical timer/ photo sensor based switching for outdoor and street lighting, to save energy.
1.12	Fittings	LED outdoor light fixtures (IP-66) of suitable lumen output / wattage for compound lighting; street lighting, Security light fittings at perimeter on boundary wall as required. The minimum illumination requirement shall be as per the NBC 2016 amended upto date. All LED outdoor lighting fixtures should be following Parameters
	Description	Parameters
	CRI	> 80

	THD	<10%
	System efficacy	minimum 120 lm/W at temp 0 to +45 deg. C and humidity 10% to 90%
	System life time (Burning Hour with driver)	50000 hours @ L70 as per LM 70 and testing as per LM 80
	Driver voltage	220-240V
	power factor	>0.9
	efficiency	> 85%
	Surge protection	(L-N) 1KV, (E-L/N) 2 KV
	Color Temperature	As per site requirement
1.13	Street Light Poles	Octagonal GI/Decorative poles/Bollard with sliding door, single / double bracketed arm along the road as per IS standards. Ornamental/ Decorative poles and bollards with LED fittings may also be provided as per the site/client requirements.
1.14	Earthing	Copper/GI plate earth pits and complete earthing system as required as per CPWD specifications.
2	Fire Fighting and Down Comer System	Fire Protection System including down comer, Yard-hydrants, internal-hydrants, first aid hose reels, Down comer pumps, electrical and diesel engine driven pumps etc. designed and provided as per provision given in NBC 2016, relevant BIS Codes, local Fire Bye Laws & CPWD Specification Part V-2020, all with up-to-date amendments; Pipe materials and other fixtures for Fire Fighting, plumbing work shall be as per CPWD specifications.
3	Fire Alarm System	As per CPWD's General specification for Electrical Works, Part - VI Fire detection and Alarm system-2018, NBC 2016, local Fire Bye Laws, IE Rules, BIS/IEC, Indian Standards amended upto date; covering Manual/ Automatic Fire Alarm System complete with items like detectors, manual call boxes, cabling/ wiring, fire panels, etc. including Integration of FAS panel with Smoke Ventilation & pressurization fans, AHUs, Lifts, sprinkler monitoring panel, water curtain system, water levels of Fire water tanks and as required for any other service.
3.1	Cables	Either copper conductor Armoured Fire survival cables of suitable size or copper wire in heavy duty MS conduit of suitable size shall be used compliant to the relevant IS.
3.2	PA System	Public Address System with Audio Amplifiers, speakers & required wiring shall be provided to cover all the areas as per the requirement of CPWD specifications, NBC 2016 and local bye laws.
4	Lifts	The lifts shall be with power operated centre opening sliding doors and AC variable voltage & variable frequency controls, Car safety & Governor, Guides, guide rails, ropes, car platform, cabin finishes (stainless steel), car & landing doors (stainless steel), hoist motors, braking system, Automatic Rescue Device (ARD) etc complete as per the CPWD Specifications and related IS standards upto date.(Lifts can be with Machine room or Machine room less depending upon site requirement)Provision of lifts will be made as per National Building Code- 2016 / relevant BIS amended upto date) considering no of floors, height of the building and speed of elevators; At least one lift in each block be made compliant to barrier free and accessibility requirements as per CPWD specifications & capacity to be minimum 16 passenger lift.
5	Diesel Generating Set	As per CPWD General Specification for Electrical Works Part VIIDG Sets – 2013, relevant code, CPCB norms and local body norms covering DG sets, AMF panel, bus ducting/ cables from DG sets to essential panel, DG set enclosure room sound insulation/ ventilation/

		smoke exhaust as required, earthing of DG set system, control cabling, fuel tank/ piping, DG set exhaust piping/ exhaust chimney as per CPCB norms, civil works connected with DG sets including foundation as required. DG sets shall be silent type, minimum rating/ capacity specified is nominal capacity at standard test conditions; The DG set capacity shall be suitable for the load covering common area lighting, outdoor lighting, lifts, water supply pumps, sewerage and drainage pumps, fire fighting essentials and other essential power supply requirements as per site requirements.
5.1	Engine	The engine shall be capable of driving the alternator continuously at its rated full load and rated speed without getting over loaded under the prevailing operating Conditions.
5.2	Alternator	Generator Sets shall be 415V, 3 Phase, 4 wire, 50Hz, 0.8 PF with acoustic enclosure
5.3	AMF Panel	Generator sets shall start automatically in the event of a power failure and shall transfer power to emergency / critical loads automatically and shall switch off after some time delay in case of restoration of power supply. Synchronization arrangement to be provided in case of parallel operations, where more than 1 DG set is installed and such arrangement is required as per site requirement.
5.4	Exhaust Pipe	As per CPWD General Specification for Electrical Works Part VII DG Sets – 2013, relevant codes and CPCB/ local body norms
5.5	Cables	Complete Electrical and control wiring for various accessories etc.
5.6	Earthing	Earthing sets as per CPWD Specs and relevant BIS
5.7	Fuel tank	As per OEM standard/ site requirement.
6	Solar PV Power Generation System	NA
7	IP based CCTV system	As per good Engineering practice or as required under the codes or by local/statutory authorities compliant to relevant Indian standard 13252; IP based CCTV system for building security comprising of PTZ/ fixed camera (Bullet and Dome), cabling, digital recording, HD display system with minimum display of 5" x 8" per camera and hardware & software support both for indoor and external surveillance IP based Closed Circuit Surveillance system is proposed to monitor the activities at critical/ desired areas. The video shall be continuously transported to existing server, using IP cameras; The items supplied by the contractor should have specifications compatible with each other required for satisfactory operation of the system. CCTV cameras and storage server/ NVR shall have ONVIF Profile-S compliance for interoperability.
8	IP BASED EPABX System	IP based EPABX System comprising of core switches & L2 switches with 10 G, 10 giga SFP modules, industry standard appliance server, cloud-based, enterprise-grade UC solution, MID/Entry level IP/SIP phone with, dual 1 gig ports, racks, CAT- 6/ 6A cable, patch panels, OFC etc.
9	Electrical Sub-station	As per CPWD General Specification for Electrical Works Part IV Substation – 2013 and relevant BIS amended upto date covering 33kV/0.433 kV or 11 kV/0.433 kV substation equipment comprising HT panel, dry type/ Oil type transformers, (Indoor/ outdoor type, as required), HT cables, bus trunking / suitable cabling from transformer to LT panel, LT panels (as per IS/IEC 61439), automatic power factor correction panel, active harmonic filters, TVSS (transient voltage suppression system), SPD (Surge protection system), essential panel, earthing, required inter- connections, substation safety equipment including LT cabling from substation to the buildings fed by the substation.

10	HT Panel	33KV or 11KV HT panel board; With required nos. of incomer (preferable to have atleast 2 incomers to have standby), bus coupler/s and required no of outgoing HT outgoing with VCBs, metering (in all panels and suitable for net metering) and protection equipment/ accessories and compatible for SCADA/ BMS as required.
11	Transformer	Dry type/ oil type Transformer, complete with online tap changer, and other accessories and safeties, cable end box etc as per IS and having maximum losses as per table of ECBC 2017 code (as per ECBC building or better) with amendments, if any. The entire work shall comply with the latest CPWD specifications for substation and electrical works.
12	LT Panel	Main LT panel (normal supply) for receiving supply from Transformers / Point of supply and having ACB's/ MCCBs as outgoing; Main LT panel (Essential supply) for receiving supply from Transformers and DG sets / Point of supply and ACB's/MCCBs as outgoing; Main Sub LT Panel for supplying power to smaller loads having MCCB's as outgoing; The incomers rating of LT breakers of Main Panel shall match with transformer rating or next available standard ratings of ACB with requisite breaking capacity of for all the incoming and outgoing switchgears as per applicable fault level.
13	APFC Panels	APFC panel shall be provided on separate sections of main LT panel and having switching off arrangement in case of DG supply in the main panel; Designed capacity to maintain overall P.F. as applicable for ECBC building category or better (presently 0.97 or better)
14	Bus trunking	All Bus trunking and rising mains shall be compact air insulated/ Sandwich type, of required breaking capacity, complete with joints, fire barriers, tap off boxes, mounting arrangement etc and of required degree of protection. Shall be used for the connections between the Transformer and main LT panel, LT panel to the rising mains, or as per specific site requirements as per directions of Engineer-in-Charge.
15	Cables	HT Panel and transformers will be connected through three core 11/33 KV HT XLPE(E) of requisite capacity based on the current carrying and breaking (fault level) capacity; HT cable end terminations shall be Cold shrink/ Heat shrinkable type; These cables shall have individually screened cores and be manufactured and tested according to IS: 7098 (Part II) - 1973 amended up to date; The power cabling should be sized to ensure that the distribution losses do not exceed 3% of total power usage.
16	Earthing	Earthing sets, connections etc. shall be as per CPWD Specifications and relevant BIS.
17	VRV/VRF AC System	Variable Refrigerant Flow (VRF) shall meet or exceed the efficiency requirements specified as per minimum "ECBC Building" as per ECBC 2017 and the work shall be executed as per CPWD General Specifications for Air Conditioning Works -2017, relevant IE rules, relevant IS; It should consist or suitable nos. or outdoor and indoor units refrigeration pipe and drain pipe, all necessary civil work for installation including 1st charging of refrigerant gas for proper and specified functioning of the VRV system, AHU, Fan coil unit, cassette, split, ductable split etc., Plumbing works, ducting, volume dampers, Thermal insulation for pipes and ducting, control, power cables, electrical panels complete as required.
18	Illuminated signages,	Exit signs shall not exceed 5 W per phase as per ECBC- 2017; Self-contained rechargeable emergency exit light with minimum 6hours

	Emergency lighting & Exit signages	battery backup; IP 20 rated; Confirms to IS : 10322 (part- 5 /sec1) : 2012, Single side / Double side facia as per requirement of location installed.
19	Street lighting & Compound Lighting	The scope of work includes design, supply, installation, testing and commissioning of street lighting, compound lighting, landscape lighting, parking lighting with LED type street lights / LED type post top lanterns , suitable size incoming and outgoing PVC/XLPE insulated PVC sheathed aluminium conductor armoured cables of appropriate rating including laying and connection, Octagonal hot dipped galvanized / Ornamental street / compound light poles with inbuilt control box, GI brackets, Base plates, Earth stud, GI studs, GI Nut Bolts, required control panel with astronomical switches, contactors, Control MCCB and MCBs, Wiring with PVC insulated FRLS copper conductor cable from terminal box to luminaires earthing etc. as required as per specifications. The execution shall be done in IIMC Campus separately as per functional requirement & specifications given in Part C and Codes.

9.0 Quality Assurance Manual (Quality Assurance Plan & Checklist for E&M Services)

1. Main contractor/ Associate Agency shall submit the required quantity of material as sample for testing from Govt./Approved Private Laboratory.
2. The decision on testing shall be as per applicable and as per the following: -
 - i) Quality Assurance Manual for Building Works 2022, Handbook
 - ii) for Total Quality Management - 2019, Quality Assurance Policy
 - iii) & Check List of E & M Services - 2016,
3. Contractor shall submit the required size and quantity of samples for the testing.
4. Department shall send the samples to the testing laboratory & the test results shall directly come to department however; the cost of all testing shall be borne by the Contractor.

LIST OF MANDATORY TESTS

Material	Test	Field/ laboratory test	Test procedure	Min. quantity of material for carrying out the test	Frequency of testing
Water	i) PH Value (ii) Limits of Acidity (iii) Limits of Alkalinity (iv) Percentage of solids (a)Chlorides (b)Suspended matter (c)Sulphates (d)Inorganic solids (e)Organic	Lab	IS 3025	-	Water from each source shall be got tested before the commencemen t of work and thereafter once in every three months till the completion of the work. Water from municipal source need be tested only once in six months. Number of Tests for each source .
Cement	(a) Physical requirement (i) Fineness (ii) Soundness (iii) Setting time (Initial & Final) (iv)Compressive strength (v)Consistenc y of standard	Lab Lab Lab Lab Lab	IS 4031 (Part II) IS 4031 (Part III) IS 4031 (Part V) IS 4031 (Part VI) IS 4031 (Part VI)	Each lot	Every 50 tonnes or part thereof. Each brand of cement brought to site shall be tested as per this frequency.
Sand	Organic impurities	Field	As per CPWD Latest Specification s	20 cum	Every 20 cum or part thereof or more frequently as decided by Engineer-in- Charge.
	Silt Content	Field		20 cum	-do-
	Particle size distribution	Field or Laboratory as decided by the Engineer- in-charge		40 cum	40 cum or part thereof

	Bulking of Sand	Field		20 cum	Every 20 cum or part thereof or more frequently as decided by Engineer-in-Charge.
Fly Ash	Total chloride in percent by mass, max.	Lab	IS 12423	10 cum	Every 10 cum or part thereof or more
	Loss of ignition in percent by mass, max.	Lab	IS 1727	10 cum	Frequency as decided by Engineer-in-charge
	Fineness, specific surface in m ² /kg	Lab/field	Blaine's permeability method	10 cum	-do-
	Compressive strength at 28days in N/mm ²	Lab	-	10 cum	Only in cases when fly ash is used as pozzolana in cement
Stone aggregate	Percentage of soft or deleterious material	Field or Laboratory Test as required	IS 2386- Part II	As required by Engineer-in-charge	For all quantities
	Particle size	Field/ Lab	As per CPWD Latest Specifications	45 cum	For every 45 cum or part thereof for RCC Work only. For rest of items as decided by Engineer-in-charge
	a) Estimation of organic impurities	Field/ Lab	IS 2386- Part II	10 cum	For every 40 cum or part thereof
	b) Surface moisture	Field/ Lab	IS 2386	10 cum	-do-
	(c) Determination of 10% fine value	Field/ Lab	IS 2386	10 cum	-do-
	d) Specific gravity	Field/ Lab	IS 2386	10 cum	-do-
	(e) Bulk density	Field/ Lab	IS 2386	10 cum	-do-

	f) Aggregate crushing	Field/ Lab	IS 2386	10 cum	-do-
	g) Aggregate impact value	Field/ Lab	IS 2386	10 cum	-do-
Concrete	Slump test	Field	As per CPWD Latest Specifications	10 cum	15 cum or part thereof
cement concrete Pavers/ Tiles	Water Absorption, Compressive strength	Laboratory	IS 15658		
Reinforced cement concrete (Nominal Mix)	(a) Slump test	Field/ Lab	As per CPWD Latest Specifications	i) 5 cum in case of column (ii) 20 cum for slabs, beams and connected columns (iii) 20 cum for other R.C.C. work for all other small items and where R.C.C. done in a day is less than 5 cum test may be carried out as required by Engineer-in-Charge	(i) Every 5 cum of part thereof (ii) Every 20 cum or part thereof (iii) -Do-
	(b) Cube Test	Lab	As per CPWD Latest Specifications	(i) 5 cum in case of column (ii) 20 cum for slabs, beams and connected columns (iii) 20 cum for other R.C.C. work for all other small items and where R.C.C. done in a day is	(i) Every 5 cum of part thereof (ii) Every 20 cum or part thereof (iii) -Do-

					less than 5 cum test may be carried out as required by Engineer-in-Charge	
Reinforced Cement Concrete (Design Mix)	Coarse Aggregates				50 cum or part thereof & also on each change of source	
	Fine Aggregates				50 cum or part thereof & also on each change of source	
	Cement				50 MT or on each change of source	
	Fresh Concrete	(a) Slump test	Field	As per CPWD Latest Specifications	10 cum	50 cum for R.C.C. work including in all other small location. R.C.C. done in a day is less than 50 cum test may be carried out as required by Engineer-in-Charge
		(b) Cube Test	Lab	As per CPWD Latest Specifications	10 cum or part thereof	50 cum or 10 batches of 5-7 cum each for R.C.C. work in all location taken together. R.C.C. done in a day is less than 50 cum test may be carried out as required by Engineer-in-Charge
Reinforced Steel for cement concrete	Nominal Mass, Elongation, Tensile Strength, Rebend test		Lab	As per CPWD Latest Specifications	Per Batch	(a) For consignment below 100 tonnes (i) under

					10 mm dia, one sample for each 25 tonnes or part thereof (ii) 10 mm to 16 mm dia one sample for each 35 tonnes or part thereof (iii) over 16 mm dia one sample for each 45 tonnes or part thereof
Bricks/ Brick Tiles/ Blocks	Testing of Bricks/Brick Tiles for dimensions, Compressive strength, Water absorption and efflorescence	Laboratory	As per CPWD Latest Specifications	As per CPWD Latest Specifications	
Granite	(i) Moisture	Laboratory	IS 1124	50 Sqm	100 Sqm or part thereof
	(ii) Specific Gravity	Laboratory	IS 1124	50 Sqm	100 Sqm or part thereof
Timber	Moisture content	Field (by moisture meter) laboratory test as required by Engineer-in- Charge	Field (by moisture meter) laboratory test as required by Engineer-in- Charge	1 cum	Every one cum or part thereof.
Flush door	End immersion Test knife test Adhesion Test	Laboratory	IS 2202	26 shutters	As per sampling and testing specified in clause 9.7.11 of CPWD Specifications.
Steel if arranged by the contractor	(a) Tensile strength (b) Bend test	Laboratory	IS 1599	20 tonne	Every 20 tonne or part thereof.
Structural Steel	(a) Tensile Test (b) Bend Test (c) Flattening Test	Laboratory	IS 1608 IS 2329 IS 2328	Every 8 tonne or part thereof	Every 8 tonne or part Thereof
Vitrified tiles/	1. Dimensions	Laboratory	IS: 13630	3000 Nos.	3000 Nos. or

ceramic tiles (for floor & wall)	and surface quality 2. Physical properties 3. Chemical				part thereof
Road works					
Coarse Aggregate	1. Lost Angles	Laboratory	IS 2386 (Part IV)		Per 200 m3
	2. Abrasion Value or Aggregate Impact Value		IS 2386 (Part V) IS 5640		
	3. Flakiness Index		IS 2386 (Part I)		
	Grading requirement	Field	IS 2386 (Part I)	Nil	Per 100 m3
Fine Aggregate	Deleterious materials	Laboratory	IS 2386 (Part III)	Nil	As required by the Engineer-in-charge
Bitumen	As prescribed in IS 73	Laboratory	As prescribed in IS 73	Nil	As required by the Engineer-in-charge
Embankment under (O.M.C. conditions)	1. Moisture content 2. Density	Laboratory / Field	IS 2720 (Part II) IS 2720 (Part XXVIII)	Nil Nil	250 m3 250 m3
	3. Control test on borrow pits.	Laboratory			
	(a) Gradation	Laboratory	IS 2720 (Part IV)	Nil	One to Two tests per 8000 m3
	(b) Plasticity	Laboratory	IS 2720 (Part V)	Nil	-Do-
	(c) Proctor Test	Laboratory	IS 2720 (Part VIII)	Nil	-Do-
	(d) Deleterious content	Laboratory		Nil	-Do-
	(e) Moisture contents	Laboratory	IS 2720 (Part II)	Nil	250 m3
Dense Bitumen Macadam	1. Quality of Binder	Laboratory	IS 73	Nil	As required
	2. Los Angles Abrasion Value/ Aggregate impact Value	Laboratory	IS 2386 (Part IV)	Nil	100 m3

	3. Stripping Value	Laboratory	IS 6241	Nil	-Do-
	4. Water Absorption	-Do-	IS 2386 (Part III)	-Do-	-Do-
	5. Flakiness Index	-Do-	IS 2386	-Do-	One test for 100 cum.
	6. Sieve Analysis for filler	-Do-	IS 2386	-Do-	One test for each consignment subject to minimum one test per 5 cum.
	7. Mix Grading	-Do-	IS 2386	-Do-	One set of test on Individual constituent and mixed aggregates from dryer for each 100 tonnes of mix subject to a minimum of two sets per plant per day
	8. Stability of Mix	-Do-	ASTM D 1559	-Do-	Three marshall specimen per 100 tonnes of mix, subject to a minimum of 2 sets being tested per plant, per day.
	9. Binder Content	Field		-Do-	One test for each 100 tonnes of mix subject to minimum of two tests per day per plant
	10. Thickness , Density of compacted layer	Field	As per latest CPWD Specifications	-Do-	One test of 3 samples per 500 sqm.
Cement concrete pavement under controlled conditions	Coarse aggregate 1. Flakiness Index	Laboratory	IS 2386 (Part I)	-	Before approval of the quarry and every subsequent change in the

					source of supply and one test per 100 cum.
	2. Impact Value	-Do-	IS 2386 (Part IV)	-	-Do-
	3. Loss Angles abrasion Value	-Do-	-Do-	-	-Do-
	4. Deleterious material	-Do-	IS 2386 (Part II)	-	Before approval of the quarry and at every subsequent change in the source of supply
	5. Moisture content	-Do-	IS 2386 (Part III)	-	Regularly as required subject to a minimum of one test per day
	Fine Aggregate				
	1. Silt content	Field	As per CPWD specifications	-	One test per 15 cum.
	2. Gradation of sand	-Do-	IS 2386 (Part II)	-	-Do-
	3. Deleterious material	-Do-	IS 2386 (Part II)	-	Before approval of the quarry and at every subsequent change in the source of supply
	4. Moisture content	-Do-	IS 2386 (Part III)	-	Regularly as required subject to a minimum of two tests per day
	5. Mix Aggregate	Field	IS 2386 (Part I)		One test per 15 cum of concrete
	6. Flexural strength	Laboratory	IS 526	-	One test consisting of 8 specimens for 30 cum. of

					concrete
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Note:

1. The procedure to be followed for testing shall be as per CPWD specifications/IS in order of preference.
2. Contractor may specifically note that the tests to be conducted is not limited to the mandatory tests above and additional tests if required and directed by the engineers on any materials including the ISI / approved Makes shall be carried out by the contractor and cost of additional tests , if any , needs to be borne by the Contractor.
3. Contractor may specifically note that the testing shall be done in Govt. / NABL accredited / Govt. Approved laboratory only.
4. Necessary instruments/equipment required to conduct field test may be arranged by the contractor at the site office at his own cost.

LIST OF PREFERRED MAKES

NOTE:

- a. For all the materials, the Contractor should produce samples and obtain approval of Engineer-In-Charge before incorporation in to the work.
- b. During the project execution stage, if the Engineer-In-Charge desires to change anyapproved make, the same shall be binding by the Contractor.

1. CIVIL WORKS

1.	Cement	Malabar, Ultra Tech, Zuari, Ramco, ACC, India Cements, Dalmia, Ambuja, Nagarjuna, J.P. Rewa, Vikram, Shri Cement, Birla Jute and Cement Corporation of India, Chettinadu, JSW Cement etc. or equivalent as approved
2.	Steel (TMT)	Tata, SAIL, TISCO, IISCO, RINL, Jindal Steel and Power Ltd, JSW Steel Ltd or equivalent as approved
3.	Structural Steel	Tata, SAIL, Jindal Steel and Power Ltd, or equivalent as approved
4.	Welding rode	ESAB, Advani, Best Arc, Solar or equivalent as approved
5.	Teak Wood	Best Quality Plantation teak
6.	Hard Wood	Best Quality treated jack, Sal Wood, Irul, Anjili, Thambakom, Cherutheku
7.	Vitrified tile	Euro Ceramic Ltd, Kerroges, Nitco, Kajaria, RAK Ceramics India, Mirage Ceramics Pvt Ltd, Orient Bell, Swastik Ceracon Ltd or equivalent as approved
8.	Ceramic Tiles	Kajaria, Regency, Nitco, Rak, Mirage ceramics pvt.ltd,orient bell, swastik ceracon Ltd or equivalent as approved
9.	Cement Concrete floor tiles	Eurocon Tiles, Excello or equivalent as approved
10.	Paver tile	Kajaria Ceramics Ltd, Pavit Ceramic pvt. Ltd orequivalent as approved
11.	Tile Breaker / Transition profile/Expansion Joint cover	Euro- Arch systems and solutions pvt. Ltd, Straton orequivalent as approved
12.	Tile Protection Sheet	Floor guard, Plystudio, Supreme or equivalent as approved
13.	Tile Adhesive / Epoxy Grout	Ardex Endura (India) Pvt. Ltd, Myk Laticrete, Kerakoll India Pvt. Ltd, Euro build construction chemicals andcoating, Bostic, MAPEI, BASF, Pidilite, Sunanda Speciality coatings, Kunal Conchem Private Ltd or equivalent as approved
14.	Self levelling compound/Bonding primer	Ardex Endura (India) Pvt. Ltd, Myk Laticrete, Piidilite Industries Limited, Kerakoll India Pvt. Ltd, Fosroc Chemical India, Sika India, BASF, Eurobuild Construction chemicals and coatings, Bostik, SunandaSpeciality coatings, Kunal Conchem Private Ltd or equivalent as approved.
15.	Vinyl Flooring	Responsive Industries LTD, Gerflor, Armstrong, Polyflor India Pvt Ltd or equivalent as approved

16.	Industrial Flooring	Euro Build, BASF, Fosroc, Sika, Pidilite or equivalent as approved
17.	Concrete Paver blocks	Basant Beton, Conwood, Automatic, Amcon, Sirex or equivalent as approved
18.	Metallic Hardener	BASF, Fosroc, Sika, Kironite, Eurobuild Construction Chemicals and Coating or equivalent as approved
19.	Integrated water proofing	India water proofing, Sika, pidilite, Structural waterproofing co pvt.ltd, BASF, MYK, Euro Build, Bostik, MAPEI, CICO Technologies Ltd, Waterproofing co. PvtLtd or equivalent as approved
20.	Water proofing compound	Roffe, Fosroc, sika, waterproofing Co. Pvt. Ltd MAPEI, CICO Technologies Ltd, Sunanda Speciality Coating Pvt Ltd, Kunal Conchem Private Ltd, the Structural Waterproofing co Pvt ltd / CHRYSO, MYK, Schomburg or equivalent as approved
21.	Water proofing membranes	Fosroc / Sika / BASF or equivalent as approved
22.	Paint & Primer, Distemper	ICI (Akzonobel), Berger, Asian, Jotun or equivalent as Approved
23.	Water proof Cement paint	Super Snowcem, Supercem or equivalent as approved
24.	Sealers	ICI (Akzonobel), Berger, Asian, Euro Build or equivalent as approved
25.	Wall Putty	Birla white, NCL, Altek, Berger, ICI or equivalent as Approved
26.	Primer	Altek, Berger, Asian, ICI, Jotun, or equivalent as approved
27.	Melamine Polish	Duco, MRF, ICI or equivalent
28.	Synthetic Enamel Paints and primer	ICI (Akzonobel), Asian Paints, Berger, Jotun or equivalent as approved
29.	Synthetic textured paint	Spectrum, Renovo or equivalent as approved
30.	Fire Retardant Paint	Akzonobel, Promat or equivalent as approved
31.	Anti-Carbonation Coating System	Euro Build, BAST, Fosroc, Sika or equivalent as approved
32.	Premium Textured Exterior Paint	Terraco India Ltd, Spectrum, Jotun or equivalent as approved
33.	Interior Textured Paint	Oikos, Spectrum, Caprol, Jotun or equivalent as approved
34.	PVC Water Stopper	Fosroc, Sika, Euro Build, BASF or equivalent as approved
35.	Door Hardware	Dorma, Ozone Overseas Pvt Ltd, Dorset kaba Security Systems Pvt Ltd, ASSA ABLOY india Pvt Ltd, Hafele, Godrej, Everite Agencies, Dyna, Door King, Hettich India Pvt Ltd or equivalent as approved
36.	Mortice locks, locks, latch	Godrej, Dorset kaba Security systems, Magnum, Ozone Overseas Pvt Ltd, Dorma or equivalent as approved

37.	Auto Closer Hinges	Blum, Hafele, Ozone overseas Pvt Ltd or equivalent as approved
38.	Draw sliders	Blum, Mepla, Grass, Ozone overseas Pvt Ltd, Hafele, Hettich India Pvt Ltd or equivalent as approved
39.	Patch Fittings	Dorma, Savex, Niki, Ozone, D-line or equivalent as approved
40.	Cabinet Handles	Kich, CHH 84 S or equivalent as approved
41.	Rolling Shutter	Popular, Gandhi Automation Pvt Ltd, Falcon, Jacob Engineering or equivalent as approved
42.	Ball Bearings	SKF, FAG, KOYO or equivalent as approved
43.	Steel doors / windows	Shakthi hormann, Indigatech, Ozone Overseas Pvt Ltd, NCL Alltek & Seccolor Ltd, Kuty Flush Doors & Furniture Co. Pvt Ltd, Madhu Industries or equivalent as approved
44.	Glazed Doors with patch fittings	Dorma, Ozone overseas Pvt Ltd, India Private Ltd, ASSA ABLOY india Pvt Ltd, Dorset Kaba Security Systems Pvt Ltd or equivalent as approved
45.	Fire rated Doors	Shakti Horman, Indigatech, Ozone overseas Pvt Ltd, Kuty Flush Doors & Furniture co. Pvt Ltd, Promat International (Asia Pacific) Ltd or equivalent as approved
46.	Flush Doors	Godrej & Boyce Mfg. Co. Ltd, Kuty Flush Doors & Furniture co. Pvt Ltd, Jacsons , Godrej or equivalent as approved
47.	SS Hand Rails	D Line, Sky Port, Q-railing India Pvt Ltd, Impact metals, Cochin tech or equivalent as approved
48.	Galvalume Roofing sheet	Tata Blue Scope, Interarch, LLOYD insulations or equivalent as approved
49.	Pre-engineered Building system	Kirby, Tata Blue Scope, Lloyd Insulations (India) Ltd, Octomec, Apex Buildsys Ltd, Alfa PEB or equivalent as approved
50.	High pressure compact laminate	Greenlam, Merino, FunderMAX India Pvt Ltd, Century, Yemang raised flooring, Alfa ICA India Ltd or equivalent as approved
51.	Laminates	Kitply, Formica, Greenlam, National, Century, Decolam, Merino or equivalent as approved
52.	Plain Float glass/ lacquered glass & mirror	Saint Gobain, Asahi, Pilkington, Modiguard or equivalent as approved
53.	Marine Plywood	Century, Kitply, Anchor, Green ply, Apple Ply or equivalent as approved
54.	Pre-Laminated Ply	Greenlam, Merino Industries Ltd, Kitply or equivalent as approved
55.	Frosted Film	Birla 3M Lumar, Garware, Ultra Mark or equivalent as approved
56.	Veneers	Century, Kitply, Anchor, Green ply, Kenwood, Jacsons or equivalent as approved
57.	Particle board	Jacsons (Exterior grade), Associate or equivalent as approved
58.	Pre-laminated MDF Boards	Scheit / Span / Euro / Century / Kitply / Green ply

59.	Aluminium composite panel	Durabuild, Alcomex (Dongshine), Alucobond, Alubond, Eurobond, Alstrong, Aludecor, Alstone International or equivalent as approved
60.	Sun control film	Garware, 3M or equivalent as approved
61.	Secondary Reinforcement	Recron 3S, Euro Build, Kamachi Sponge & Power Corporation Ltd or equivalent as approved
62.	Silicon Sealant	Dow corning, GE Silicon, Euro Build, CIPY Polyurethanes Pvt Ltd, MYK Schomburg or equivalent as Approved
63.	Epoxy chemical for anchoring grout	HILTI India Pvt Ltd, Fisher, Euro Build, Pidilite or equivalent as approved
64.	Chemical / Mechanical Anchor Fasteners	HILTI India Pvt Ltd, Fisher, MKT (Germany), Black & Decker India Ltd, MAPEI, SS Fasteners pvt.ltd or equivalent as approved
65.	Plasticisers, non-shrink grout	Fosroc, BASF, Krishna conchem, Fairmate, Eurobuild Construction Chemicals & coating, MYK Schomburg or equivalent as approved
66.	Admixtures	Fosroc, Polygon, STP, BASF, CERACHEM, Don Chemicals, Sika, Euro Build, MAPEI, BOSTIK, MYK Schomburg, Sunanda Speciality coatings Ltd, Kunal Conchem Private Ltd, The Structural Waterproofing co.Pvt Ltd. / CHRYSO or equivalent as approved
67.	Glazed Doors / Automatic Bi-Parting Sliding Door	Dorma GMBH & Co. KG, Kaba Gilgen SA, Geze GmbH L.L.C or equivalent as approved
68.	Composite Doors	Suraksha
69.	High Performance Glass / Clear Glass/Reflective glass	Saint Gobain, ASAHI India Glass Ltd, Emirates Glass L.L.C or equivalent as approved
70.	Insulation	UP TWIGA, Lloyd, Owens Corning, Paramount, Rockwool (India) Ltd Or equivalent as approved
71.	EPDM Gaskets	Osaka Rubber Private Limited, Ameer Rubber Industries Pvt. Ltd or equivalent as approved
72.	Glass Processing	AIS Glass Solutions Ltd, Gold Plus group, GSC Toughened Glass Pvt Ltd, Sejal Architectural Glass Limited, Impact Safety Solution Ltd, TPRS Enterprise Pvt.Ltd, FUSO Glass India Pvt Ltd, or equivalent as approved
73.	Perforated / Non Perforated Metal Ceiling	Hunter Douglas India Pvt Ltd, Saint Gobain, Gyproc India Ltd, Anutone, Armstrong, Fameline or equivalent as approved
74.	Gypsum Board False ceiling	Gyproc of Saint gobain / USG Boral / Approved equivalent
75.	Metallic Ceiling	Hunter Douglas India Pvt Ltd, Armstrong World Industries (India) Pvt. Ltd, Linder Middle east L.L.C., Chicago Metallic Corporation or equivalent as approved

76.	Aluminium Curtain Glazing / Structural glazingFabricators	Sherin Hifab Contracts India pvt Ltd, Sherbrooke Aluminium Products, Creative Building Solutions Pvt Ltd,GEFAB FAÇADE SOLUTION private solution or equivalent as approved
77.	Testing of Facade	Façade India Testing
78.	Hermetically sealed double glazing	Saintgobain, Asahi or equivalent as approved
79.	Aluminium Structural Members	Hindalco Industries Ltd, Jindal Aluminium Ltd, IndalcoAlloys or equivalent as approved
80.	Fibre Glass Wool	UP Twiga, KIMCO, Rock wool or equivalent as approved
81.	GI Section for Partitions & False Ceiling	Saint Gobain Gyproc or equivalent as approved
82.	Glass Partitions System	Intact, Deko, Dorma or equivalent as approved
83.	Aluminium Louvers	Hunterdouglas, Euro Building or equivalent as approved
84.	Stainless steel	Saleem Steel, Hindalco, Jindal or equivalent as approved
85.	Expansion Joint Filler	Sil Fex of Supreme Industries Ltd. or equivalent asapproved
86.	Floor hardeners	Fosroc, Roffe, Fairmate, STP Ltd, Apurva India Ltd, Eurobuild Construction Chemicals & Coating, Pidilite or equivalent as approved
87.	Bolts / Screws (SS 316)	HILTI, Fischer, MKT (Germany), S.S. Fasteners Pvt Ltdor equivalent as approved
88.	Fixtures for structural glazing	Dorma, Giesse, Cottswold or equivalent as approved
89.	Glass Hardware	Dorma , Ozone, Hafele, Kinlong or equivalent as approved
90.	Sanitary Fixtures	Duravit, Kohler, Queo, Artize, Jaquar, Toto, Hindware,Cere, Roca, Kerovit or equivalent as approved
91.	Plumbing Fittings	Kohler, Jaquar, Artize, Grohe or equivalent as approved
92.	Concealed Flush	Geberit, Grohe, Artize, Jaquar, Kohler, Queo or equivalent as approved
93.	Toilet accessories	Bobrick, Euronics, Simex, Jaquar, Best practice wash Room Pvt Ltd or equivalent as approved
94.	Fixtures for specially abled	Hindware, Toto, Cera, Roca or equivalent as approved
95.	Eye wash & shower	Simex, Euronics or equivalent as approved
96.	Stainless Steel Sink & fittings	Nirali, Diamond, Cera or equivalent as approved
97.	Polyethylene water storage tank	Hycount, Sintex Aqua Tech, Ion exchange or equivalentas approved
98.	Water purifier	Eureka Forbes, Kent, Purit or equivalent as approved
99.	Water Heater	Racold, V-guard, Jaquar or equivalent as approved
100.	Manhole cover	Neco/Thermodrain or equivalent as approved

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101.	MS/GI Pipe & specials	Tata, Zenith, Jindal or equivalent as approved
102.	PVC, CPVC, PPR Pipe specials	Astral/Ajay/Ashirvad/Sudhakar or equivalent as approved
103.	CI pipe & specials	Electro steel, SAIL or equivalent as approved
104.	CI valve	Kirloskar, L & T. Leader or equivalent as approved
105.	Polyethylene Storage Tank	Syntex, Hycount or equivalent as approved
106.	AAC blocks	Siporex of BG Shirke Group / Aerocon of HIL / Xtralite of Ultratech
107.	PVC doors / windows	Fenesta / Venster / Elgin / Rehau
108.	Aluminium Hardware fittings	Argent / Classic (ISI marked) / Approved equivalent
109.	SS Hardware	Dorma / Hafele / Ozone
110.	Spider Glazing, Frameless glass Partitions & Pivoted doors	Dorma / Hafele / Ozone
111	Epoxy Paint	Fosroc/ Sika/ BASF
112	Lifts	Johnson/ Kone/ Shindler/ OTIS or Approved equivalent

2. PHE

S.No	Item	Makes
1	PVC/UPVC/CPVC Pipes & fittings	Astral/Ajay/Ashirvad/Sudhakar or equivalent
2	MS Pipes	Jindal / Sail/ TATA/Zenith or equivalent
3	M.S Fittings	AVR / HB/ R' Brand or equivalent
4	PVC Pipes & Fittings	Ashirwad /Astral / Finolex / Prince / Supremeor equivalent
5	Sand / Spun Cast Iron pipes & fittings	BIC / NECO / Nyco or equivalent
6	Manhole Covers - Cast Iron	BIC / NECO / Nyco / Thermodrain or equivalent
7	Manhole Covers - SFRC	Indian Hume Pipe /NECO/ Sudharsan/ Thermodrain or equivalent
8	R.C.C Hume Pipes	Indian Hume Pipe, Thirunelveli Spun Pipe or equivalent
9	Hot water insulation (Nitrile Rubber)	Armaflex /Vidoflex or equivalent
10	PVC Connection Pipes	Jaquar/Kohinoor /Parryware or equivalent
	Valves	Audco/Leader/ Advance/ equivalent
11	Butterfly valve	Advance /Audco/ Crawley & Ray/ Intervale/ Saunders or equivalent
12	Ball valves (Upto 40mm Dia and above)	Audco / RB / Leader or equivalent

13	Non-Return Valve	Advance / Intervale/ Saunders or equivalent
14	Y Strainer	Advance/Leader /Sant / Zoloto or equivalent
15	Pressure Reducing Valve	Cimberio or equivalent
16	Monoblock Pumps	ITT/ Kirlosker/ Grundfos / Wilo/ Armstrong or equivalent
17	Submersible Pumps	ITT/ Kirlosker/ Grundfos / Wilo or equivalent
18	STP	Ion Exchange/ SEPCON SYSTEM/ SEAMAKHI - TECH PRODUCTS/ Thermaxor equivalent
19	WTP	Ion Exchange/ SEPCON SYSTEM/ SEAMAKHI - TECH PRODUCTS/ Thermax/ Grundfos/ Green Eco Water Systems or equivalent
20	Sanitary Fixtures	Cera/Hindustan/Jaquar/Parryware/ Duravit, Kohler, Queo, Hindware, Grohe, Roca or equivalent
21	Plumbing Fittings	Cera/Grohe/Hindustan/Jaquar/Parryware/Kohler, Artize, Toto or equivalent
22	Galvanized Iron Pipe / MS Pipe - IS : 1239	Jindal / Tata Steel or equivalent
23	Expansion Bellow	Easyflex/Kinetics/Resistoflex or equivalent
24	Pressure Gauge	H-Guru / Fiebig or equivalent
25	HDPE Pipes	Supreme/Kodoor, Hasti or equivalent
26	Electromagnetic Water meter	Krohne marshall, Sitrans or equivalent
27	Rails and clamps	Mupro/Hi tech supports or equivalent

3. ELECTRICAL

S.No	Description of Items	Preferred Makes
1.	MS Conduit & Accessories (IS Marked)	AKG/BEC/RM CON/ Steel Craft/ NIC
2.	PVC Insulated FRLSH Copper Conductor Wires 1.1 KV Grade (ISI Marked)	Polycab/Havells/RRKable/Finolex/KEI
3.	Telephone/RJ-45/RJ-11/Modular Socket/Bell	Legrand (Arteor)/ABB(Tvisha)/MK(Orna) /Crabtree (Athena)/L&T(Entice)/Wipro North West (Stylus)
4.	MCB/Isolator/DB/RCCB/ELCB	Legrand/Schneider/Havells/ABB/Siemens /L&T/ C&S
5.	LED Luminaires/Light Fittings	Philips/Trilux/ Havells/LT/Wipro
6.	Ceiling Fan/Exhaust Fan/ Wall fan (IS Marked)	Havells/Crompton /Usha/Orient/Atomberg/Bajaj
7.	Telephone Cables/Wires (IS Marked)/ CoAxial TV Cables	Finolex/ Polycab / Havells / RR Kable / Delton /Batra Hanley
8.	Cat6/Cat6a/Fiber Optic cable & Associated Items	Schneider/Siemon/ Legrand / Beldon/DLink/Commscope/ Delton
9.	Telephone Tag Block DP Boxes	Krone (Germany)/ DLink /ITL

10.	Double Wall Corrugated (DWC) HDPE Pipe (ISMarked)	Rex/Duraline/Tirupati/Gemini/Universal/SFMC/As tral
11.	1.1KV XLPE(Alumium /Cu) Cable (IS Market)	Polycab/Havells/RRKable/KEI/Gloster
12.	Occupancy & Daylight Sensor	Philips/Wipro /ABB/HoneyWell/Legrand
13.	Electrical Control Panels	SPC Electrotech Pvt Ltd/Krypton/Precision S ystem Control/ Tricolite/ Adlec/Pristine/Application control panel.
14.	FeederPillars	SPC Electrotech Pvt Ltd/Krypton/Precision System Control/ Tricolite/Adlec/Pristine/Application Control Panels/RP Control
15.	ACBs	L&T (U-Power)/ Siemens (3WA)/ Schneider (Master Pact–MVS)/ABB(EMax)/Equivalent model of Legrand/Havells
16.	MCCBs	Siemens(3VL)/Schneider (Master Pact– NSX)/Equivalent Model in ABB/ L & T/ Equivalent model of Legrand/Havells/ C& S
17.	Data Rack & Accessories	Schneider/Rittal/Valrack/D-Link
18.	Current Transformers	Automatic Electric/Kappa/L&T
19.	Potential Transformer	Automatic Electric/Kappa/L&T
20.	Digital Indicating Measuring Instruments/Multifunction Meters/Digital Energy Meter	L&T/Conzerv/AutomaticElectric/ABB
21.	Selector Switch	L&T/Kaycee/Siemens /ABB/Automatic Electric/Kappa/Schneider
22.	Indication Lamp & Push Button/ Relay/Timer/Contactor	L&T/Kaycee/Siemens /ABB/Automatic Electric/Kappa/Schneider
23.	APFC Relays	L&T/ Siemens/Schneider/Ducati/ ABB/ Automatic Electric/Kappa
24.	Bus Trunking, Rising Main &Tap-Off Box	Legrand/L&T/Schneider/ C&S /ABB/ Siemens/ Schnieder Electric
25.	Cable Trays & Accessories	Legrand/Venus/Slotco/Pilco/Reeco/Rmco n/ Solution system/ Storrox
26.	Capacitor (IS Marked)	Neptune/Siemens/L&T/Schneider
27.	Cable Gland/Lugs/Thimbles	Commet/Dowells/ Raychem/ Gripwell Jainson/ HEX/ Comex
28.	Motor Pump set	Kirlosker/KSB/CRI/Maxwell/ Crompton greaves
29.	G.I. PIPE	Jindal/TATA/Prakash Surya
30.	Electrical cable	RR Kabel/polycab/Finolex/Havells/ Grandlay
31.	Starter Panel	BCH/LK(L&T)/Crompton/ CRI
32.	Valves	Kirlosker/Sant/Zoloto/Leader
33.	UV + RO Water Purifier	Thermax/EurekaForbes/Bluestar/Usha

4. FIRE FIGHTING MATERIAL

S.No	Description of Items	Preferred Makes
1.	Pressure Switches	Honeywell / Tyco/ Ansul/ Hygood / Pyroquench/ Ceasefire Pvt Ltd/ Safex/ASES
2.	Nozzles	Honeywell/ Tyco / Ansul/ Hygood / Pyroquench/ Ceasefire Pvt Ltd/ Safex/ASES

3.	Pipes (MS/GI)	Tata/Jindal (Hisar)/SAIL/Prakshsurya
4.	RRL First Aid Hose Reel	Jyoti/Newage /Padmini /Omex/Minimax/ Safe guard
5.	Single headed Hydrant valves, four-way Fire Brigade inlet & shut off nozzle	Newage /Minimax/Superex/safeguard/Omex
6.	Sprinkler system, Heads, Alarm valves, flexible pipe etc	HD/TYCO/Viking/Omex
7.	CI Sluice Valve, Non return valve, Ball Valve, Butterfly valve & Y-strainer	Kirloskar/Sant/castle/Kartar/Zoloto/H-Guru
8.	Pressure switch	Fiebig/H-Guru/Switzer/Danfoss
9.	Pressure Gauges	Fiebig/H-Guru/Switzer/Danfoss
10.	Fire Pump/ drinking water pump/de-watering pump	Mather&Platt/Kirloskar/KSB/Grundfos/ Wilo
11.	Motor	ABB/Siemens/CromptonGreaves/ Kirloskar
12.	Hook	JaquarACN-CHR-1161N, Hindware
13.	PaintPrimer	Asian/Jenson Nicholson/Berger/Nerolac
14.	Flow switch	System sensor/plotter/Johnson control/ H-guru
15.	Branch pipe	Newage/Superex/Omex
16.	Vibration Eliminator Connector/ Metallic Expansion Bellows	Resisto Flex/EasyFlex/D.wren
17.	Installation Control Valve	HDfire/Newage/Padmini /Safex/Tyco
18.	First Aid Hose Drum	Newage Bombay/ Minimax/ Superex/CRC/ Getech/Kanex/Safeguard
19.	Signaling Line Circuit Cable	RRkabel/polycab/Finolex/Havells
20.	CO2 and ABC type Fire extinguisher	Minimax/Safex/Ceasefire/Kanex
21.	Firealarmpanel	Honeywell/ Notifier/ Bosch/Siemens/Edwards/simplex
22.	Repeater panel	Honeywell/ Notifier/ Bosch/ Siemens/Edwards/simplex
23.	Intelligent FA panel	Honeywell/ Notifier/ Bosch/ Siemens/Edwards/simplex
24.	Photo thermal Detector	Honeywell/ Notifier/ Bosch/ Siemens/Edwards/simplex
25.	Response Indicator	Ravel/ Honeywell/ Bosch/ Siemens/Edwards/simplex
26.	Programmable Sounder	System Sensor/ Spectralert/ Honeywell/ Bosch/Siemens/Edwards/simplex
27.	Horn-Cum-Strobe	SystemSensor, SYS-HSR-FIRE/ Honeywell/Bosch/Siemens/Edwards/ simplex
28.	Address able control module/ relay Module	Honeywell, Notifier, /Bosch/ Siemens/Edwards/simplex
29.	FireFighterTelephone handset	Honeywell/Bosch/Siemens/Edwards/ simplex
30.	FireFighter Phone Jack	Honeywell/Bosch/Siemens/Edwards/ simplex
31.	Voice Alarm Controller (with button Paging station)	Honeywell/Bosch/Siemens/Edwards/ simplex
32.	1.5/3/6WCeilingSpeaker	Honeywell, /Bosch/Siemens/Edwards/ simplex
33.	1.5/3/6WwallSpeaker	Honeywell/EN/Bosch/Siemens/Edwards/ simplex
34.	Voice Command KeyPad	Honeywell, /Bosch/Siemens/Edwards/ simplex
35.	Digital	Honeywell/Bosch/Siemens/Edwards/ simplex
36.	Beam detector	Honeywell/Bosch/siemens/Edwards/ simplex

37.	Fire survival armored cable	RRKabel/Polycab/Lapp/Bonton/Avocab
38.	PC in whole as per Tender	IBM/DELL/HP/LENOVO
39.	Audio Speaker Cable	RRKabel/Lapp/Polycab/Belden/Havells/ Plutron

5. LIFT

S.No.	Description of Items	Preferred Makes
1	16 Passenger lift	Manufacturers meeting the eligibility criteria as mentioned in the tender documents.

6. DG SET

S.No.	Description of Items	Preferred Makes
1	Diesel Engine	Cummins/Perkins /Caterpillar/ Kirloskar/ Baudouin/ Mitsubishi
2	Alternator	Crompton/Stamford/LeroySomer/Kirloskar/ Mitsubishi/ Siemens
3	Anti Vibration Mountings	As Per Manufacturer Standard Practice
4	Synchronization Panel With AMF	As per OEM Of DG Sets/As Per EI Works
5	MS Angle/Channel/ISMB	Tata/Jindal (Hisar)/Sail
6	PLC	Siemens/L&T/Cummins/Mitsubishi/ AllenBradly
7	Battery	AMCO/Exide/Prestolite/Amaraja/Luminous
8	Acoustic Enclosure	OEM/OEA of DG set
9	Cable glands double compression with earthing links	COSMOS/ Comet/3D/3M/TrinityTouch/Dowells
10	Cushy Foot	Dunlop or equivalent
11	Others Item	As Per Sub-Station/IEI Work

7. SOLAR

S.No.	Description of Items	Preferred Makes
1.	Solar PV module	Reneways /Goldi/ BHEL/BOSCH/ Vikram Solar/Tata Solar/Adani/ Luminous/Jackson
2.	Grid Tied Solar Inverter	Delta/SMA/ABB/TMEIC/Sungrow/Hua wei/ Hitachi/ Schneider/ Luminous/ Microtek
3.	Solar PV Controller	As per OEM standards
4.	LT Cable/DC Cable/FRLSH PVC Insulated Wire (IS Marked)	As per IEI Work
5.	Electrical Panel/ MCCB/ MCB/ ACB/ ELCB/ Isolator/Allother Accessories	As per IEI work
6.	Maintenance free batteries	Exide/Prestolite/Amaraja/Luminous/ AMCO

8. MEDIUM VOLTAGE EQUIPMENT

S.No.	Description of Items	Preferred Makes
1.	OPTICAL FIBER CABLE	Belden/CommScope/Tyco/Legrand/ Panduit/Molex
2.	LIU	Belden/CommScope/Tyco/Legrand/ Panduit/Molex
3.	Fiber patch Cord	Belden/CommScope/Tyco/Legrand/ Panduit/Molex
4.	Cat 6UTP Cable	Belden/CommScope/Tyco/Legrand/ Panduit/Molex
5.	Jack Panel	Belden/CommScope/Tyco/Legrand/ Panduit/Molex
6.	Patch Cord	Belden/CommScope/Tyco/Legrand/ Panduit/Molex

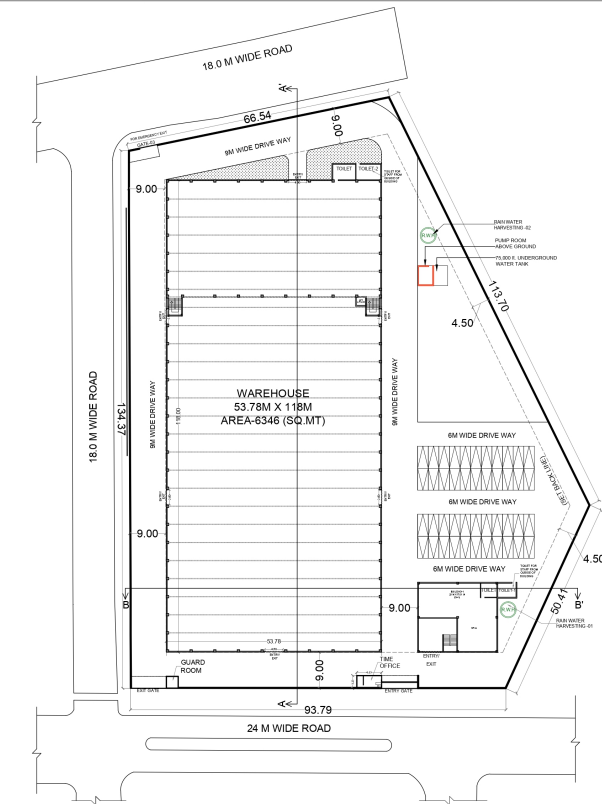
7.	Face Plate	Belden/CommScope/Tyco/Legrand/ Panduit/Molex
8.	Outdoor Encloser Network rack	Rittal/APW/Valrack/Netrack/Legrand
9.	PTZ	Avigilon/Axis/Bosch/Pelco/Honeywell
10.	Hard Disk	WD/Seagate/Toshiba/HP
11.	LED Professional Display Panel	Panasonic/LG/Sony/Samsung
12.	L2 Switch/L3Switch/Industrial Grade Switch	Cisco/HP/Extreme/Arista/Netgear
13.	Fiber Module	Cisco/HP/Extreme/Arista/Juniper/ Netgear
14.	Work Station PC	HP/Dell/IBM/Lenovo

9. Sub-Station Equipment and LT Distribution

S.No.	Description of Items	Preferred Makes
1.	HT VCB Panel	Siemens/Areva/Crompton/ABB/ Schneider/L&T/ Electricfield
2.	Transformers	Voltamp/Schneider/ABB/CG/PVJ Power/Areva (Alsthom)/Kirloskar
3.	LT Panels/Sub LTPanels/Essential panels/ TTA Panels	Tricolite /Adlec / SPC Electrotech Pvt Ltd / Application Control Panel/Precision System Control /Pristine/ TTA Panel Seimens (Sivacon)/Schneider (Prisma)/L&T (Ti)/ABB(ArTuK)/Vinayak Engineering Projects HPFC Panel – Siemens/ABB/Schneider/L&T/Legrand.
4.	Air Circuit Breakers	L&T(U-Power)/ Siemens(3WL)/ Schneider (MVS) / ABB(E-Max)/ Equivalent Models of Legrand.
5.	MCCBs	As per IEI Work
6.	MCBs /ELCBs	As per IEI Work
7.	Current Transformers	Automatic Electric/Kappa/L&T
8.	Potential Transformers	Automatic Electric/Kappa/L&T
9.	Digital Meter/Multifunction Meters/Digital Energy Meter	As per IEI Work
10.	Selector Switch	As per IEI Work
11.	SMF Batteries	AMCO/Exide/Prestolite/Amaraja/ Luminous
12.	Indication Lamp & Push Button	As per IEI Work
13.	APFC Relays	L&T/C & S/Siemens/ Schneider/ABB
14.	Sandwich Bus Trunking, Rising Main & Tap Off Box	Legrand/L&T/Schneider/ABB/C&S
15.	HT Cable (ISI Marked)11KV	Polycab/Havells/RR Cable/Batra Hanley
16.	LT Cable(ISI Marked)1.1 KV	Polycab/Havells/RR Cable/Batra Hanley
17.	Fire Extinguisher	CeaseFire/Lifeguard/Padmini/Superex/ Omex/ Minimax
18.	CableTrays &Accessories	As per IEI Work
19.	Capacitor (Ismarked)	Neptune/ Siemens/L&T
20.	Cable Gland/Lugs/Thimbles	As per IEI Work
21.	Heat Shrinkable HT Cable End Termination	Denson/Raychem/ Capseal

TENDER DRAWINGS

WAREHOUSE PROPOSAL FOR APPROVAL



SITE PLAN

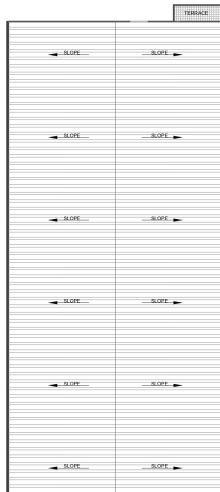
AREA CHART			
S.NO	LENGTH	BREATH	AREA
1	118.00	53.78	6346.04
2	4.08	12.42	50.97
0	TOTAL		6397.01

AREA CHART			
S.NO	LENGTH	BREATH	AREA
1	6.05	3.53	21.36
0	TOTAL		21.36

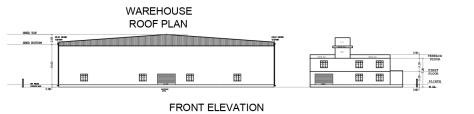
AREA CHART			
S.NO	LENGTH	BREATH	AREA
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0	TOTAL		1598.06



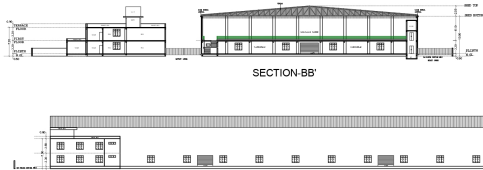
SECTION-AA'



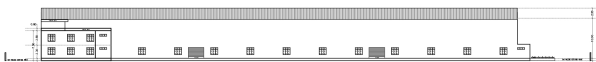
MEZZANINE FLOOR PLAN
(WAREHOUSE BLOCK)



FRONT ELEVATION



SECTION-BB'



SIDE ELEVATION



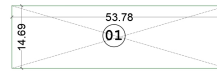
GROUND FLOOR PLAN
BUILDING BLOCK-1



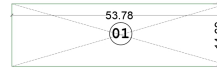
FIRST FLOOR PLAN
BUILDING BLOCK-1



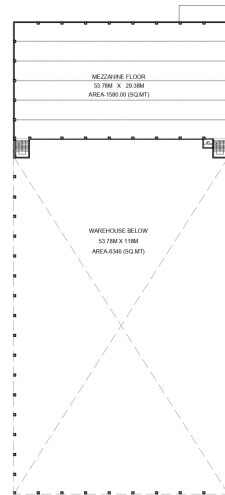
MUMMY PLAN
BUILDING BLOCK-1



MEZZANINE FLOOR BLOCK PLAN-01



MEZZANINE FLOOR BLOCK PLAN-02



WAREHOUSE
BLOCK PLAN

AREA CHART			
GROUND FLOOR			
S.NO	LENGTH	BREATH	AREA
1	11.47	20.00	229.4
2	6.05	6.38	38.48
3	6.0500	3.53	21.5565
4	6.05	10.11	61.1655
5	3.96	4.80	19.0080
TOTAL			350.4

AREA CHART			
FIRST FLOOR			
S.NO	LENGTH	BREATH	AREA
1	11.47	20.00	229.4
2	6.05	6.38	38.48
3	6.0500	3.53	21.5565
4	6.05	10.11	61.1655
5	3.96	4.80	19.0080
TOTAL			350.4

CERTIFICATE

THIS IS CERTIFY THAT THERE IS NO
PROP OR ROAD MAINTAINED BY ANY
GOVT. DEPT. OR BELONGING TO RAILWAY
ADM. WITH IN 30.48 MTS. FROM THE SITE IN
QUESTION

SPECIFICATION

1. P.C.C. IN FOUNDATION & FL.	1:4:8
2. D.P.C. 50 MM THICK	1:1:3
3. BRICK WORK IN C. MORTAR	1:6
4. ALL R.C.C. WORK	1:2:3
5. CEMENT PLASTER	1:6

AREA STATEMENT

TOTAL PLOT AREA	- 13532.82 (SQ.MT)
GROUND COVERAGE @ 50%	- 6766.41 (SQ.MT)
PERMISSIBLE FAR	- 0.65
	- 0.65(13532.82(SQ.MT))
	= 8796.33(SQ.MT)

COVERED AREA STATEMENT

FLOOR	AREA(SQ.M)
WAREHOUSE BUILDING BLOCK AREA	- 6370.75 (SQ.MT)
MEZZANINE FLOOR AREA	- 1580.00(SQ.MT)
BUILDING BLOCK - 1 GROUND FLOOR AREA	- 350.00(SQ.MT)
BUILDING BLOCK - 1 FIRST FLOOR AREA	- 350.00(SQ.MT)
TOILET BLOCK-1	- 19.00 (SQ.MT)
TOILET BLOCK-2	- 26.09 (SQ.MT)
TOTAL COVERED AREA	- 8695.75 (SQ.MT)

ACHIEVED FAR	- 98.85% (SQ.MT)
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ACHIEVED GROUND COVERAGE	- 99.99%
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PARKING

REQUIRED PARKING @ 1 CAR PER 100.0 SQMT PERMISSIBLE FAR	
DOUBLE LEVEL MECHANIZED PARKING (44X2)	- 88 NOS.

PROJECT-

PROPOSED WAREHOUSE AT
PLOT NO Y-9, INDUSTRIAL AREA,
TRANS GANGA CITY, UNNAO, U.P

OWNED BY-

ARTIFICIAL LIMBS MANUFACTURING
CORPORATION OF INDIA, NARAMAUM
KANPUR, U.P

ORG TITLE-

SUBMISSION DRAWING

SHEET NO.-01

TOTAL SHEET- 01

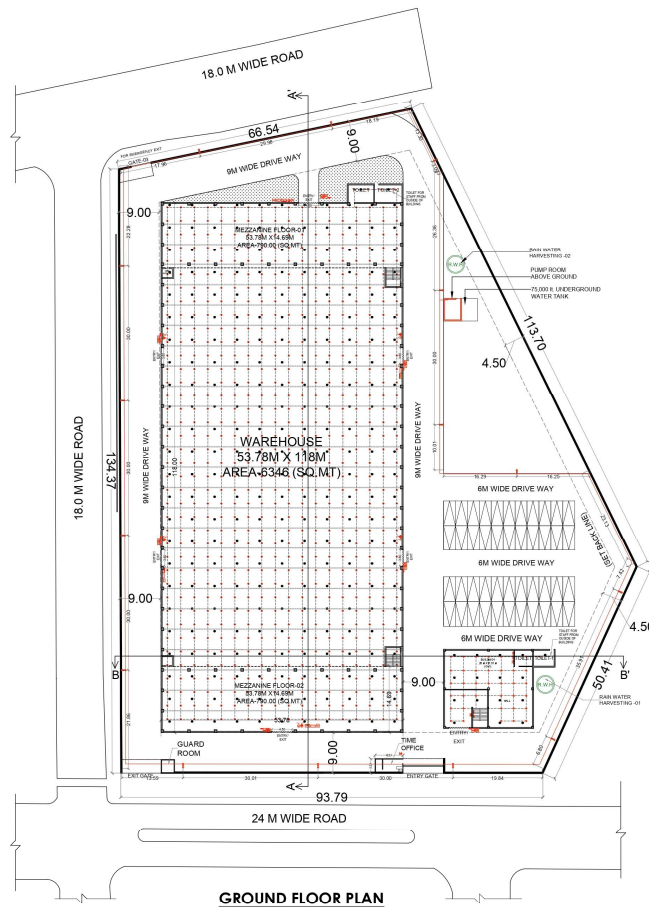
ALL DIMENSIONS ARE IN METERS
SCALE- 1:800

ARCHITECTS -

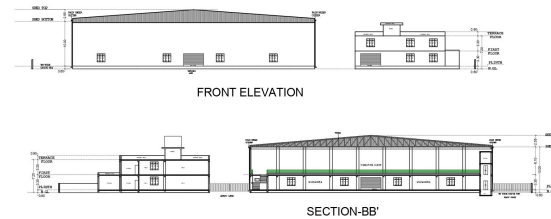
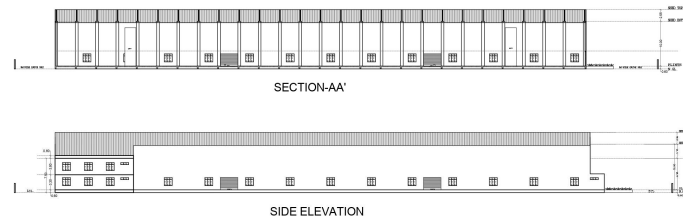
M/S Design Incorporates
ARCHITECTS & ENGINEERS
6/22/ PAND. KUNDA AREA, KANPUR (U.P.)
Phone: 0512-3066753

OWNER'S SIGN.

ARCHITECT'S SIGN.



S.NO	SYMBOL	NOMENCLATURE
1		MANUAL CALL POINT
2		HOOTER
3		SPRINKLER
4		SPRINKLER HEADER
5		FIRE EXTINGUISHER (CO ₂ 4.5kg)
6		FIRE EXTINGUISHER (W.C. 4.5kg)
7		FIRE ALARM CONTROL PANEL
8		HOSE REEL
9		WET RISER SHAFT + SPRINKLER RISER
10		DOWN COMER
11		AUTOMATIC CALL POINT
12		SMOKE DETECTOR
13		YARD HYDRANT



CERTIFICATE	
THIS IS CERTIFY THAT THERE IS NO PROP. OR ROAD MAINTAIN BY ANY GOVT. DEPT. OR BELONGING TO RAILWAY ADM. WITH IN 30.48 MTS. FROM THE SITE IN QUESTION	
SPECIFICATION 1. P.C.C. IN FOUNDATION & FL. 1:4:8 2. D.P.C. 50 MM THICK 1:1:3 3. BRICK WORK IN C. MORTAR 1:6 4. ALL R.C.C. WORK 1:2:3 5. CEMENT PLASTER 1:6	
AREA STATEMENT	
TOTAL PLOT AREA	- 13532.82 (SQ.MT)
GROUND COVERAGE @ 50%	- 6766.41 (SQ.MT)
PERMISSIBLE FAR	- 0.65
	= 0.65 x 13532.82 (SQ.MT)
	= 8796.33 (SQ.MT)
COVERED AREA STATEMENT	
FLOOR	AREA (SQ.M)
WAREHOUSE BUILDING BLOCK AREA	- 6370.75 (SQ.MT)
MEZANINE FLOOR AREA	- 1860.00 (SQ.MT)
BUILDING BLOCK -1 GROUND FLOOR AREA	- 350.00 (SQ.MT)
BUILDING BLOCK -1 FIRST FLOOR AREA	- 350.00 (SQ.MT)
TOILET BLOCK-1	- 19.00 (SQ.MT)
TOILET BLOCK-2	- 24.09 (SQ.MT)
TOTAL COVERED AREA	- 8695.75 (SQ.MT)
ACHIEVED FAR	- 98.85% (SQ.MT)
ACHIEVED GROUND COVERAGE	- 99.99%
PARKING	
REQUIRED PARKING @ 1 CAR PER 100.0 SQMT PERMISSIBLE FAR	
DOUBLE LEVEL MECHANIZED PARKING - (44X2) - 88 NOS.	
PROJECT-	
PROPOSED WAREHOUSE AT PLOT NO Y-9, INDUSTRIAL AREA, TRANS GANGA CITY, UNNAO, U.P	
OWNED BY:	
ARTIFICIAL LIMBS MANUFACTURING CORPORATION OF INDIA, NARMAUM KANPUR, U.P	
DRG TITLE-	FIRE SUBMISSION DRAWING
	SHEET NO. -01
NOTES	TOTAL SHEET- 01
	ALL DIMENSIONS ARE IN METERS
SCALE- 1:700	
ARCHITECTS - M/S Design Incorporates ARCHITECTS & INTERIOR DESIGNERS 6/220 JHANSI ROSSBY AREA, VARANASI, KANPUR (U.P.) Phone: 0512-3062753	
OWNER'S SIGN.	ARCHITECT'S SIGN.